~~Discrete~~ Probability Theory

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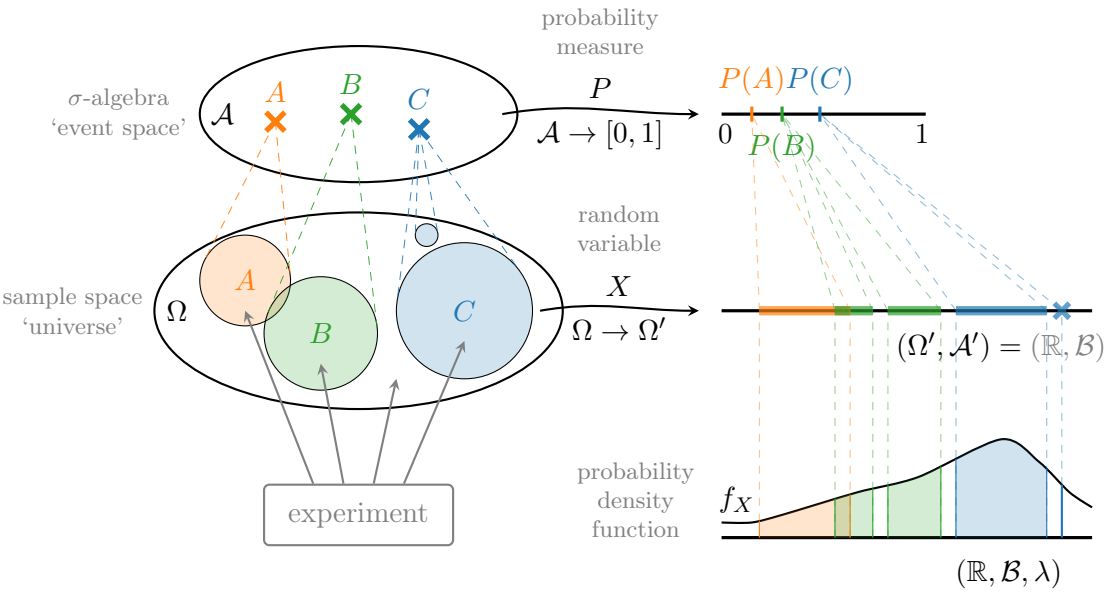

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Literature and further reading

There is no single book or script that we follow in this course. The books mentioned below have served as inspirations for various parts.

Probability theory

- E. T. Jaynes. *Probability theory: The logic of science*, Cambridge University Press, 2003 [book] (Jaynes, 2003) [Available here.]
- A. Klenke. *Probability theory*, Springer, 2013 [book] (Klenke, 2013)
- H.-O. Georgii. *Stochastik: Einführung in die Wahrscheinlichkeitstheorie und Statistik*, Walter de Gruyter GmbH & Co KG, 2015 [book] (Georgii, 2015)
- D. P. Bertsekas and J. N. Tsitsiklis. *Introduction to probability*, Athena Scientific, 2000. [book + lecture notes] [Available here.]
- A.F. Siegel. *Counterexamples in Probability*, Routledge, 2017 (Siegel, 2017)

0. What is probability?

This question has kept scholars in many disciplines busy over centuries if not millennia. In this lecture, we will introduce the mathematical machinery to sensibly talk about and perform computations involving probability. We sometimes provide glimpses into philosophical or practical aspects whenever relevant. These come in the form of "philosophical interludes" or "(practical) excursions". At its core, it's a theoretical course teaching concepts.

— Start Philosophical Interlude —

Remark 0.1 (historical developments). Although gambling dates back thousands of years, the birth of modern probability is often considered to be a 1654 letter from the French aristocrat and notorious gambler Chevalier de Méré to the mathematician and philosopher Blaise Pascal. This concerned the "problem of points", which is about how to split a pot when a repeated game of chance has to end prematurely fairly with respect to the current win-lose statistics. Blaise Pascal discussed this problem with Pierre de Fermat in his correspondence and they developed some of the key concepts underlying modern probability. In the same year (1654), Pascal reportedly had a religious vision and largely abandoned all scientific work.

Other key steps were Pierre-Simon Laplace's 1812 work on "Théorie Analytique des Probabilités," which unified much of probability theory at the time and introduced Bayes' theorem (independently of Bayes) and the axiomatization by Andrey Kolmogorov in 1933 (which we will build upon).

Unlike Pascal, Laplace seemed not to be too religious. Napoleon once asked him why he had not talked about God in his book to which he replied: "I had no need of that hypothesis."

Remark 0.2 (historical developments of this lecture). Friedrich L. Bauer started the first study course on "Informationsverarbeitung" at TUM in 1967. For at least a few decades, computer science was thought of as inherently discrete, leading to core modules such as "discrete structures" and "discrete probability theory" early on (potentially in the 80s or 90s). It made sense to only consider *discrete* probability theory. Since then, a lot has changed. In particular, the advent of randomized algorithms, machine learning, AI, etc have rendered *continuous* probability theory equally important also for computer scientists. The content of the course has been adapted—the name has not.

Example 0.3 (Will they wake up? (Adapted from lecture notes by Dimitri P. Bertsekas and John N. Tsitsiklis)**.** After a complicated surgery, the patient is still in a coma. The following conversation between the doctor and a relative of the patient unfolds.

> Relative: *What's the probability they will wake up?*
> Doctor: *I hope they do. At the moment we can only wait and see.*
> Relative: *Yes, but what is the probability that they will?*
> Doctor: *Each case is different, we have to wait.*
> Relative: *Well, but out of a hundred coma patients after similar procedures, how many would you expect to wake up again?*
> Doctor: *I told you, every person and situation is different. Some wake up, some don't.*
> Relative: *Ok, if you had to bet, would you bet they'll wake up?*
> Doctor: *Yes, I'd bet they will.*
> Relative: *Good. Would you be willing to lose two dollars if you're wrong, and gain one dollar if you're right?*
> Doctor: *What a sick thought, this is not a game.*

Example 0.4 (the card game)**.** I have two cards in a closed opaque box, one is red on both sides and the other is red on one and blue on the other side. I pull out one card and place it on the table and a red side faces up. What's the probability that the other side is also red?

Example 0.5 (Russian roulette)**.** You play Russian roulette with a six shooter revolver and there are precisely two bullets in neighboring chambers. Your opponent goes first, spins the barrel, pulls the trigger and the gun does not fire. Now it is your turn. Should you spin the barrel again or pull the trigger right away?

Example 0.6 (sleeping beauty)**.** On Sunday evening you are put to sleep with a potent pill that also wipes your entire memory. You are told that after you fall asleep, a coin will be tossed. If it comes up heads, you will be woken up on Monday and your life continues normally. If it comes up tails, you will be woken up on Monday, put back to sleep (wiping your memory) and woken up again on Tuesday after which you continue your life. In any case, each time you're woken up, you will be asked whether you believe the coin came up heads or tails. Next thing you know, you wake up and are being asked about the probability of the coin having come up heads.

Example 0.7 (predictions)**.** Let's see what's trending on polymarket today...

Example 0.8 (random numbers (adapted from Hennig et al. (2022)))**.** Which of the following sequences of numbers are random?

(i) `433444666644444111111`

(ii) `69468398352`

(iii) `1000111111011111111100101000001`

(iv) `2849607135`

(v) `0868308472`

Here's how we got them.

(i) We got this from throwing a regular die 6 times and repeating the i-th result i times. The sequence is unpredictable, but it would not pass any standard test of randomness.

(ii) The 2016th to 2026th decimals of π. The decimals of π follow no structure and are fine for use in randomized algorithms, but are far from "random."

(iii) "These are digits taken from a CD-ROM published by George Marsaglia containing random numbers generated using a variety of physical random number generators. These digits (perhaps) were once 'random'. But, now that you know where we got them, they are obviously deterministic. But wait! Marsaglia's CD is actually not available online anymore. Most original copies have long been lost as retiring academics cleared out their offices. We will admit that we do not own a physical copy either. Does this revelation make those digits above random again?" (Hennig et al., 2022)

(iv) This was generated by ChatGPT with the prompt "a sequence of 10 random digits." While ChatGPT is reportedly bad at providing random numbers "following a given distribution," since I do not know the seed/temperature used for this generation, these digits are random. Yet, with the exact model weights and seed in hand, they are perfectly deterministic. But then again, hardware non-determinism may have made them random again.

(v) These digits were generated via `[random.randint(0, 9) for _ in range(10)]` using Python's built in `random` library. These are based on a Mersenne twister pseudo random number generator, which generally provides "good randomness." However, I first seeded it with 2849607135, so this sequence has turned deterministic a half-sentence ago.

Remark 0.9 (where does randomness come from?).

- In example 0.4, the randomness comes from me "choosing blindly between the two cards," in example 0.6 from "tossing a coin," and in example 0.5 from "spinning the chamber." We assume these mechanisms will result in an equal probability for the different possible events. However, once the precise positioning and movement of my hands are fixed, the result could be predicted and is thus deterministic rather than random in all these examples.

- Example 0.8 revolves around random number generators and how/where they get their "randomness" from. It highlights that a slight change in "ignorance" (e.g., knowing a seed) can make an otherwise unpredictable sequence entirely deterministic. We will not dive into the subtleties of (pseudo) random number generation in this lecture.
- In example 0.3, it is more subtle, we do not directly invoke a mechanism that is supposed to induce randomness. Instead, the randomness stems from our imperfect understanding of the system (a human). Again, one could argue that "if we knew more" (essentially all details of how a human "functions"), we should be able to predict whether they will wake up or not.
- In example 0.7, it's a crowd-sourced (heavily biased), subjective, collective opinion/belief/hope.
- Quantum mechanics is often invoked for a source of "true randomness." When measuring well prepared quantum states, the measurement outcome can indeed not be predicted perfectly, yet, in principle, the probabilities of different measurement outcomes can be predicted perfectly.
- In the mathematical models we'll introduce, e.g., of a coin toss, probabilities represent uncertainty or ignorance about certain aspects of a system. We want to express consistently and quantitatively what we believe about uncertain states before observation. See also viewpoints 0.10, 1.29 and 1.30.
- This can be summarized as "there is no probability of something happening, but only a probability of something happening *given what I currently know*". This podcast with Andrew Jaffe gives a nice account of how models and assigning probabilities to them help us understand the world.

Viewpoint 0.10 (determinism, free will, compatibilism). In remark 0.9, one is tempted to conclude that if only we knew enough (up to quantum effects) nothing is ever "truly random" and uncertainty would be banished. In practice, we can't, so we encode ignorance with probabilities. The idea that the universe is deterministic at its core is called "determinism." Taking quantum mechanics into account, this means that if one knew the wave function of the universe, its evolution—and thus the probability of any future event happening—can be perfectly predicted (if the laws of physics are known perfectly).

Aside from leaving little room for "real randomness, serendipity, or luck" the deterministic viewpoint for many people also clashes with the subjective experience of free will and agency. However, determinism and free will are not logically incompatible. In philosophy, various versions of "compatibilism" describe how and why a deterministic universe and the capacity for free will are not contradictory. Randomness, determinism, and free will are subject of ongoing debate in philosophy and physics alike.

— End Philosophical Interlude —

1. Basic objects of probability theory

— Start Motivation —

Often, our intuition about probability is quite good for simple scenarios, but as we encounter more complex situations, especially involving infinity and continuous spaces, we need a robust mathematical framework to ensure consistency and to handle the subtleties. In simple finite or discrete settings, the mathematical machinery appears to be an overkill. In these motivation section, we aim at providing intuitive examples for why it is needed.

The finite world: a deck of cards

Imagine a standard, well-shuffled deck of 52 playing cards. Clearly, there is probability involved in many card games like poker, so probability theory should be a useful tool to talk about different events happening with that set of cards. Let's consider first, drawing cards from a well-shuffled deck. There are 52 different possible outcomes. The set of outcomes, we call Ω, so $|\Omega| = 52$. It is also called *sample space*. Note that we deliberately choose Ω to be "the types of outcomes we care about". We do not include in the "outcome" where on the table the card comes to lie, who picked it, or which corner it was picked up at. These are all properties of the actual "drawing", we just choose not to care about those aspects.[1] In this card drawing setting, talking about probabilities is simple: If we draw one card, there's a certain probability we draw the ace of spades, $A\spadesuit$. Intuitively, since the deck is well-shuffled, each card is equally likely and since there's 52 of them, we conclude $P(\{A\spadesuit\}) = \frac{1}{52}$, where we freely use $P(\ldots)$ to just stand for "probability of...".[2] This simple idea of assigning a number (a probability) to each individual outcome is the essence of a *probability mass function (pmf)*.[3] For our deck of cards, the pmf $p : \Omega \to [0, 1]$ is $p(\omega) = \frac{1}{52}$ for every card $\omega \in \Omega$. Each possible card "is assigned" probability $\frac{1}{52}$. We could also imagine a not so well shuffled deck that results in another distribution over the cards. Let's say a seasoned magician has shuffled the deck in an attempt to put all aces to the top. In this case, $P(A\spadesuit)$ may be larger than $\frac{1}{52}$. That can only work as long

[1] In certain card games, the color/symbol of the cards may be irrelevant and only their values count. Then it would make sense to choose $\Omega = \{2, 3, 4, \ldots, 10, J, Q, KA\}$ instead with $|\Omega| = 13$ even though nothing has changed about the physical process of drawing cards and the deck of cards itself.

[2] You wonder why I put curly braces around $A\spadesuit$? You should!

[3] At this point, the name pmf appears overly complicated. Why not just call it a "probability function" or simply "probability"?

as another card (or some other cards) are now less likely. In the end, the probabilities should add to 1. It wouldn't really work out to have a probability of more than $\frac{1}{52}$ for each card, when there's exactly 52 of them. So besides the choice of "what outcomes do we care about", we also have to specify the probabilities of each outcome.

Notice two key properties that our intuition demands whenever we assign probabilities to each of a finite collection of outcomes:

(i) $p(\omega) \geq 0$ for all $\omega \in \Omega$, probabilities can't be negative.

(ii) $\sum_{\omega \in \Omega} p(\omega) = 1$, the probabilities of all possible individual outcomes must sum to one. (Here, $52 \times \frac{1}{52} = 1$).

Any such function is a pmf of some distribution, which is just to say that it fully describes the distribution. So for finite sample spaces, the pmf "is the distribution and all you need to know." Let's see what this means.

So far, we only have probabilities for individual cards, so called "singleton events". But what's the probability of drawing a King (any King, that is)? In your head, you will likely immediately say: "There are 4 kings, each has a probability of $\frac{1}{52}$ so it's $\frac{4}{52}$." Spot on! You have added the probabilities of the individual events. If we were to write this down, we could write $P(\text{drawing a king})$ or we denote by E_K the event "a king is drawn" and write $P(E_K)$. This event corresponds to the set of outcomes $\{K\spadesuit, K\clubsuit, K\heartsuit, K\diamondsuit\}$. So whenever you talk about "aggregate outcomes," like a king is drawn, or a 7 is drawn, or a black card is drawn, it's easiest to write down the set of outcomes that constitute this event. So to repeat, *outcomes/samples* are the individual things that can happen (here the 52 cards), whereas *events* can be collections of outcomes we are interested in (for example all kings).

Since the four outcomes making up this event are mutually exclusive (4 different cards), our intuition (and the axioms we'll formalize) tells us to sum their probabilities: $P(E_K) = p(K\spadesuit) + p(K\clubsuit) + p(K\heartsuit) + p(K\diamondsuit) = \frac{4}{52} = \frac{1}{13}$. Similarly, for the event E_R "a red card is drawn" (26 cards), $P(E_R) = \frac{26}{52} = \frac{1}{2}$. Note how we now assign probabilities not just to individual outcomes, but full events. The pmf only assigned probabilities to individual outcomes. Therefore we distinguish between p for the pmf, which takes as input individual $\omega \in \Omega$ and P, which will be the *probability measure*, which takes as input events (i.e., sets of outcomes). Therefore, everything you plug into P should be a subset of Ω and everything you plug into p should be an element of Ω. Trivially, we get for singleton events that $P(\{7\clubsuit\}) = p(7\clubsuit)$. Again, this distinction appears superfluous until we hit the infinite world.

This process of adding outcome probabilities to get event probabilities works for any subset of Ω. That is, we could be interested in "any possible event." We can ask for "what's the probability of drawing a red card or the $7\spadesuit$ or any 8." The probability of that is simply the sum of probabilities of each of the distinct(!) outcomes within that event. You must not add the probability for $8\heartsuit$ twice, once as a red card and once as an 8. Assume again the not so well shuffled deck, where aces are more likely to be near the

top. Everything works exactly the same way here. The distribution is not uniform, that is, not all cards have a probability of $\frac{1}{52}$, but we can still sum the individual probabilities to get the probability of the event: $P(\text{an ace is drawn}) = P(\{A\spadesuit, A\clubsuit, A\heartsuit, A\diamondsuit\}) = p(A\spadesuit) + p(A\clubsuit) + p(A\heartsuit) + p(A\diamondsuit) > \frac{4}{52}$. The only difference here is that we can't just count the outcomes within the event and divide by 52, which was a convenient shortcut for the uniform distribution, but we actually have to sum up the values.

Since we can define the probability of any event $A \subseteq \Omega$ as $P(A) = \sum_{\omega \in A} p(\omega)$, we can not only map individual outcomes to probabilities (like each card mapping to $\frac{1}{52}$), but we can directly assign probabilities to entire events (sets of outcomes). In this finite case, we can assign a probability to *every* possible subset of Ω. The collection of all subsets is the *power set*, $\mathcal{P}(\Omega)$. So here we can extend the "probability function" $P(\bullet)$ to

$$P : \mathcal{P}(\Omega) \to [0, 1] \,, \qquad A \mapsto P(A) := \sum_{\omega \in \Omega} p(\omega) \,.$$

The lower case p on the right hand side is used to compute probabilities of any event using P. We now call P a *probability measure* (it "measures" probabilities). We can always define it on all possible subsets (the power set $\mathcal{P}(\Omega)$) so that's what we'll do. For finite sample spaces, we now have a probability measure P, which we can feed with any element of the entire power set $\mathcal{P}(\Omega)$ over the outcomes Ω. The triple $(\Omega, \mathcal{P}(\Omega), P)$ forms a *probability space*. Again, here it seems that we shouldn't need to specify $\mathcal{P}(\Omega)$ explicitly, and P appears to be a much more complex object than p. Note how P is in a sense the "more fine grained" or "more complete" distribution, but in the finite case all computations are done with the pmf and it's the one natural object to work with. So intuitively, a probability space for finite sets of outcomes should just be the tuple (Ω, p). This is absolutely correct, but unfortunately, does not scale to uncountably infinite spaces Ω. In the discrete case: give me a pmf and everything is said. From there on, every question is essentially summing the pmf over different inputs (the event of interest).

The nice infinities: countably infinite outcomes

This whole idea also extends nicely to *countably infinite* sample spaces, like the set of natural numbers $\mathbb{N}$, whole numbers $\mathbb{Z}$ or rational numbers $\mathbb{Q}$. If instead of drawing one card from a well-shuffled deck, let's instead ask the following: I shuffle the deck, you draw a card, I re-shuffle the entire deck, you draw a card, and so on. We are interested in the probability of us needing k iterations of that game until you draw the $A\heartsuit$. Theoretically, the set of possible outcomes is $\Omega = \mathbb{N} \setminus \{0\}$, I could draw it in the first draw, in the second, etc, but I may also get very unlucky and need 923578723 draws, or any number $k \in \mathbb{N} \setminus \{0\}$ for that matter.[4] We already know $p(1)$, the probability that you get it in the first try is $\frac{1}{52}$. (I'm already using p instead of P as this is going to be a pmf. I could

[4] Again, we design Ω to capture what we care about. The number of draws until $A\heartsuit$. So Ω does not need to capture all the possible sequences of cards we could have drawn until the $A\heartsuit$ shows up.

have equivalently written $P(\{1\})$ instead.) Now we would like to go on and define $p(k)$ for all $k \in \{2, 3, \ldots\}$ as well. Note that we now do not have a finite space of possible outcomes. While it gets increasingly unlikely, it could actually be that we need 100 or 1,000 or 10,000 trials or even more until you draw $A\heartsuit$. However, we still have "discrete outcomes," so in principle it sounds good to define a probability for each number. And we can actually reason our way to $p(k)$.

Getting it right (drawing $A\heartsuit$) in the second draw means we got it wrong in the first one, which happens with probability $\frac{51}{52}$ (any other card) and then right in the second draw, which happens with probability $\frac{1}{52}$. Our intuition (possibly thinking about a tree of possible sequences) tells us that we should multiply these probabilities to get $p(2) = \frac{51}{52}\frac{1}{52}$. Getting it in the third try means getting it wrong twice and then right, so $p(3) = (\frac{51}{52})^2\frac{1}{52}$. Extending this argument further, we get the pmf $p(k) = (\frac{51}{52})^{k-1}\frac{1}{52}$. This is exactly the pmf of a geometric distribution $\mathrm{Geo}(p)$ with the success probability $p = \frac{1}{52}$. We have introduced $\mathrm{Geo}(p)$ as the distribution that models the number of trials until the first success.[5] It is defined on $\Omega = \{1, 2, \ldots\}$ and, hopefully, if our reasoning wasn't flawed somehow, we should have actually gotten a valid pmf. It is clearly nonnegative and one can also check that $\sum_{k=1}^{\infty} p(k) = 1$. So again, we can play the same game and use this pmf for individual outcomes (note how outcomes here are now more involved objects, not directly related to where the underlying probability came in, namely in shuffling the deck). For example, we can ask for the probability that it takes more than 3, fewer than 10, or more than 20 trials for you to draw $A\heartsuit$. This would be $P(\{k \in \Omega \mid 3 < k < 10 \text{ or } k > 20\})$ and we get it by summing the pmf $p(k)$ over those k. So again, the recipe works for every possible subset of $\Omega = \{1, 2, \ldots\}$ we are interested in and we get a probability measure on the full power set $\mathcal{P}(\Omega)$ as

$$P : \mathcal{P}(\{1, 2, \ldots\}) \to [0, 1]\,, \quad A \mapsto P(A) = \sum_{k \in A} p(k)\,.$$

The fact this still works here is because the series of pmf values $\{p(k)\}_{k=1}^{\infty}$ consists of nonnegative numbers whose infinite sum converges to a finite number (namely 1). Analysis then tells us that any subseries (i.e., the sum over any subset of those values) also converges. So we can actually compute $P(A)$ for *all* $A \in \mathcal{P}(\Omega)$. Note that $\mathcal{P}(\Omega)$ is *very large*. All possible subsets of infinitely many outcomes is indeed a large infinity. It actually has the same cardinality as $\mathbb{R}$, the set of real numbers.

Main takeaway: The key concepts we have derived are the *sample space* or *outcomes* Ω, containing the possible outcomes "we care about." Certain subsets of outcomes we call *events*. The main player is then a *probability measure* P assigning probabilities to events. That is, for an event $A \subseteq \Omega$ we want to assign $P(A) \in [0, 1]$. When Ω is discrete,

[5] The order is important here: We have derived the pmf of a scenario we care about modeling and then decided to give it a name. One easily gets the perception that "special distributions" like normal, Bernoulli, Laplace, Weibull, chi-squared, and what have you, dropped from the heavens and later people realized: "Oh look, how convenient, the geometric distribution happens to model this process". Certain distributions have names, because they do indeed come up in many settings.

we have seen that we can do this for any possible subset $A \in \mathcal{P}(\Omega)$. Specifically in this discrete setting, we can make our lives even simpler by only defining the *probability mass function (pmf)* p, which assigns a probability to each individual outcome $p(\omega) \in [0,1]$ for each $\omega \in \Omega$ such that $\sum_{\omega \in \Omega} p(\omega) = 1$, i.e., the sum of all probabilities is 1. We then compute $P(A) := \sum_{\omega \in A} p(\omega)$ for any event $A \subseteq \Omega$.

The uncountable world: it's getting nasty

What happens if our sample space Ω is *uncountably infinite*, for instance, we model the event that the bus arrives at any time within the next hour and every time is equally likely. This means our outcome space of interest is $\Omega = [0,1]$ and we would like a "uniform distribution" on Ω such that each time is equally likely. Let's try to do what we did before, assign a probability to each possible waiting time. The equivalent of the pmf would be some function $p(x) \in [0,1]$ for all $x \in [0,1]$ which is the probability that the bus arrives at time x. Since each time is equally likely, we want $p(x) = p(x')$ for all $x, x' \in [0,1]$. Intuitively, each arrival time should also have a nonzero probability $p(x) = c > 0$ for every $x \in [0,1]$. But then $\sum_{x \in [0,1]} p(x) = \sum_{x \in [0,1]} c$ would be infinite, which violates $\sum_x p(x) = 1$. (Mathematically this "sum" over an uncountable set is itself problematic, but the intuition holds).

You may note that this would have also been problematic for a uniform distribution over a countably infinite set $\mathbb{N}$. If each number had a non-zero and equal probability the sum would also not converge. However, we have seen that at least there exists sequences $(p(k))_{k \in \mathbb{N}}$ that sum to one and each $p(k) > 0$ (for example, the pmf of Binomial or Geometric distribution). In the continuous case not even that works anymore. The interpretation of $p(x)$ as a probability becomes subtle. In a sense, we are forced to get comfortable with the idea of "the probability of the bus arriving *exactly* at time x" being zero, i.e., $P(\{x\}) = 0$ for all $x \in [0,1]$. However, we still want to maintain that the event "the bus arrives at some point in the next hour" to be one, i.e., $P([0,1]) = 1$. Similarly, intuitively we want $P([0, \frac{1}{2}]) = \frac{1}{2}$ and $P([0.15, 0.46]) = 0.31$. That is, we want to still assign nonzero probabilities to *events* (sets of times) even though each individual time has probability zero.

This leads to the idea of a *probability density function (pdf)*, $p(x)$. On continuous spaces, the density $p(x)$ is *not* $P(\{x\})$, so the pdf does not give probabilities of something happening directly. Instead, we can compute probabilities of collections of outcomes (events) such as intervals using the pdf, for example via $P([a,b]) = \int_a^b p(x)dx$ for $[a,b] \subseteq \Omega$. The density $p(x)$ tells us how "densely" probability is packed around the point x. A higher $p(x)$ means that X is more likely to fall into a small interval around x. For the uniform distribution on $[0,1]$, $p(x) = 1$ for $x \in [0,1]$ and 0 otherwise. So we still maintain that $p(x) \geq 0$ for all $x \in \Omega$ and the "sums to 1" constraint becomes $\int_{-\infty}^{\infty} p(x)dx = 1$.

Compare this to an actually density (mass per volume). A density function of an object assigns to each individual point in 3D space a density, how dense the object is at that

point. Similarly to a pdf, the value of such a density function at a single point does not tell us how "heavy that point" is, the mass of any point in space is zero. Instead, it tells us how denseley mass is distributed at this point. Once we integrate the density function over a region of space, we get the mass that is sitting in that region. This is precisely analogous to $P(A) = \int_A p(x)\,dx$ for events $A \subseteq \Omega$.

Main takeaway: In continuous (uncountably infinite) outcomes spaces, we still maintain our notions of the sample space Ω and our probability measure P assigning probabilities to events $A \subseteq \Omega$. Our notion of the pmf had to be changed to the pdf. Any single outcome now has probability zero, but we describe the "density of probability" around that point at get the probability of events by integrating the density. We have not yet talked about which types of events, i.e., subsets of Ω, we can assign probabilities to...

Not all subsets are created equal

In the finite (deck of cards) and countable (Geometric distribution) cases, we happily used the power set $\mathcal{P}(\Omega)$ as our *event space* (the collection of all events to which we assign probabilities). We could assign a probability $P(A)$ to *any* subset $A \subseteq \Omega$ without any problem. Intuitively, we would of course also like to do this in the continuous case.

Question: For an uncountable space like $\Omega = [0, 1]$, can we still define a probability measure P on $\mathcal{P}([0, 1])$, that is, assign a probability to *any* subset of $[0, 1]$, that is consistent with our intuitions and axioms? Let's say we want our "uniform choice" P on $[0, 1]$ to satisfy $P([a, b]) = b - a$ (the length of the interval). This seems natural. We also want P to satisfy some basic axioms:

(i) $P(A) \geq 0$ for all $A \in \mathcal{P}([0, 1])$.

(ii) $P([0, 1]) = 1$.

(iii) *σ-additivity:* If $A_1, A_2, \ldots$ are disjoint sets in $\mathcal{P}([0, 1])$, then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

It turns out – and this is a deep result in mathematics (related to the work of Vitali, Banach, Tarski) – that it's **impossible** to define such a probability measure P on *all* subsets $\mathcal{P}([0, 1])$ that satisfies these natural conditions together with highly intuitive criteria, for example, translation invariance, which means $P(A) = P(A + c)$ if A and $A + c$ are in $[0, 1]$. There exist "exotic" or "pathological" subsets of $[0, 1]$ (often called *non-measurable sets*) for which a consistent "length" or "probability" cannot be assigned to without leading to contradictions.

This means we cannot simply continue to use $\mathcal{P}(\Omega)$ as our event space when Ω is continuous. While at first this appears like a smallish inconvenient bug, it requires us to build up an abstract machinery of *σ-algebras* for a rigorous treatment of probability. When trying to define a probability measure for continuous outcome spaces, we must restrict it to a certain set of events (not all subsets of Ω go). At first sight, this often feels like an

unnecessary annoyance, but ultimately we'll be pragmatic about it and just "demand what we need" for things to work.

Instead of trying to assign probabilities to *all* subsets, we restrict ourselves to a smaller (but still very rich) collection of "well-behaved" or "measurable" subsets. A σ-algebra $\mathcal{A}$ is a collection of subsets of Ω that is "nice enough" for probability theory:

(i) It contains Ω itself (and $\emptyset$).

(ii) It's closed under complements (if $A \in \mathcal{A}$, then $A^{\mathsf{c}} \in \mathcal{A}$).

(iii) It's closed under countable unions (if $A_1, A_2, \ldots \in \mathcal{A}$, then $\bigcup A_i \in \mathcal{A}$). (And also countable intersections).

Main takeaway: To build a coherent theory that does not need a constant distinction between discrete versus continuous, the fundamental object, a *probability space*, is thus a triple $(\Omega, \mathcal{A}, P)$ consisting of the outcome space Ω, a σ-algebra $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ containing all events we want to assign probabilities to, and the probability measure P, assigning a probability to all events in $\mathcal{A}$. We can talk about these three parts the same way regardless of whether we are in a discrete or continuous setting. The pmf and pdf, while often the much more intuitive objects, are "non-fundamental" in the sense that they only "work" in certain settings.

Warning. Since we do not want to have to develop concepts multiple times, we will often aim for the most general representation that works for all cases, i.e., finite, countably infinite, uncountably infinite, all at the same time. When things feel too complex, try to understand what the statements would mean in the finite setting.

— End Motivation —

1.1. Probability spaces

Section goal. Introduce a concise mathematical framework to talk about "probability," and "distributions."

Convention 1.1 (0 is a natural number). Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$, i.e., 0 is a natural number.

Definition 1.2 (set operations). We generally assume familiarity with common concepts and notation from set theory—this is a short reminder. For a more detailed refresher, see Section A.1. For any set Ω, we call the set of all subsets, i.e., $\mathcal{P}(\Omega) := \{A \mid A \subseteq \Omega\}$, the *power set of* Ω. For any $A \subseteq \Omega$, we call $A^{\mathsf{c}} := \Omega \setminus A \subseteq \Omega$ the *complement of* A *in* Ω. For a countable (i.e., finite or countably infinite) collection of sets Ω_i for $i \in \mathbb{N}$ we call

- $\bigcup_{i\in\mathbb{N}} \Omega_i = \Omega_0 \cup \Omega_1 \cup \ldots = \{\omega \mid \exists i \in \mathbb{N} : \omega \in \Omega_i\}$ the *union of* $(\Omega_i)_{i\in\mathbb{N}}$, and
- $\bigcap_{i\in\mathbb{N}} \Omega_i = \Omega_0 \cap \Omega_1 \cap \ldots = \{\omega \mid \forall i \in \mathbb{N} : \omega \in \Omega_i\}$ the *intersection of* $(\Omega_i)_{i\in\mathbb{N}}$.

Sometimes, we want to highlight that a union is over disjoint sets, for which we use "$\mathring{\cup}$." We write $[n] := \{1, \ldots, n\}$ for $n \in \mathbb{N}_{>0}$. We use $\subseteq$ for *subset* (which allows for equality) and $\subsetneq$ for *proper subset* (equality not allowed). When we do use $\subset$ (we try to avoid it), it "probably" means $\subseteq$, but you better call me out for it.

How do the above two points include the "finite collection" case?

The power set is also often denoted by 2^Ω, which usually stands for the set of all mappings $\Omega \to \{0, 1\}$. There is a one to one mapping to $\mathcal{P}(\Omega)$ by interpreting each map as deciding for each element whether it is included (1) or excluded (0) in the subset. More formally, the map is $\mathcal{P}(\Omega) \to 2^\Omega : A \mapsto \chi_A$, where χ_A is the *indicator function* with $\chi_A(a) = 1$ if $a \in A$ and $\chi_A(a) = 0$ otherwise.

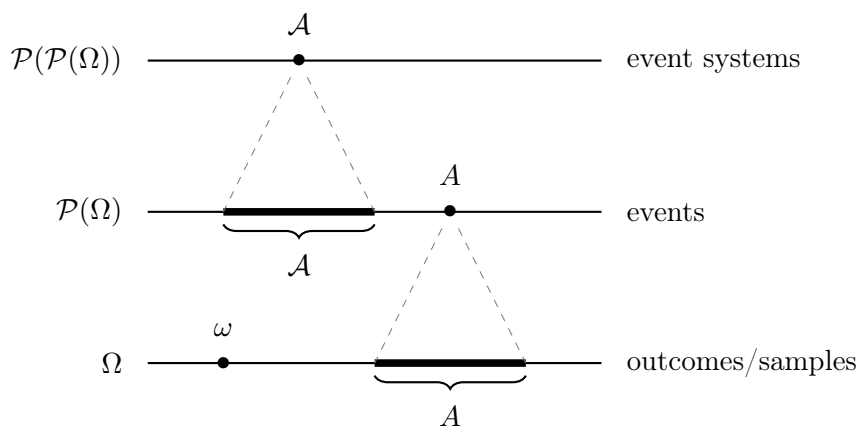
Lemma 1.3 (de Morgan's laws). *Let $\{A_i\}_{i\in I}$ be a family of sets indexed by a set I. Then:*

$$\Big(\bigcup_{i\in I} A_i\Big)^{\mathsf{c}} = \bigcap_{i\in I} A_i^{\mathsf{c}}\,, \qquad \textit{and} \qquad \Big(\bigcap_{i\in I} A_i\Big)^{\mathsf{c}} = \bigcup_{i\in I} A_i^{\mathsf{c}}\,.$$

Proof. Exercises. □

Convention 1.4 (non-empty Ω). Throughout the lecture, unless stated otherwise, let Ω be a non-empty set.

Remark 1.5 (samples, events, event systems). The following illustration adapted from Georgii (2015) shows the relationships between samples (individual outcomes of experiments), events (which we want to assign a probability to), and event systems (all events of interest that can be "measured").



Definition 1.6 (σ-algebra). We call $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ a *σ-algebra on Ω*, if

(i) $\emptyset \in \mathcal{A}$,

(ii) for each $A \in \mathcal{A}$ we have $A^{\mathsf{c}} \in \mathcal{A}$, and

(iii) for each sequence $(A_i)_{i\in\mathbb{N}}$ in $\mathcal{A}$ we have $\bigcup_{i\in\mathbb{N}} A_i \in \mathcal{A}$.

Proposition 1.7 (intersection of σ-algebras). *Let I be a nonempty index set (finite, countable or even uncountable) and let $\mathcal{A}_\alpha$ be a σ-algebra over Ω for every $\alpha \in I$. Then $\mathcal{A} := \bigcap_{\alpha\in I} \mathcal{A}_\alpha$, the intersection of all the sigma-algebras, is also a σ-algebra over Ω*

Proof. Exercises. □

Lemma 1.8 (generator of a σ-algebra). *For a system of subsets $\mathcal{E} \subseteq \mathcal{P}(\Omega)$,*

$$\sigma(\mathcal{E}) := \bigcap \{\mathcal{A} \mid \mathcal{A} \text{ is a } \sigma\text{-algebra on } \Omega \text{ and } \mathcal{E} \subseteq \mathcal{A}\}$$

is the smallest σ-algebra on Ω containing $\mathcal{E}$. We call $\mathcal{E}$ the generator of $\sigma(\mathcal{E})$ *and $\sigma(\mathcal{E})$ the* σ-algebra on Ω generated by $\mathcal{E}$.

Example 1.9 (σ-algebras).

(i) The system $\{\emptyset, \Omega\}$ is a σ-algebra on Ω, called the *trivial σ-algebra on Ω*.

(ii) The power set $\mathcal{P}(\Omega)$ is a σ-algebra on Ω.

(iii) For every $A \subseteq \Omega$, $\sigma(\{A\}) = \{\emptyset, \Omega, A, A^{\mathsf{c}}\}$ is a σ-algebra on Ω.

Corollary 1.10 (σ-algebras are "closed under set operations"). *Let $\mathcal{A}$ be a σ-algebra over Ω and let $A, B \in \mathcal{A}$. Then Ω, $A \cup B$, $A \cap B$, $A \setminus B$, $A\Delta B := (A \setminus B) \cup (B \setminus A)$ are in $\mathcal{A}$. For a countable index set I and $\{A_i \mid i \in I\} \subseteq \mathcal{A}$, we have $\bigcap_{i\in I} A_i \in \mathcal{A}$.*

Definition 1.11 (measurable space, measurable sets). A *measurable space* is a tuple $(\Omega, \mathcal{A})$, where Ω is a set and $\mathcal{A}$ is a *σ-algebra* on Ω. The elements of $\mathcal{A}$ are then called *measurable sets*.

Overview 1.12 (why "measure"). The idea behind "measurability" is literally to be able to "measure" mathematical objects. We have a good intuitive grasp of what it means to "measure the size" of

(i) subsets of $\mathbb{R}$: the length of intervals.

(ii) subsets of $\mathbb{R}^2$: the area of surfaces.

(iii) subsets of $\mathbb{R}^3$: the volume of solids or bodies.

However, a rigorous formalism is required to capture all the corner cases: What "volume" should a line in 3D space have? What about the volume of an uncountable collection of points?

Example 1.13 (Borel-σ algebra on $\mathbb{R}^n$). Let $\Omega = \mathbb{R}^n$ for some $n \in \mathbb{N}_{>0}$. Then the σ-algebra generated by

$$\{(a_1, b_1) \times (a_2, b_2) \times \ldots \times (a_n, b_n) \mid a_i < b_i \text{ for all } i \in [n]\} \subseteq \mathcal{P}(\mathbb{R}^n)$$

is called the *Borel-σ-algebra on $\mathbb{R}^n$* and denoted by $\mathcal{B}^n$. For $n = 1$, we simply write $\mathcal{B}$.

Remark 1.14 (topological view on $\mathcal{B}$). If you know what a "topology $\mathcal{O}$ on a set Ω" is, then you can try to prove that the Borel-σ-algebra $\mathcal{B}^n$ on $\mathbb{R}^n$ is given by $\sigma(\mathcal{O}^n)$, where $\mathcal{O}^n$ is the standard topology on $\mathbb{R}^n$.

Lemma 1.15 (induced σ-algebra). *Let $(\Omega, \mathcal{A})$ be a measurable space and $B \subseteq \Omega$ a non-empty subset. Then $\mathcal{A}|_B := \mathcal{A} \cap B := \{A \cap B \mid A \in \mathcal{A}\}$ is a σ-algebra on B called the* induced σ-algebra by B *or* restriction of $\mathcal{A}$ on B. *For $\mathcal{E} \subseteq \mathcal{P}(\Omega)$ we have $\sigma(\mathcal{E})|_B = \sigma(\mathcal{E}|_B)$.*

Convention 1.16 (σ-algebras). In this course, we always assume nonempty "continuous spaces" $A \subseteq \mathbb{R}^n$ to be equipped with the Borel-σ-algebra (restricted to A) $\mathcal{B}^n|_A$ and countable spaces to be equipped with their power set as a σ-algebra.

Definition 1.17 (measure, measure space). Let $(\Omega, \mathcal{A})$ be a measurable space. We call $\mu : \mathcal{A} \to [0, \infty]$ a *measure* if

(i) $\mu(\emptyset) = 0$,

(ii) it is non-negative, i.e., for all $A \in \mathcal{A}$ we have $\mu(A) \geq 0$, and

(iii) it is *σ-additive*, i.e., for all countable collections $(A_i)_{i \in \mathbb{N}}$ of pairwise disjoint sets in $\mathcal{A}$, we have
$$\mu\Big(\bigcup_{i \in \mathbb{N}} A_i\Big) = \sum_{i \in \mathbb{N}} \mu(A_i)\,.$$

The triple $(\Omega, \mathcal{A}, \mu)$ is then called a *measure space*.

Proposition 1.18 (basic properties of measures). *Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $A, B \in \mathcal{A}$, and $(A_i)_{i \in \mathbb{N}} \subseteq \mathcal{A}$. Then*

(i) $\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B)$,

(ii) $\mu(B \setminus A) = \mu(B) - \mu(A)$, *if $A \subseteq B$ and $\mu(A) < \infty$,*

(iii) μ *is* monotone, *meaning that whenever $A \subseteq B$ we have $\mu(A) \leq \mu(B)$,*

(iv) μ *is* continuous from below, *i.e., $\mu(A_k) \uparrow \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right)$ for $A_0 \subseteq A_1 \subseteq A_2 \subseteq \ldots$,*

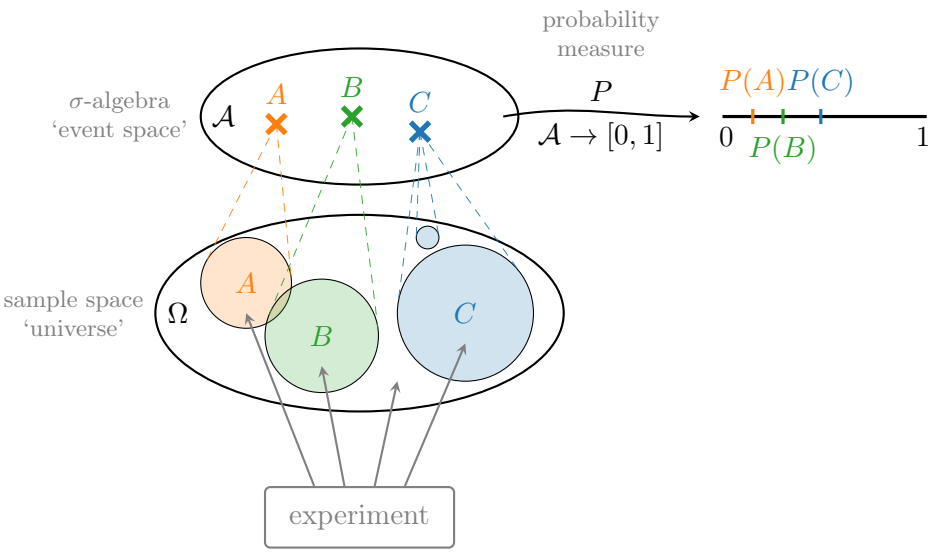
(v) μ *is* continuous from above, *i.e., $\mu(A_k) \downarrow \mu\left(\bigcap_{i \in \mathbb{N}} A_i\right)$ for $A_0 \supseteq A_1 \supseteq A_2 \supseteq \ldots$ with $\mu(A_0) < \infty$,*

(vi) μ *is* σ-subadditive, *meaning that*
$$\mu\Big(\bigcup_{i \in \mathbb{N}} A_i\Big) \leq \sum_{i \in \mathbb{N}} \mu(A_i)\,.$$

Definition 1.19 (probability measure, probability space). Let $(\Omega, \mathcal{A})$ be a measurable space. A measure $P : \mathcal{A} \to [0, \infty]$ is called *probability measure* if $P(\Omega) = 1$. The triple $(\Omega, \mathcal{A}, P)$ is then called *probability space*. In this case, $\mathcal{A}$ is called the *event space* or *set of events of interest*.

Overview 1.20 (probability as measure). Intuitively, the fact that probability is essentially captured by a measure, is the idea that we "measure the likelihood" of something happening. The concepts developed for mathematically measuring subsets of arbitrary spaces is just what we need to consistently assign probabilities to events, i.e., measure the "size of subsets" according to their probability.

What we have so far can be illustrated as follows



Theorem 1.21 (inclusion–exclusion principle). *Let $(\Omega, \mathcal{A}, P)$ be a probability space and let $A_1, \dots, A_n \in \mathcal{A}$ be any finite collection of events ($n \in \mathbb{N}_{>0}$). Then*

$$P\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{k=1}^{n} (-1)^{k-1} \sum_{1 \le i_1 < \dots < i_k \le n} P\big(A_{i_1} \cap \dots \cap A_{i_k}\big).$$

Proof. Exercises. □

Definition 1.22 (cumulative distribution function (cdf)). Let P be a probability measure on $(\mathbb{R}, \mathcal{B})$. The *cumulative distribution function (cdf) of P* is

$$F_P : \mathbb{R} \to [0,1],\ x \mapsto P\big((-\infty, x]\big)$$

and is often just denoted by F when P is clear.

Lemma 1.23 (properties of cdfs). *Let $(\mathbb{R}, \mathcal{B}, P)$ be a probability space and let F be the cdf of P. Then*

(i) F is monotonically increasing, i.e., for all $x, y \in \mathbb{R}$ with $x \leq y$ we have $F(x) \leq F(y)$.

(ii) F is right-continuous, i.e., for all $x_n \downarrow x$ in $\mathbb{R}$ we have $F(x_n) \downarrow F(x)$.

(iii) F has a left-hand limit, i.e., for all $x_n \uparrow x$ in $\mathbb{R}$ we have $F(x_n) \uparrow F(x-) := P((-\infty, x))$.

(iv) the discontinuities of F correspond to the probabilities of singletons, i.e., $F(x) - F(x-) = P(\{x\})$ for all $x \in \mathbb{R}$.

(v) we have $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$.

Theorem 1.24 (correspondence of cdfs). *For any $F : \mathbb{R} \to [0, 1]$ that satisfies (i), (ii), (v) of lemma 1.23, there exists a unique probability measure λ_F on $(\mathbb{R}, \mathcal{B})$ such that $F_{\lambda_F} = F$. For any probability measure P on $(\mathbb{R}, \mathcal{B})$, we have $P = \lambda_{F_P}$.*

Caveat 1.25 (measure problem). Generally, we cannot construct "useful" (probability) measures on the power set of infinite spaces. Therefore, we need σ-algebras to meaningfully talk about probability theory.

Theorem 1.26 (Vitali, 1905). *Let $\Omega = \{0, 1\}^{\mathbb{N}}$ be the sample space of infinitely many repeated coin tosses. For $n \in \mathbb{N}$, define $T_n : \Omega \to \Omega$ by*

$$\omega = (\omega_n)_{n\in\mathbb{N}} \mapsto T_n(\omega) := (\omega_1, \ldots, \omega_{n-1}, 1 - \omega_n, \omega_{n+1}, \ldots)$$

the map that flips the result of the n-th coin toss. For $A \in \mathcal{P}(\Omega)$, let $T_n(A) := \{T_n(\omega) \mid \omega \in A\}$ be the image of A under T_n. Then there does not exist a probability measure P on the power set $\mathcal{P}(\Omega)$ that satisfies the following invariance property: For all $A \in \mathcal{P}(\Omega)$ and $n \in \mathbb{N}$, we have $P(T_n(A)) = P(A)$.

Remark 1.27 (Vitali interpretation). The invariance property in theorem 1.26 states that every sequence of coin tosses should be equally probable as the one where the n-th coin flip came up differently. Intuitively this property should hold true, i.e., any "useful" probability measure on Ω should satisfy this property. Theorem 1.26 says that if we want that property to hold, we cannot use $\mathcal{P}(\Omega)$ as σ-algebra on Ω. While $\mathcal{P}(\Omega)$ *is* a σ-algebra on Ω, it does not allow us to construct a "useful" probability space on top of it–we have to use a smaller one. On some level, this is the reason we introduced σ-algebras in the first place.

— Start Philosophical Interlude —

Remark 1.28 (interpretations of probability). Connecting to remark 0.9, the initial probability space $(\Omega, \mathcal{A}, P)$ does not (necessarily) represent a physically real source of randomness, but rather our ignorance or limitation of our knowledge. The measure P is our best understanding and description and the space Ω a convenient representation of the outcomes of experiments. "Randomness" is thus (typically) not "in nature" but "in our model." Probability theory is a quantitative framework to reason consistently about uncertainty, not an explanation of some intrinsic source of randomness. Our premise from here on is that probability is useful.

In practice, we call $A \in \mathcal{A}$ an event, and $P(A)$ the probability, that event A happens. Note that A can still consist of multiple "samples" or "experiment outcomes." For example, "rolling a 1 or a 3" is a valid event (with probability $\frac{1}{3}$ for a fair die) even though only one of the two can happen in any given experiment. Common interpretations of "what randomness, i.e., P, actually is" include

(i) the *naive interpretation*: "Nature is not sure," what it does and $P(A)$ is the degree of uncertainty, with which it decides for event A happening.

(ii) the *frequentist interpretation*: $P(A)$ is the relative frequency, with which A tends to occur under the same external conditions.

(iii) the *subjective interpretation*: $P(A)$ is the degree of certainty with which, based on my personal assessment of the situation, I am willing to bet on the occurrence of A.

Viewpoint 1.29 (probability from gambling: the Dutch book argument). We have started directly with Kolmogorov's axioms for what a probability measure has to satisfy. Besides "falling from the heaven's" they can also be derived from other principles. The Dutch book arguments for example, collectively show that if a person's subjective probabilities violate the axioms of probability, a better can come up with a list of bets that guarantee that person to lose money. As a consequence, rational belief assignments must obey the axioms of probability to avoid inconsistencies.

Viewpoint 1.30 (probability from logic and partial beliefs). Jaynes (2003) builds up the axioms of probability by extending classical logic to settings where there is uncertainty due to a lack of complete information. Leaning on basic principles from rational reasoning like consistency and ranking plausibility, he shows that any system for manipulating degrees of belief must obey the rules of probability. This motivates the axioms of probability as ultimately yielding the unique rational calculus for reasoning under uncertainty.

I would have loved to take this approach in the lecture as well, but due to a lack of time we just dumped the final framework onto you.

On another note, this recent podcast on the "Logic and Use of Probability" with philosopher Branden Fitelson gives a nice account of how probabilities can (not) be used to assess the strength of evidence for or against scientific claims.

Remark 1.31 (Bayes vs. frequentist). It is often said that there are "two approaches" to probability, the frequentist and the Bayesian one. In a nutshell, the frequentist approach interprets probabilities as long run relative frequencies of events in repeated trials. The parameters of the underlying experiment are considered fixed but unknown. The Bayesian approach interprets probability as a degree of belief, allowing uncertainty about the parameters to be modeled directly with probability distributions. They have less trouble talking about events that by their very nature can only happen once. Bayesians rely on prior beliefs (one must already start with certain credences), whereas frequentists rely on repeatedly sampling from (identical) distributions (at least conceptually) and avoid prior beliefs. Hopefully, we will have time to go into more details about this.

— End Philosophical Interlude —

1.2. Discrete probability spaces and distributions

Section goal. Introduce common discrete probability distributions and learn how to perform computations in discrete probability spaces.

Definition 1.32 (discrete probability spaces, probability mass function (pmf)). Let Ω be an at most countable set. A probability measure P on $(\Omega, \mathcal{A} := \mathcal{P}(\Omega))$ is called *discrete probability measure* and the probability space $(\Omega, \mathcal{A}, P)$ is called *discrete probability space*. The map

$$p : \Omega \to [0,1]\,, \quad \omega \mapsto p(\omega) := P(\{\omega\})$$

is called *probability mass function (pmf) for P*.

Lemma 1.33 (correspondence between pmf and P).

(i) Let $(\Omega, \mathcal{A}, P)$ be a discrete probability space. Then P is uniquely defined via its pmf p.

(ii) Let Ω be countable and let $p : \Omega \to [0,1]$ be such that $\sum_{\omega\in\Omega} p(\omega) = 1$. Then

$$P : \mathcal{P}(\Omega) \to [0,1],\ A \mapsto \sum_{\omega\in A} p(\omega)$$

is a discrete probability measure on $(\Omega, \mathcal{P}(\Omega))$ with pmf p.

Lemma 1.34 (classical urn models). *We consider an urn with $n \in \mathbb{N}_{>0}$ distinct objects and $k \in \mathbb{N}_{>0}$ draws.*

(i) With replacement, ordered draws. *The space of possible outcomes is*

$$\Omega = [n]^k \quad \textit{with} \quad |\Omega| = n^k\,.$$

(ii) Without replacement, ordered draws. *The space of possible outcomes is*

$$\Omega = \left\{ (\omega_1, \dots, \omega_k) \in [n]^k \,\middle|\, \omega_i \neq \omega_j \text{ for } i \neq j \right\} \quad \text{with} \quad |\Omega| = \frac{n!}{(n-k)!} \,.$$

(iii) Without replacement, unordered draws. *The space of possible outcomes is*

$$\Omega = \{ S \subseteq [n] \,|\, |S| = k \} \quad \text{with} \quad |\Omega| = \binom{n}{k} \,.$$

(iv) With replacement, unordered draws. *The space of possible outcomes is*

$$\Omega = \left\{ (a_1, \dots, a_n) \in \mathbb{N}_0^n \,\middle|\, \sum_{i=1}^n a_i = k \right\} \quad \text{with} \quad |\Omega| = \binom{n+k-1}{k} \,,$$

where a_i counts how often object i is drawn.

Remark 1.35 (applications of urn models). The urn models apply in a large variety of sampling schemes like the following.

- Repeated independent trials like wheels of fortune or generating random passwords (with replacement and ordered).
- Sequentially picking distinct items like assigning prizes to distinct winners (without replacement and ordered).
- Simple random sampling without replacement like drawing lottery numbers (without replacement and unordered).
- Distributing indistinguishable balls into distinguishable bins like distributing identical resources among agents (with replacement and unordered).

Remark 1.36 (discrete probability measures in $\mathbb{R}$). Let $\Omega \subseteq \mathbb{R}$ be countable and let p be a pmf on Ω. The probability measure P on $(\mathbb{R}, \mathcal{B})$ given by

$$P : \mathcal{B} \to [0,1],\ A \mapsto \sum_{\omega \in \Omega \cap A} p(\omega)$$

and the corresponding cdf

$$F : \mathbb{R} \to [0,1],\ x \mapsto \sum_{\omega \in \Omega \cap (-\infty, x]} p(\omega)$$

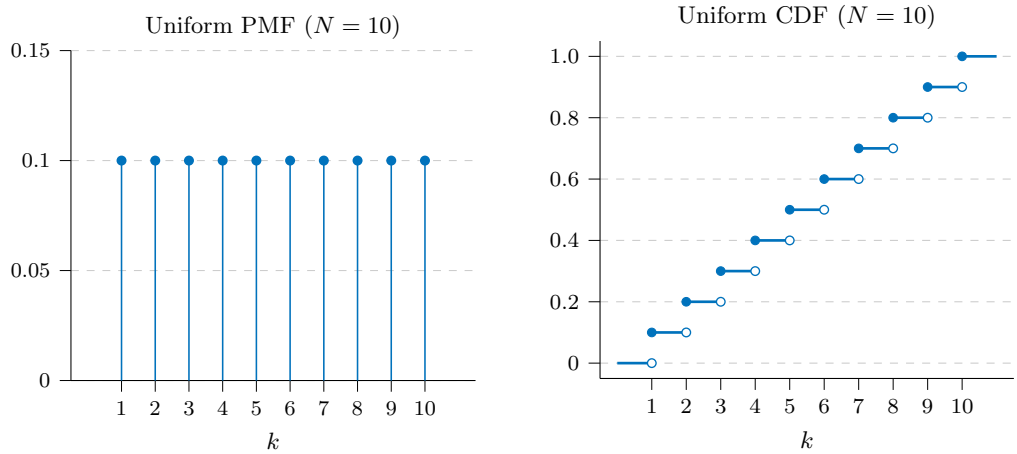
are called *discrete*.

Example 1.37 (common discrete probability measures). The following pmfs describe frequently occurring discrete distributions.

(i) *Laplace or uniform distribution:* For a finite sample space Ω, we define the Lap_Ω distribution via the pmf

$$p_{\mathrm{Lap}_\Omega} : \Omega \to [0,1],\ \omega \mapsto \frac{1}{|\Omega|}\,.$$

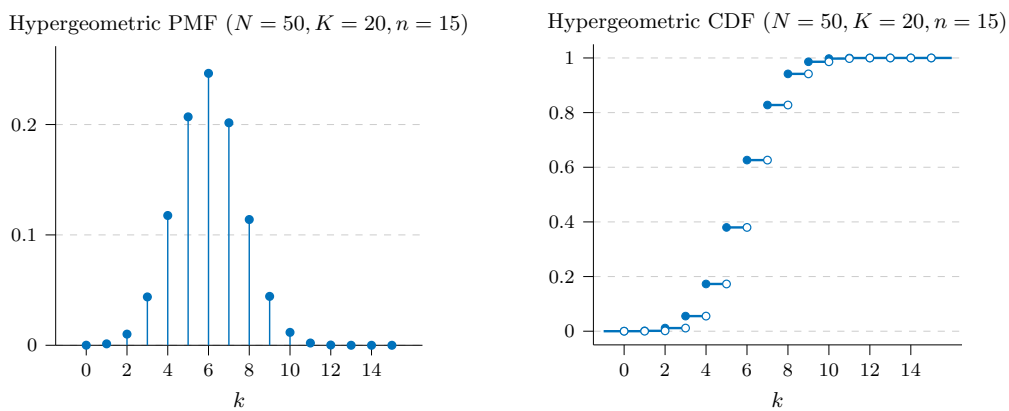
The Laplace model Lap_Ω applies when every outcome in a finite set is equally likely, for example, drawing a card from a well-shuffled deck.



(ii) *Hypergeometric distribution:* Given parameters $N \in \mathbb{N}$, $K \in \{0,\dots,N\}$, and $n \in \{0,\dots,N\}$, the $\mathrm{Hyp}(N,K,n)$ distribution is defined on the sample space $\Omega = \{\max(0, n+K-N),\dots,\min(n,K)\}$ via the pmf

$$p_{\mathrm{Hyp}(N,K,n)} : \Omega \to [0,1],\ \omega \mapsto \frac{\binom{K}{\omega}\binom{N-K}{n-\omega}}{\binom{N}{n}}\,.$$

The model $\mathrm{Hyp}(N,K,n)$ describes sampling without replacement, for example counting defectives in a quality-control sample of size n drawn from a batch of N items containing K defectives.



(iii) *Bernoulli process:* For $p \in [0,1]$ and $n \in \mathbb{N}_{>0}$, the $\mathrm{Ber}(n,p)$ distribution is defined on the sample space $\Omega = \{0,1\}^n$ via the pmf

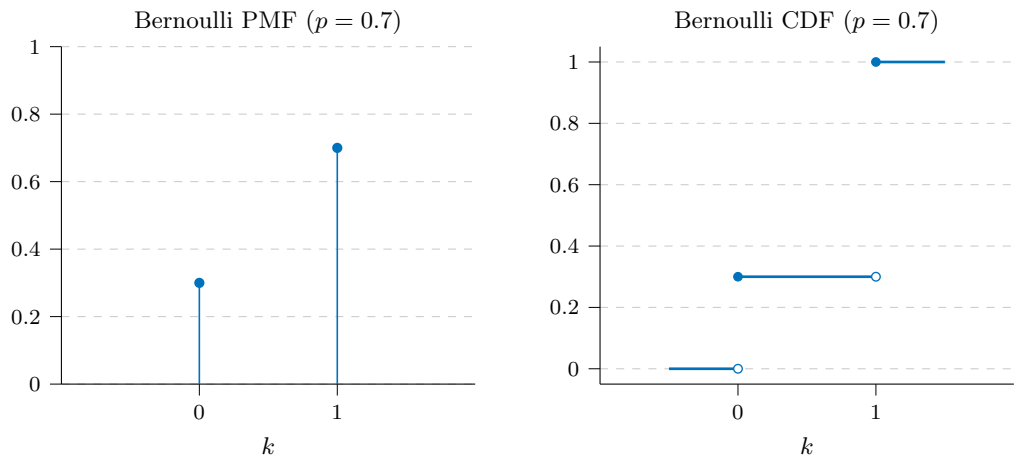
$$p_{\mathrm{Ber}(n,p)} : \Omega \to [0,1],\ \omega = (\omega_i)_{i\in[n]} \mapsto p^{\sum_{i\in[n]}\omega_i}(1-p)^{n-\sum_{i\in[n]}\omega_i} .$$

Note that we overload the symbol "p" to stand both for the pmf as well as for the probability parameter $p \in [0,1]$—also in the following pmfs. We stick to p for the latter despite the possible confusion due to the widespread use of p for these kinds of parameters. The meaning is typically clear from the context.

For $n = 1$, we also write $\mathrm{Ber}(p) := \mathrm{Ber}(1,p)$, which is called *Bernoulli distribution* and its pmf is simply

$$p_{\mathrm{Ber}(p)} : \{0,1\} \to [0,1],\ \omega \mapsto \begin{cases} p & \text{if } \omega = 1 , \\ 1-p & \text{if } \omega = 0 . \end{cases}$$

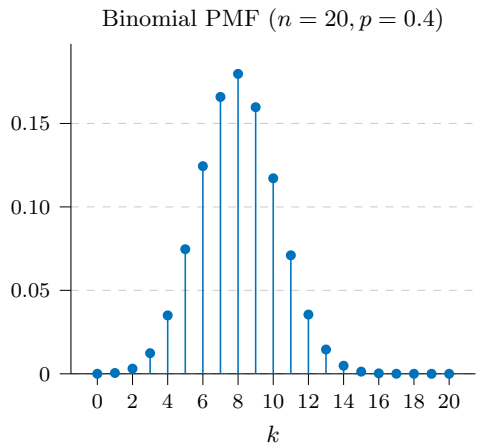
The model $\mathrm{Ber}(p)$ captures a single binary trial, such as a biased coin flip or testing whether a manufactured part passes inspection. The model $\mathrm{Ber}(n,p)$ captures, for example, repeated coin tosses. The Bernoulli process can be extended to trials with more than two outcomes (e.g., repeatedly rolling a die), which is then called a *Bernoulli scheme*.

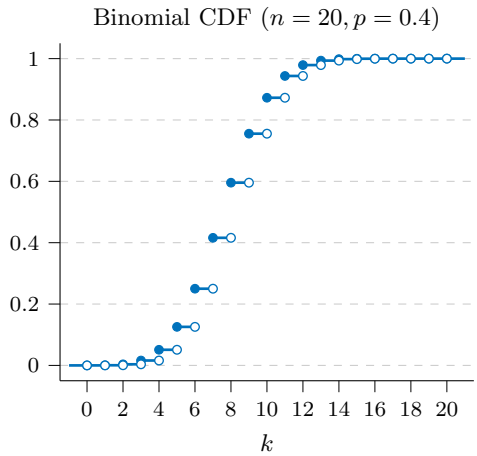


(iv) *Binomial distribution:* For $p \in [0,1]$ and $n \in \mathbb{N}_{>0}$, the $\mathrm{Bin}(n,p)$ distribution is defined on the sample space $\Omega = \{0,\dots,n\}$ via the pmf

$$p_{\mathrm{Bin}(n,p)} : \Omega \to [0,1],\ \omega \mapsto \binom{n}{\omega} p^{\omega}(1-p)^{n-\omega} .$$

The model $\mathrm{Bin}(n,p)$ counts the number of successes in n independent trials, for example the number of heads in ten flips of a coin with success probability p.

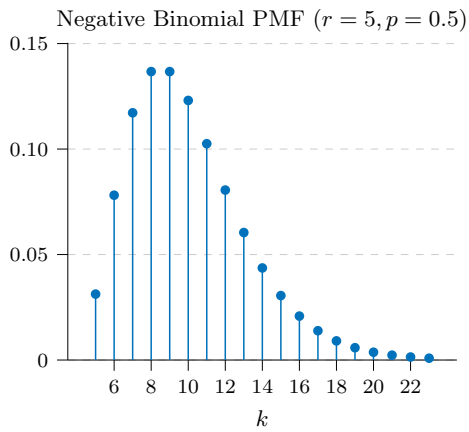


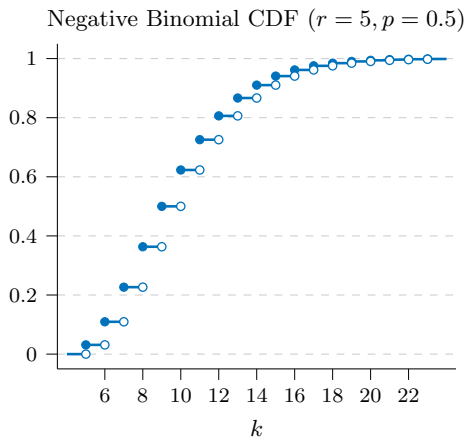


(v) *Negative binomial distribution:* For $p \in [0,1]$ and $r \in \mathbb{N}_{>0}$, the $\mathrm{NegBin}(r,p)$ distribution is defined on the sample space $\Omega = \mathbb{N}_{\geq r}$ via the pmf

$$p_{\mathrm{NegBin}(r,p)} : \Omega \to [0,1],\ \omega \mapsto \binom{\omega - 1}{r - 1} p^r (1-p)^{\omega - r}\,.$$

Here, ω denotes the total number of trials needed to observe r successes. The model $\mathrm{NegBin}(r,p)$ measures the total number of trials needed to observe r successes, for example the number of free-throw attempts until a basketball player makes r baskets.

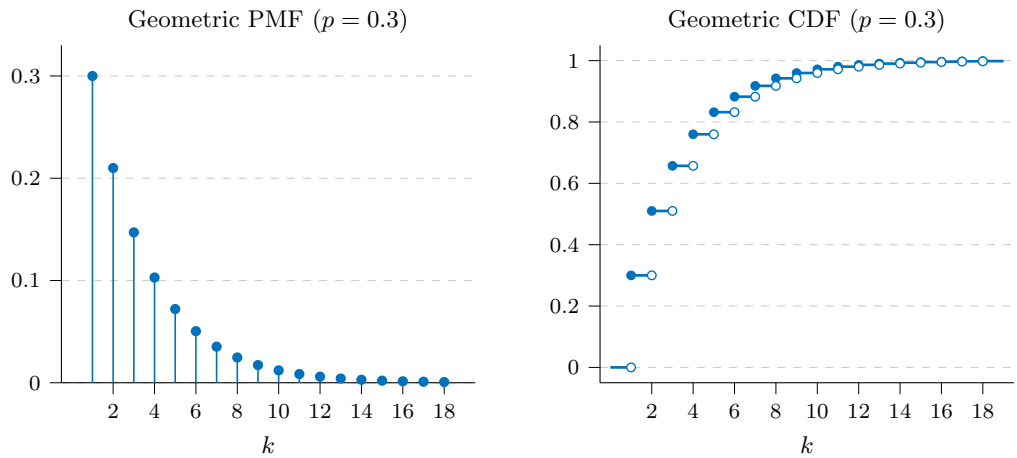




(vi) *Geometric distribution:* For $p \in [0,1]$, the $\mathrm{Geo}(p)$ distribution is defined on the sample space $\Omega = \mathbb{N}_{>0}$ via the pmf

$$p_{\mathrm{Geo}(p)} : \Omega \to [0,1],\ \omega \mapsto (1-p)^{\omega - 1} p\,.$$

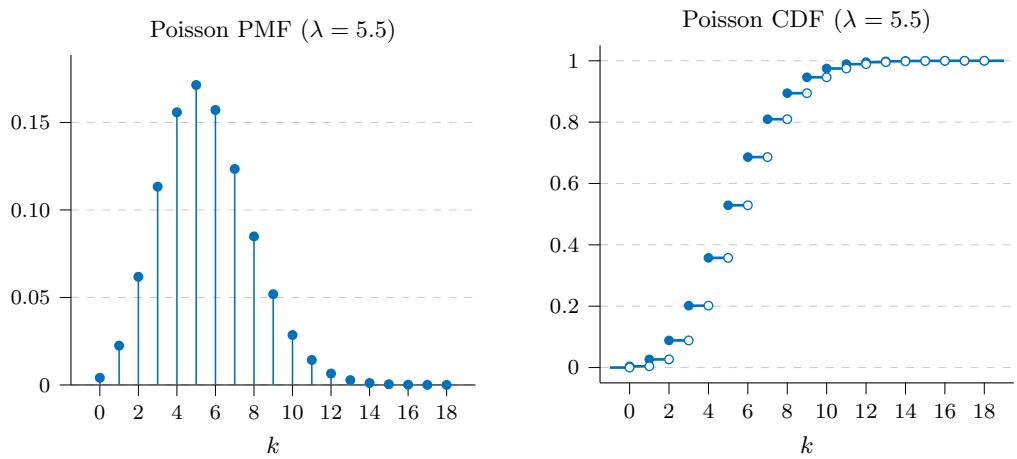
The model $\mathrm{Geo}(p)$ measures the number of trials until the first success, for example the number of customers arriving until the first sale is made.



(vii) *Poisson distribution:* For $\lambda \in \mathbb{R}_{>0}$, the $\mathrm{Poi}(\lambda)$ distribution is defined on the sample space $\Omega = \mathbb{N}$ via the pmf

$$p_{\mathrm{Poi}(\lambda)} : \Omega \to [0,1],\ \omega \mapsto \frac{e^{-\lambda}\lambda^{\omega}}{\omega!}\ .$$

The model $\mathrm{Poi}(\lambda)$ approximates the count of rare events occurring at a fixed rate in a fixed interval, for example the number of phone calls received by an operator in one hour when calls occur at rate λ.



Example 1.38 (urn model applications). Urn models are typically paired with the uniform distribution, i.e., we assume p_{Lap_Ω} on $(\Omega, \mathcal{P}(\Omega))$ for the different sample spaces in lemma 1.34. In this case, any event $A \subseteq \Omega$ has probability $P(A) = \frac{|A|}{|\Omega|}$ since all urn model sample spaces are finite. In short, computing probabilities reduces to "counting sizes of sets," which in turn often comes down to combinatorics.

(i) You and your friend each generate a password of 8 alphanumeric characters (62 symbols). What is the probability that the passwords are identical? We sample with replacement and ordered draws, i.e., $\Omega = [62]^8$. Then the probability of a collision is simply $1/|\Omega| \approx 4.58 \cdot 10^{-15}$ (assuming a uniform distribution).

(ii) You distribute 5 probability exercises among a study group of 5 students. What is the probability that the student first in the alphabet receives exercise 1 and the student last in the alphabet receives exercise 5? This corresponds to drawing without replacement and ordered draws and a sample space $\Omega = \{(\omega_1, \dots, \omega_5) \in [5]^5 \mid \omega_i, \omega_j \text{ pairwise different}\}$. The event A we are interested in consists of all tuples $(\omega_1, \dots, \omega_5) \in \Omega$ with $\omega_1 = 1$ and $\omega_5 = 5$. Once these two are fixed, the remaining three ω_i can receive the remaining 3 exercises in any order, i.e., $|A| = 3! = 6$. Assuming a uniform distribution we have $P(A) = |A|/|\Omega| = 1/20$. We will later see how we can arrive at the same answer by "conditioning."

(iii) PhD students randomly leave their 6 (identical) coffee mugs in any of the 4 meeting rooms. At the end of the day, what is the probability that exactly one room has no mug in it? This can be modeled by drawing with replacement and unordered draws (like balls into bins). The sample space is $\Omega = \{(a_1, a_2, a_3, a_4) \in \mathbb{N}^4 \mid \sum_{i=1}^4 a_i = 6\}$ with the total number of constellations being $\binom{4+6-1}{6} = \binom{9}{6} = 84$. For the event A of interest (exactly one room empty):

a) Choose which of the 4 rooms is empty. This can be done in $\binom{4}{1} = 4$ ways.

b) The remaining 6 mugs must be distributed among the other 3 rooms, with the condition that each of these 3 rooms must contain at least one mug. Let b_1, b_2, b_3 be the number of mugs in these three rooms. We need $b_1 + b_2 + b_3 = 6$ with $b_i \geq 1$. This is equivalent to first placing one mug in each of these three rooms, and then distributing the remaining $6 - 3 = 3$ mugs among these three rooms without restriction. The number of ways to do this is $\binom{3+3-1}{3} = \binom{5}{3} = 10$.

Therefore, the number of constellations in event A is $|A| = 4 \cdot 10 = 40$. The probability (assuming uniformity of these constellations) of finding exactly one room without a mug is

$$P(A) = \frac{|A|}{|\Omega|} = \frac{40}{84} = \frac{10}{21}\,.$$

Example 1.39 (gotcha! model misspecification). In the mug example in example 1.38, the modeling choice is subtle. If we directly take the full occupancy tuple $(a_1, \dots, a_4)$ as the outcome, then we are in the "with replacement, unordered draws" setting from lemma 1.34, and the computation $\frac{10}{21}$ is correct under the additional (typically left implicit) assumption that all occupancy vectors are equally likely.

If instead we interpret the story literally as "each of the 6 students independently chooses one of the 4 rooms," the proper urn model is "with replacement, ordered draws." This means we work on $\Omega = [4]^6$ with the uniform distribution. Equivalently, if $X_i(\omega) := \omega_i$

denotes the room chosen by student i, then $X_1, \dots, X_6 \overset{\text{iid}}{\sim} \mathrm{Lap}_{[4]}$. The occupancy numbers are then derived random variables

$$N_j := \sum_{i=1}^{6} \chi_{\{j\}}(X_i)\,, \qquad j \in [4]\,.$$

Let A be the event that exactly one room is empty. We choose the empty room in $\binom{4}{1} = 4$ ways. Fix one empty room, say room 4. Then all 6 students must choose among rooms $1, 2, 3$, giving 3^6 ordered outcomes. Among these, we must exclude those where at least one of the rooms $1, 2, 3$ is also empty. By the inclusion–exclusion principle,

$$3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6 = 540$$

counts the ordered outcomes in which all of the rooms $1, 2, 3$ are used at least once. Hence,

$$|A| = \binom{4}{1}\left(3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6\right) = 2160$$

and therefore

$$P(A) = \frac{|A|}{|\Omega|} = \frac{2160}{4^6} = \frac{135}{256} \approx 0.527\,.$$

Both the earlier answer $\frac{10}{21} < \frac{1}{2}$ as well as this new answer ($> \frac{1}{2}$) are "correct", they simply belong to different modeling choices that have to be made very explicit. Under the "students place mugs iid" model, occupancy tuples are not equally likely; for example, $(3, 1, 1, 1)$ can arise in many more ordered ways than $(6, 0, 0, 0)$.

Theorem 1.40 (Poisson limit theorem). *Let $(p_n)_{n\in\mathbb{N}} \subseteq [0, 1]$ with a fixed $\lambda := \lim_{n\to\infty} n\,p_n \in \mathbb{R}_{>0}$. Then for all $\omega \in \mathbb{N}$*

$$\lim_{n\to\infty} p_{\mathrm{Bin}(n,p_n)}(\omega) = p_{\mathrm{Poi}(\lambda)}(\omega)\,.$$

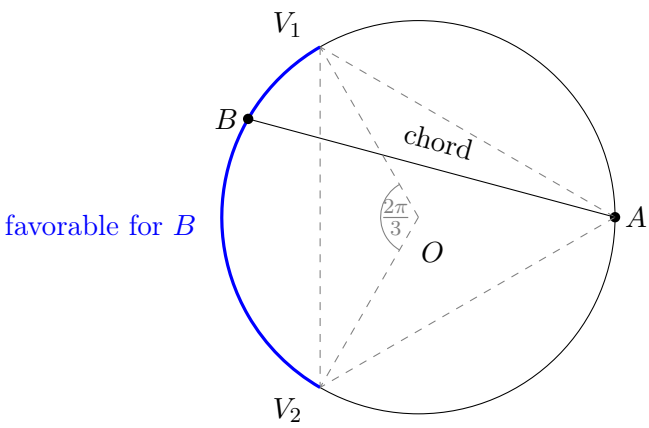
In words, this means that the pmf of the binomial distributions converges to the Poisson distribution for large n while $n\,p$ remains finite, i.e., p simultaneously has to become small as n grows in a controlled way. This controlled "dependence of p on n" (not an actual dependence) is such that, in the limit, on average we should expect λ successes among the n trials.

— Start Philosophical Interlude —

Example 1.41 (Bertrand's paradox: the ambiguity of "random"). Consider a circle of radius R. What is the probability that a randomly chosen chord of this circle is longer than the side of an inscribed equilateral triangle? This seemingly simple question, posed by Joseph Bertrand in 1889, reveals a profound ambiguity in what "randomly chosen" means. Depending on the method of "random selection," we arrive at different answers.

Let s be the side length of an inscribed equilateral triangle, i.e., $s = R\sqrt{3}$. We are interested in $P(\text{chord length} > R\sqrt{3})$.

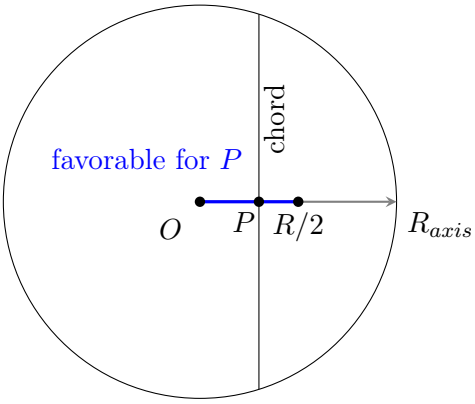
Method 1: random endpoints. Choose two points uniformly and independently on the circumference of the circle and join them to form a chord. Fix one endpoint A. For the chord to be longer than s, the second endpoint B must lie on the arc between vertices V_1 and V_2 of an equilateral triangle AV_1V_2, where this arc does not contain A. This favorable arc for B subtends an angle of $2\pi/3$ at the center O, and thus has length $(2\pi/3)R$.



$$P_1 = \frac{(2\pi/3)R}{2\pi R} = \frac{1}{3}$$

The probability is $P_1 = \frac{(2\pi/3)R}{2\pi R} = \frac{1}{3}$.

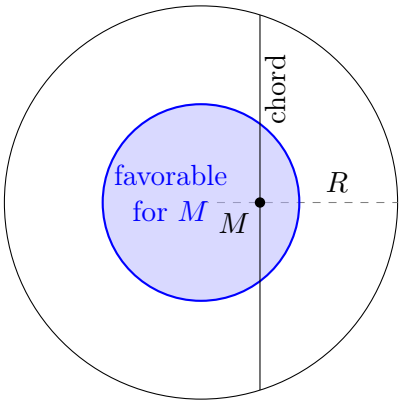
Method 2: random radial point. Choose one radius uniformly at random (uniform angle), then choose a point P uniformly along this radius. Each cord is uniquely determined by a midpoint within the circle, by constructing the corresponding chord perpendicular to the radius through P. For the chord to be longer than s, its midpoint (which is P) must be within a distance $R/2$ from the center O (since the apothem of the inscribed equilateral triangle is $R/2$).



$$P_2 = \frac{R/2}{R} = \frac{1}{2}$$

The probability is $P_2 = \frac{R/2}{R} = \frac{1}{2}$.

Method 3: random midpoint. Choose a point M uniformly at random from within the area of the circle to be the midpoint of the chord (uniquely determining the chord). For the chord to be longer than s, its midpoint M must lie within a concentric circle of radius $R/2$.



$$P_3 = \frac{\pi(R/2)^2}{\pi R^2} = \frac{1}{4}$$

The probability is $P_3 = \frac{\pi(R/2)^2}{\pi R^2} = \frac{1}{4}$.

We obtained three different probabilities (1/3, 1/2, 1/4) for the same event and all of them seem to be justified in that violating their respective assumptions would appear to violate the problem description. Bertrand's paradox highlights that when talking about infinite spaces, the phrase "randomly choose one of X" is ambiguous until the precise mechanism of randomness (i.e., the probability distribution over the sample space of X) is specified. Each method implicitly defines a different probability measure over the set of chords. See this video and this video for more details and animations

— End Philosophical Interlude —

1.3. Densities, continuous distributions, and random variables

Section goal. Introduce a concise mathematical framework to talk about "random variables" and their distributions or probability densities.

Definition 1.42 (probability density function). Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. A *(probability) density function (pdf) on* $(\Omega, \mathcal{A}, \mu)$ is a μ-integrable function $p : \Omega \to \mathbb{R}_{\geq 0}$ such that $\int p \, d\mu = 1$.

For $(\Omega, \mathcal{A}, \mu) = (\mathbb{R}^n, \mathcal{B}^n, \lambda^n)$, where λ^n is the n-dimensional Lebesgue measure, this reduces to $p : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ with

$$\int_{\mathbb{R}^n} p(x) \, dx = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} p(x_1, \ldots, x_n) \, dx_1 \ldots dx_n = 1 \, .$$

Definition 1.43 (density functions simplified). A *(probability) density function (pdf)* on $\mathbb{R}^n$ is a function $p : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ such that

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} p(x_1, \ldots, x_n) \, dx_1 \ldots dx_n = 1 \, .$$

Here, we leave out the technicalities around (Lebesgue-)integrability and assume the function "can be integrated" and integrates to 1.

Remark 1.44 (densities with respect to λ^n, Riemann vs Lebesgue). In this lecture, all probability density functions will be on $(\mathbb{R}^n, \mathcal{B}^n, \lambda^n)$, i.e., we always consider the Lebesgue measure as the underlying measure on $\mathbb{R}^n$.

People are most often familiar with the "Riemann integral" that can be derived from partitioning the area under a curve into many vertical rectangles and letting the width of these rectangles/bars go to zero. Unfortunately, the Riemann integral does not give good answers to integrals of functions with many discontinuities, i.e., only specific functions are "Riemann-integrable." The Lebesgue integral is more general, i.e., every function that is Riemann-integrable is also Lebesgue-integrable and in that case they agree. However, there are functions that are Lebesgue-integrable, but not Riemann-integrable. Unfortunately, we do not have time to go into the measure theoretic details of integration (with respect to a measure).

Proposition 1.45 (density induced probability measure). *Let $p : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ be a probability density function. Then*

$$\mathcal{B}^n \to [0,1],\ A \mapsto \int_{\mathbb{R}^n} p(x) \cdot \chi_A(x)\, dx$$

uniquely defines a probability measure on $(\mathbb{R}^n, \mathcal{B}^n)$ called the probability measure induced by the probability density function p (with respect to λ^n). Here χ_A is the indicator function with $\chi_A(x) = 1$ for all $x \in A$ and $\chi_A(x) = 0$ otherwise.

Theorem 1.46 (Radon-Nikodym (for $\mathbb{R}^n$)). *Let P be a probability measure on $(\mathbb{R}^n, \mathcal{B}^n)$. Then the following are equivalent:*

(i) We have $P \ll \lambda^n$ (read as P is absolutely continuous with respect to λ^n), i.e., for all $A \in \mathcal{B}^n$ with $\lambda^n(A) = 0$ we also have $P(A) = 0$. Note that $\lambda^n(A) = 0$ can (essentially) be interpreted as $\int_{\mathbb{R}^n} \chi_A(x)\, dx = 0$.

(ii) There exists a probability density function $p : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ inducing P.

When these hold, we call $(\mathbb{R}^n, \mathcal{B}^n, P)$ a continuous probability space.

Lemma 1.47 (relationship between pdf and cdf).

(i) If p is the pdf of a probability measure P on $(\mathbb{R}^n, \mathcal{B}^n)$ with cdf F, then

$$F(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} p(y_1, \dots, y_n)\, dy_1 \dots dy_n\,.$$

(ii) If the cdf F of a probability measure P on $(\mathbb{R}, \mathcal{B})$ is (weakly) differentiable, then $p := F'$ is the corresponding probability density of P.

In words: "the cdf is the integral of the pdf and the pdf is the derivative of the cdf."

Definition 1.48 (measurable map, random variable). Let $(\Omega, \mathcal{A})$ and $(\Omega', \mathcal{A}')$ be measurable spaces. A map $X : \Omega \to \Omega'$ is *measurable* (with respect to $\mathcal{A}$ and $\mathcal{A}'$), if $X^{-1}(A') \in \mathcal{A}$ for all $A' \in \mathcal{A}'$. Let $(\Omega, \mathcal{A}, P)$ be a probability space and $(\Omega', \mathcal{A}')$ a measurable space. A *$(\Omega', \mathcal{A}')$-valued random variable on $(\Omega, \mathcal{A}, P)$* is a measurable (with respect to $\mathcal{A}$ and $\mathcal{A}'$) map $\Omega \to \Omega'$.

For $(\Omega', \mathcal{A}') = (\mathbb{R}, \mathcal{B})$ we speak of a *real-valued random variable* and for $(\Omega', \mathcal{A}') = (\mathbb{R}^n, \mathcal{B}^n)$ we speak of a *random (real) vector*.

Convention 1.49 ((RV)RV). We abbreviate (real-valued) random variables with *(RV)RV*. For a random variable X as in definition 1.48, we write for $A' \subseteq \Omega'$ and $c' \in \Omega'$

$$\{X \in A'\} := \{\omega \in \Omega \mid X(\omega) \in A'\} = X^{-1}(A') \text{ and}$$
$$\{X = c'\} := \{\omega \in \Omega \mid X(\omega) = c'\} = X^{-1}(\{c'\})\,.$$

For a real-valued random variable X and $c, c' \in \mathbb{R}$ we analogously define

$$\begin{aligned}
\{X \leq c\} &:= \{\omega \in \Omega \mid X(\omega) \leq c\} = X^{-1}(\{x \mid x \in \mathbb{R}, x \leq c\}) \,, \\
\{X \geq c\} &:= \{\omega \in \Omega \mid X(\omega) \geq c\} = X^{-1}(\{x \mid x \in \mathbb{R}, x \geq c\}) \,, \\
\{X < c\} &:= \dots \\
&\dots \\
\{c' \leq X \leq c\} &:= \{\omega \in \Omega \mid c' \leq X(\omega) \leq c\} = X^{-1}(\{x \mid x \in \mathbb{R}, c' \leq x \leq c\}) \,, \\
&\dots
\end{aligned}$$

We will also slowly increase our sloppiness level and write $P(X \leq c)$ instead of $P(\{X \leq c\})$, or even use plain text to describe the event of interest like $P(\text{first throw hits the red target})$. Always remember that under the hood this describes an event $A \subseteq \Omega$.

Example 1.50. Continuous functions are measurable with respect to their Borel-σ-algebras.

Definition 1.51 (induced distribution)**.** Let $(\Omega, \mathcal{A}, P)$ be a probability space, $(\Omega', \mathcal{A}')$ a measurable space, $X : \Omega \to \Omega'$ a $(\Omega', \mathcal{A}')$-valued random variable on $(\Omega, \mathcal{A}, P)$. Then

$$P_X : \mathcal{A}' \to [0,1],\ A' \mapsto P\big(X^{-1}(A')\big) = P\big(\{X \in A'\}\big)$$

is a probability measure on $(\Omega', \mathcal{A}')$, the *(probability) distribution of X*. In short, $P_X = P \circ X^{-1}$. We then say *X follows P_X* or *X is distributed like/according to P_X* and denote it by $X \sim P_X$. You will often see notation like $X \sim \mathrm{Lap}_\Omega$ or $X \sim \mathcal{N}(0, \sigma^2)$ (which we yet have to define).

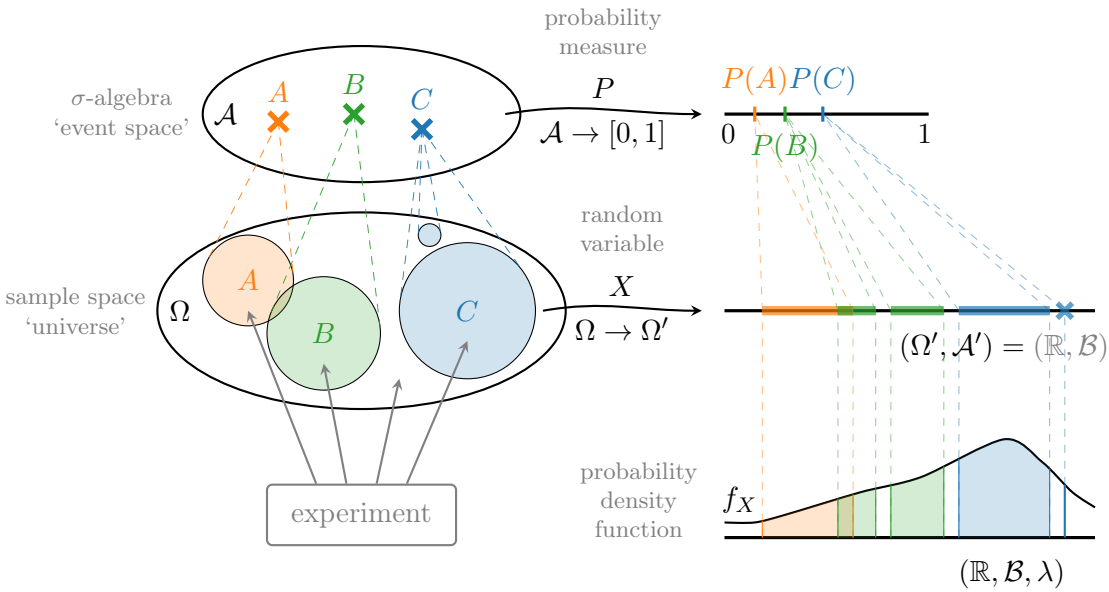
Definition 1.52 (identically distributed)**.** Let $(X_i)_{i \in I}$ be $(\Omega, \mathcal{A})$-valued random variables for an index set I. Then the X_i are called *identically distributed*, if for all $i \in I$ we have $P_{X_i} = P$ for some probability measure P on $(\Omega, \mathcal{A})$. Note that the base probability spaces of the random variables can be different.

Definition 1.53 (non-degenerate random variables)**.** We call a real-valued random variable or random vector X *non-degenerate* if there exists no $x \in \mathbb{R}$ (or $x \in \mathbb{R}^n$) such that $P(X = x) = 1$.

Definition 1.54 (cdf and pdf of a random variable)**.** Let X be an RVRV. The *cumulative distribution function (cdf) of X* is $F_X := F_{P_X}$ and (if F is (weakly) differentiable, i.e., P_X is continuous), the *probability density function (pdf) of X* is $p_X := F'_X$.

Corollary 1.55. *Real-valued random variables are identically distributed if and only if they have the same cumulative distribution function (or, if they exist, the same probability density function).*

Viewpoint 1.56 (common view of random variables)**.** What we have so far can be illustrated as follows



In practice, once we talk about random variables, we typically do not care about the specifics of the underlying probability space and where it came from. Even more, the specifics of the random variable itself are often irrelevant. All that matters is its distribution. In fact, we often "start from the assumed distribution/measure" and then posit that there is a random variable following that distribution.

That also means that all theoretical results we have for probability measures, directly transfer to random variables in the sense that the random variable naturally induces a probability measure.

Many problem setup sections in machine learning start with something like "let X be a random variable following distribution $P \ldots$" without any mention of $\Omega, \mathcal{A}, P$. In a sense, a random variable provides another "layer of abstraction" on top of the model $(\Omega, \mathcal{A}, P)$.

Remark 1.57 (pdfs and cdfs are all we need).

- The general version of Theorem 1.46 is formulated in terms of a generic σ-finite measure space (Ω, S, μ) and ensures the existence of a density function for a probability measure P if and only if $P \ll \mu$.
- Theorem 1.46 specifies that many "reasonable" probability measures can be fully characterized by their probability density function.
- Together with viewpoint 1.56 this is the reason we usually think of a real-valued random variable X (as well as $\mathbb{R}^n$-valued ones) directly in terms of the probability density function of the distribution induced by X with respect to the Lebesgue measure. We also call the density function the *Radon-Nikodym derivative* of P_X with respect to the Lebesgue measure.

- Probability mass functions are then the Radon-Nikodym derivative of P_X with respect to the counting measure.
- Being sloppy about notation, you may also encounter $X \sim p(x)$ or $X \sim F(x)$.

Definition 1.58 (Gamma and Beta functions). The *Gamma function* is

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx, \quad \alpha > 0\,.$$

The *Beta function* is

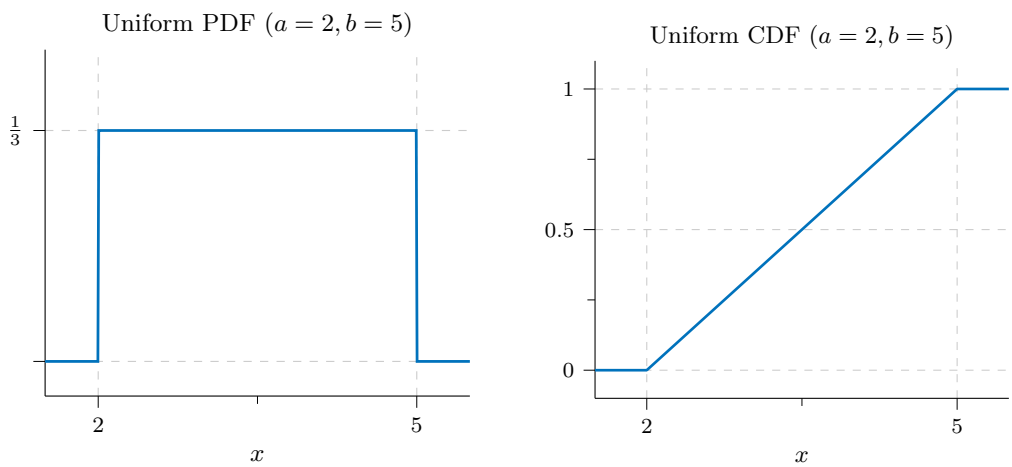
$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\, dx = \frac{\Gamma(\alpha)\,\Gamma(\beta)}{\Gamma(\alpha+\beta)}, \quad \alpha, \beta > 0\,.$$

Example 1.59 (common continuous probability measures). The following pdfs describe frequently occurring continuous probability measures, i.e., on (subsets of) $\mathbb{R}$.

(i) *Uniform distribution:* For $a, b \in \mathbb{R}$ with $a < b$, the $\mathrm{Unif}(a,b)$ distribution on $\Omega = [a,b]$ is characterized via the pdf

$$p_{\mathrm{Unif}(a,b)} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \chi_\Omega(x)\frac{1}{b-a} = \begin{cases} \dfrac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise}\,. \end{cases}$$

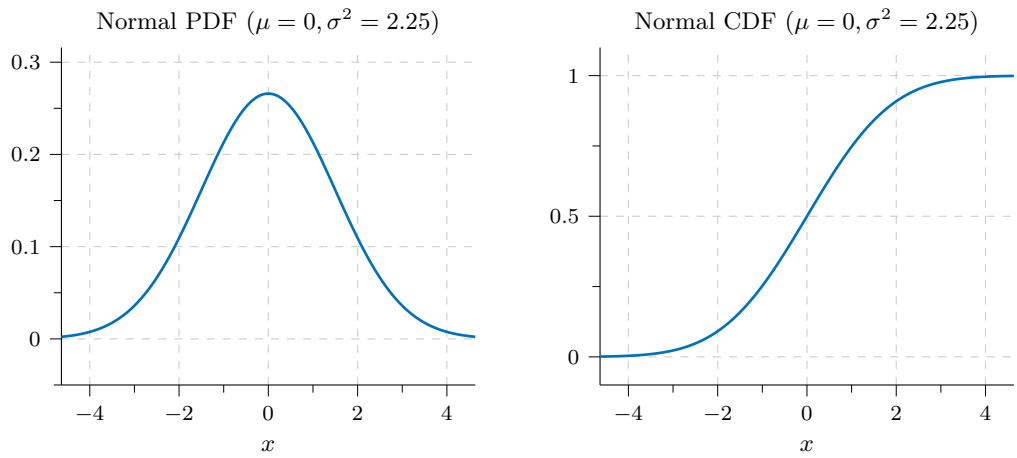
It models a random draw where every point in $[a,b]$ is equally likely, for example picking a random time uniformly between two deadlines.



(ii) *Normal distribution:* For $\mu \in \mathbb{R}$, $\sigma > 0$, the $\mathcal{N}(\mu, \sigma^2)$ distribution on $\Omega = \mathbb{R}$ is characterized via the pdf

$$p_{\mathcal{N}(\mu,\sigma^2)} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \frac{1}{\sqrt{2\pi}\,\sigma} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)\,.$$

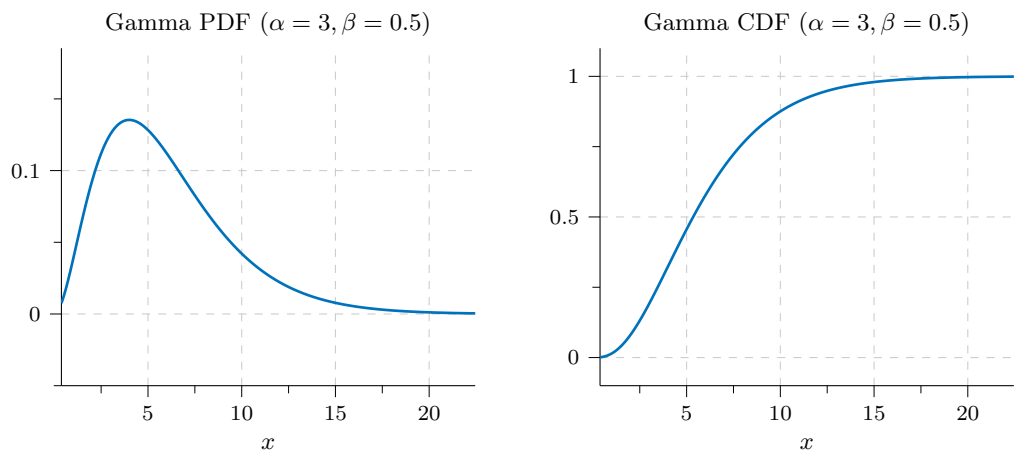
It models natural variability or measurement noise, for example the distribution of heights in a large population. It is arguably the most important (continuous) distribution of them all. A bit like the number π, it pops up over and over again in unexpected places and is in (specific mathematically rigorous) ways "natural."



(iii) *Gamma distribution:* For $\alpha > 0$, $\beta > 0$, the $\Gamma(\alpha, \beta)$ distribution on $\Omega = \mathbb{R}_{>0} = (0, \infty)$ is characterized via the pdf

$$p_{\Gamma(\alpha,\beta)} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \chi_\Omega(x) \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} .$$

It models the waiting time for α events in a Poisson process of rate β, for example the time until α emails arrive at rate β per hour.

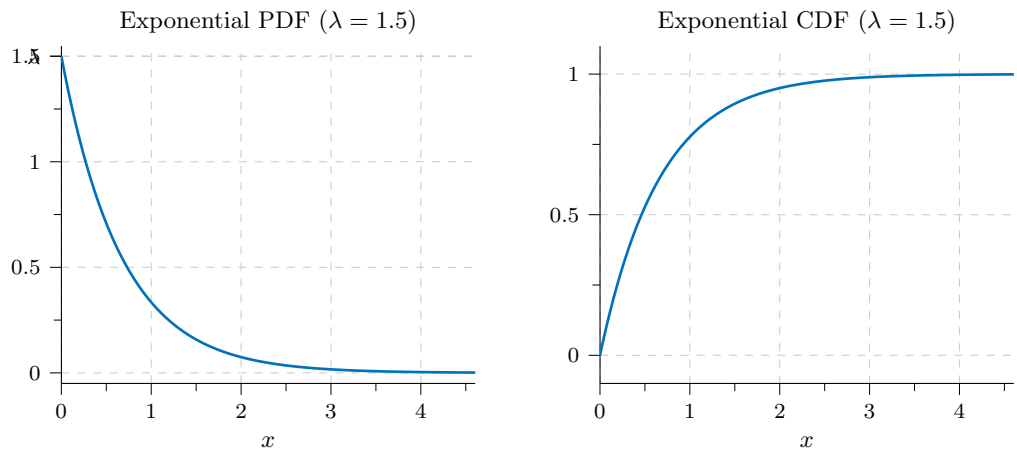


The Gamma distribution has other important distributions as special cases.

(iv) *Exponential distribution:* The special case $\mathrm{Exp}(\lambda) = \Gamma(1, \lambda)$ with $\lambda > 0$ on $\Omega = \mathbb{R}_{>0}$ is characterized via the pdf

$$p_{\mathrm{Exp}(\lambda)} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \chi_\Omega(x)\, \lambda e^{-\lambda x}\ .$$

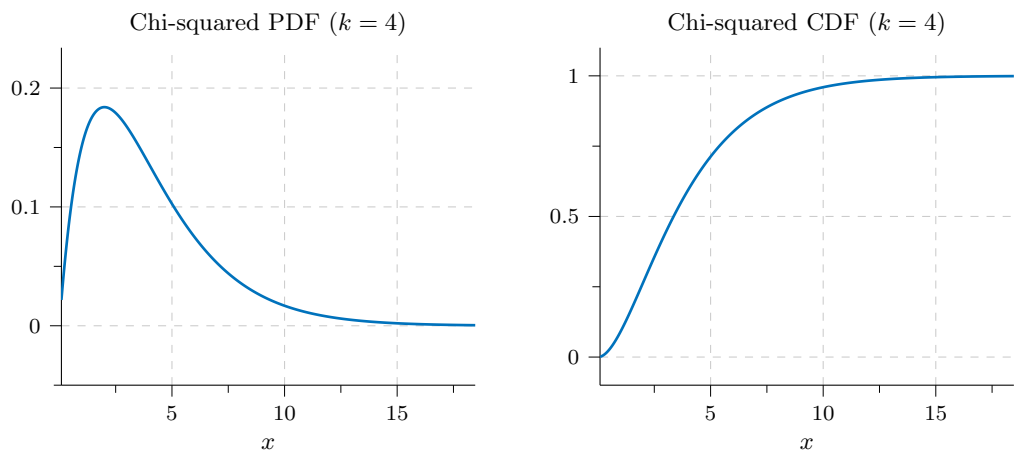
It models the time until the first event in a Poisson process of rate λ, for example the time until the next bus arrives or the length of a phone call in a call center.



(v) *Chi-squared distribution:* The special case $\chi^2_k = \Gamma(\frac{k}{2}, \frac{1}{2})$ with $k > 0$ on $\Omega = \mathbb{R}_{>0}$ is characterized via the pdf

$$p_{\chi^2_k} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \chi_\Omega(x)\, \frac{1}{2^{k/2}\Gamma(k/2)}\, x^{k/2-1} e^{-x/2}$$

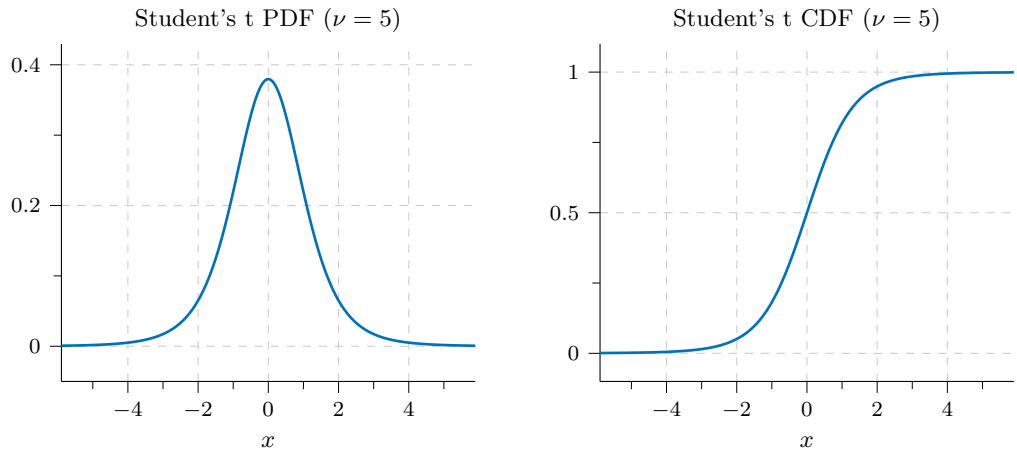
and called the *Chi-squared distribution with k degrees of freedom*. It appears as the sum of squares of k independent standard normals, for example in variance-based hypothesis testing.



(vi) *Student's t distribution:* For $\nu > 0$, the t_ν distribution on $\Omega = \mathbb{R}$ is characterized via the pdf

$$p_{t_\nu} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}} .$$

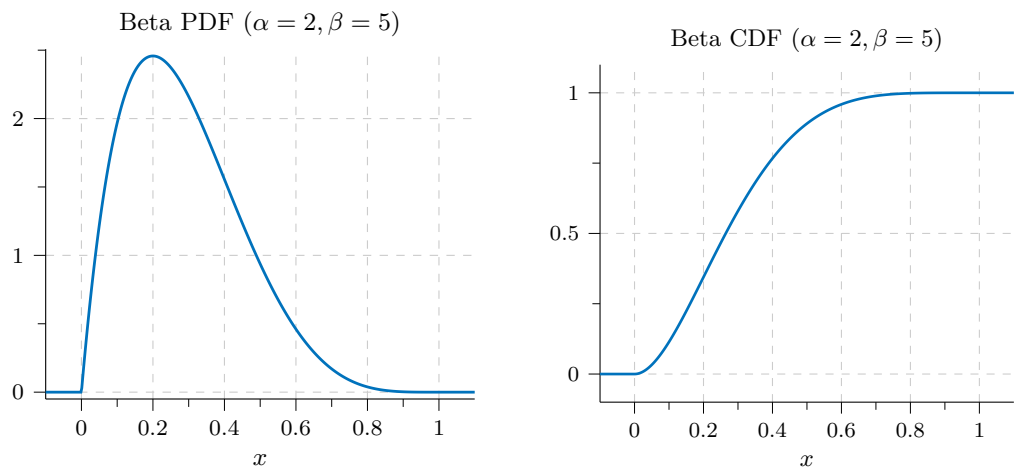
It models heavier-tailed data than the Gaussian, for example small-sample inference on a mean when variance is unknown. Like the χ^2_k it is important in statistical testing.



(vii) *Beta distribution:* For $\alpha > 0$, $\beta > 0$, the $\mathrm{Beta}(\alpha, \beta)$ distribution on $\Omega = [0, 1]$ is characterized via the pdf

$$p_{\mathrm{Beta}(\alpha,\beta)} : \mathbb{R} \to \mathbb{R}_{\geq 0},\ x \mapsto \chi_\Omega(x)\, \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} .$$

It models random proportions, for example the fraction of defective items in a batch. The uniform $\mathrm{Unif}(0, 1)$ is the special case $\mathrm{Beta}(1, 1)$.



Note that we have deliberately defined all densities on all of $\mathbb{R}$, even though many distributions are only supported on a subset, e.g., $\mathbb{R}_{>0}$. This is taken into account in the density via the indicator function χ_Ω. Instead, one often defines the density to only "exist" on this subspace, e.g., $p_{\mathrm{Beta}(\alpha,\beta)}(x) : [0,1] \to \mathbb{R}_{\geq 0}$.

Remark 1.60 (distributions as models). The student's t distribution t_ν arises as the ratio of a standard normal and the square-root of an independent χ^2_ν/ν. Many other continuous distributions—such as the log-normal, Cauchy, Weibull, Pareto, Rayleigh, and more—can be obtained via transformations, mixtures, or limits of those above. Again real-world accuracy, simplicity, computational feasibility, theoretical guarantees, or deliberate ignorance about certain aspects can all be valid reasons to choose any of these distributions to model a given situation.

As the (real) quote from George Box goes: "Remember that all models are wrong; the practical question is how wrong do they have to be to not be useful." This is often paraphrased as "All models are wrong, but some are useful." (Sometimes this quote is used to indicate that one shouldn't trust science too much, but that's not at all its intended meaning.) For example, a statistician may choose to model people's height via a random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ following a Gaussian distribution. This model is clearly "wrong" as any Gaussian has $P_X(X < 0) > 0$, i.e., there is a non-zero probability of a person having negative height. Also, $P(X > 5) > 0$, i.e., the probability of a person being taller than 5 meters i non-zero. Yet, if the ultimate goal is to estimate the fraction of people globally being taller than 1.9 meters from measuring the height of 100 representative humans, this model may be accurate and totally nail that inference.

Similarly, when modeling arrival times of customers at your store, you could use a (discrete) Poisson distribution with a fixed λ. This ignores that the store has opening times or that lunchtime or the time shortly before closing may be more popular. At the same time, it may give you a highly informative estimate to decide how many sales staff you should hire overall. If you split the day into segments (morning, lunchtime, pre-closing, etc) and use a time-dependent $\lambda(t)$ in response, you would still be ignoring events like a sudden rainfall, or pre-Christmas time. Even if you then decide to take into account such covariates Z capturing these settings via $\lambda(t, Z)$, one would still assume customers to arrive independently from one another, i.e., one is not properly capturing groups of people arriving at once. As you can roll out in your mind, if you pushed this further and further you would likely end up with a large scale simulation including weather, city dynamics (e.g., construction sites), weather, people's behavior, fashion trends, and—ultimately—the entire evolution of the universe. Maybe the $\mathrm{Poi}(\lambda)$ was actually quite good in answering the question you were originally interested in.

Example 1.61 (continuous distribution applications). Probabilities for continuous random variables are typically computed by integrating their probability density function (pdf) over the region of interest. For a random variable X with pdf $p_X(x)$, the probability that X falls into an interval $[a, b]$ is given by $P(a \leq X \leq b) = \int_a^b p_X(x)\, dx$.

(i) **Bus arrival time:** A bus is scheduled to arrive at a bus stop at any time between 10:00 AM and 10:30 AM, with all times equally likely. If you arrive at the bus stop exactly at 10:00 AM, what is the probability that you have to wait for the bus for less than 10 minutes?

Let T be the arrival time of the bus in minutes past 10:00 AM. Since the arrival time is uniformly distributed over a 30-minute interval, T follows a uniform distribution $\text{Unif}(0, 30)$. The pdf of T is

$$p_T(t) = \begin{cases} \frac{1}{30-0} = \frac{1}{30} & \text{for } 0 \leq t \leq 30\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

We are interested in the probability that the waiting time is less than 10 minutes, which means the bus arrives between 10:00 AM and 10:10 AM. So, we want to calculate

$$P(0 \leq T < 10) = \int_0^{10} p_T(t)\,dt = \int_0^{10} \frac{1}{30}\,dt = \frac{1}{30}[t]_0^{10} = \frac{10-0}{30} = \frac{10}{30} = \frac{1}{3}\,.$$

Thus, there is a $1/3$ chance of waiting less than 10 minutes.

(ii) **Component lifetime:** The lifetime of a certain type of electronic component, in years, is modeled via an exponential distribution. The average lifetime of these components is 5 years. What is the probability that a randomly selected component lasts for more than 10 years? Let L be the lifetime of the component in years, then L follows an exponential distribution $\text{Exp}(\lambda)$. The mean of an exponential distribution $\text{Exp}(\lambda)$ is $1/\lambda$ (more on expectations/means later; we'll simply use this now for the example already). Knowing that the average lifetime is 5 years implies $\lambda = 1/5 = 0.2$. Hence, the pdf of L is

$$p_L(l) = \begin{cases} 0.2e^{-0.2l} & \text{for } l \geq 0\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

The probability that the component lasts for more than 10 years is then

$$P(L > 10) = \int_{10}^{\infty} 0.2e^{-0.2l}\,dl = \left[-e^{-0.2l}\right]_{10}^{\infty} = e^{-2} \approx 0.1353\,.$$

There is approximately a 13.53% chance that the component will last more than 10 years. At this point, as an actual product developer, you'll likely have tons of follow up questions: How many returns should I expect as a function over time. How would revenue be affected if this part was more durable, let's say if the average lifetime was extended by 6 months?

(iii) **Standardized test scores:** The scores on a nationwide standardized mathematics test are found to be normally distributed with a mean $\mu = 100$ points and a standard deviation $\sigma = 15$ points. (Why is this model "wrong"?) What is the probability

that a randomly selected student scores between 90 and 120 points? (Why is this still a meaningful question to pose to the model?)

Let X be the score of a randomly selected student following a normal distribution $\mathcal{N}(100, 15^2)$. The pdf of X is

$$p_X(x) = \frac{1}{15\sqrt{2\pi}} \exp\left(-\frac{(x-100)^2}{2 \cdot 15^2}\right) .$$

We want to calculate

$$P(90 \leq X \leq 120) = \int_{90}^{120} \frac{1}{15\sqrt{2\pi}} \exp\left(-\frac{(x-100)^2}{2 \cdot 15^2}\right) dx .$$

This integral does not have a closed-form elementary solution. To evaluate it, we typically ask a computer. Since this is a lecture, we bend over backwards and try to do it analytically. First, we standardize the random variable X to a standard normal random variable $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$ (transformation of densities yet to come). The lower limit $x_1 = 90$ becomes $z_1 = \frac{90-100}{15} = -\frac{2}{3}$. The upper limit $x_2 = 120$ becomes $z_2 = \frac{120-100}{15} = \frac{4}{3}$. Hence,

$$P(90 \leq X \leq 120) = P\Big(-\frac{2}{3} \leq Z \leq \frac{4}{3}\Big) .$$

The cdf of a standard normal distribution $\mathcal{N}(0,1)$ is often denoted by $\Phi(z)$. Values of $\Phi(z)$ are typically found using statistical tables or software (e.g., `scipy.stats.norm.cdf` in Python). Using this notation,

$$P\Big(-\frac{2}{3} \leq Z \leq \frac{4}{3}\Big) = \Phi\Big(\frac{4}{3}\Big) - \Phi\Big(-\frac{2}{3}\Big) .$$

Since $\Phi(-z) = 1 - \Phi(z)$, we have $\Phi(-\frac{2}{3}) = 1 - \Phi(\frac{2}{3})$. Numerically, this evaluates to

$$P(90 \leq X \leq 120) \approx 0.6563 .$$

Thus, approximately 65.63% of students score between 90 and 120 points.

— Start Motivation —

Why random variables

In the introductory motivation, you may have noticed that the modeling of the probability space itself came quite natural in the first example—drawing a single card from a well shuffled deck. We understand where the "randomness" comes from and it directly links to the probabilities of the outcomes we are interested in (drawing any specific card). When

we moved on to the number of draws until $A\heartsuit$ is drawn, the underlying mechanism of randomness remained the same, but the space of outcomes was a somewhat complex construction on top of that underlying randomness mechanism. The original "randomness" was over 52 abstract symbols with no ordering $2\clubsuit, 2\spadesuit, \ldots, A\diamondsuit, A\heartsuit$, but now the outcomes of interest are integers $1, 2 \ldots$. Those are entirely different spaces. Intuitively, it was clear how we can take the underlying distribution (uniform over the 52 cards, each happening with probability $\frac{1}{52}$) to the probability of needing k trials until drawing the $A\heartsuit$. But we had to somehow directly write down what $p(k)$ has to be, intuitively. One would be hard pressed to justify formally where these values come from. For example, why do you multiply the probability of failure and non-failure? For $k > 2$, do we need to somehow account for the order of failures in the term $(\frac{51}{52})^{k-1}$ by some combinatorial factor?

In other cases, it is often somewhat impossible to reason our way to the distribution over the outcomes of interest. For example, when we model the number of customers arriving at a store within a fixed time window, where does the "original randomness" come from. What is the equivalent of the uniform distribution over the 52 cards? We would have to make assumptions about a pool of potential customers somehow making up their mind "randomly," whether to walk into the store at any given time, which appears awkward only to then model on top of this fundamental randomness the outcomes of interest (the total number of people arriving in a given time interval).

This is one of the main reasons for using *random variables*. We can capture the process of "building outcomes of interest on top of basic random event" in random variables. For example, in our card drawing example, we can take $\mathbb{N}$ (independent) copies of $(\Omega, \mathcal{P}(\Omega), P)$, which would represent our iterative game of shuffling and drawing until infinity. This *product probability space* captures the "basic underlying randomness." On top of this product space, we could build a random variable $X : \Omega^{\mathbb{N}} \to \{1, 2, \ldots\}$ that simply looks for the first trial that gave a $A\heartsuit$. For example $X(7\spadesuit, 2\heartsuit, K\diamondsuit, 2\heartsuit, A\heartsuit, \ldots) = 5$. Of course, that is not the only input that gives a 5. When we take all possible series of cards that have the first $A\heartsuit$ in the fifth position, we get an event in the *product space* and can apply the product measure to it to obtain the probability of that event. Guess what, it's going to be $p(5) = (\frac{51}{52})^4 \frac{1}{52}$.

Of course, this would be completely impractical to actually compute for every k—*but we can do it*! Instead, we worked with our (well informed) intuition to just directly write down what the distribution of X is going to be like by directly specifying its pmf. In many situations, it will be much simpler and a more elegant modeling approach to say: What I am interested in here is a "number of trials until success" problem, so I will model my outcome of interest with a random variable $X \sim \mathrm{Geo}(p)$. Since we know the pmf of X, everything is said and we don't need any further information. The whole probability space, the problem where randomness comes from and how to model it, the random variable X as an actual measurable map, all of that has been pushed into the background. By just saying *I decide to model my outcome of interest via* $X \sim \mathrm{Geo}(p)$, we have specified Ω (implicit in $\mathrm{Geo}(p)$), the σ-algebra (always the power set anyway, just because we can), and the probability measure (defined fully by the pmf $p_{\mathrm{Geo}(p)}$).

In the above game, I could have also said I am interested in the number of aces you draw in the first $n = 50$ draws. Intuitively, we would again sit down, think hard, and try to directly specify $p(0)$: What's the probability I'll see no ace at all? All 50 draws would have to be no ace, which happens with $q := \frac{48}{52}$, so $p(0) = q^n$. To see exactly one ace, it could be in any of the $n = 50$ positions, there it happens with probability $1 - q$, in all other positions I want no ace (probability q for each) giving $p(1) = 50(1 - q)q$. If you continue like this, you'll likely generalize this to the Binomial pmf $p_{\mathrm{Bin}(n,q)}(k) = \binom{n}{k}(1-q)^k q^{n-k}$. (Note how success and failure probability have reversed their roles here, but this is just semantics.) And again, if I were to push you for rigorous mathematical reasoning you would construct the product probability space $(\Omega^{50}, \mathcal{P}(\mathbb{N}^{50}), \mathrm{Lap}_{\Omega}^{50})$ and introduce a function counting the number of aces in each sequence of 50 cards and then compute the probability of the pre-images of each natural number. Tedious! Instead, we recognize that this is a "number of successes in n independent trials" problem and directly say: We model the number of aces drawn via a Binomial distribution with success probability $\frac{4}{52}$.

Main takeaway: In practice, we typically directly model our outcome of interest by a random variable with a prescribed/pre-assumed distribution.

Our obsession with real-valued random variables

Often, the outcomes in Ω aren't numbers themselves (like cards), but we're interested in some numerical aspect of these outcomes. This is also a way in which random variables come into play. A RVRV is a function that maps outcomes from Ω to real numbers. This is often obvious in the example of dice. Each abstract "face" of the die with its dot pattern is equated with the corresponding number. It makes sense to ask about what the average rolled number is (we all intuitively know that it's 3.5). We can further ask for the variance, that is, how much the numbers typically vary or how broad the spread of numbers is. For abstract discrete probability spaces, like the deck of cards, there is no such notion. What is the "average card" being drawn? Hence, we often care about outcomes that take numerical values which we achieve either implicitly (dice rolls, counting numbers of events or trails until success) or explicitly by defining suitable RVRVs.

Let's stick with our card example. Suppose we're playing a simplified Blackjack where cards have point values. Let X be an RV representing the point value of a drawn card:

- $X(\text{Ace}) = 11$
- $X(\text{King}), X(\text{Queen}), X(\text{Jack}), X(\text{Ten}) = 10$
- $X(\text{Nine}) = 9, \ldots, X(\text{Two}) = 2$

Now, instead of asking about $P(A\spadesuit)$, we might ask: what's $P(X = 10)$? The outcomes $\omega \in \Omega$ that map to $X(\omega) = 10$ are the 16 face cards (K, Q, J, 10 of each suit). So, $\{\omega \in \Omega \mid X(\omega) = 10\}$ is an event in our original space Ω. Its probability is $P(X =$

$10) = P(\{\text{any K, Q, J, or 10}\}) = \frac{16}{52}$. Similarly, $P(X = 11) = P(\{\text{any Ace}\}) = \frac{4}{52}$. For $k \in \{2, \dots, 9\}$, $P(X = k) = P(\{\text{any card of rank k}\}) = \frac{4}{52}$.

The RV X induces a new probability distribution on the set of possible numerical values $\Omega_X = \{2, 3, \dots, 9, 10, 11\}$. This new distribution has its own pmf, say $p_X(k) = P(X = k)$. Once we have numerical values and their probabilities, we can compute important summaries:

- *expectation (mean):* $\mathbb{E}[X] = \sum_{k \in \Omega_X} k \cdot p_X(k)$. This gives the "average" value of X. For our card example: $\mathbb{E}[X] = \sum_{k=2}^{9} k \frac{4}{52} + 10 \frac{16}{52} + 11 \frac{4}{52} = \dots$ This tells us the average point value we'd expect if we drew many cards (repeatedly from the full deck).
- *variance:* $\mathrm{var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$. This measures the "spread" of the values around the mean.

In abstract discrete settings, the notions of expectation and variance are sometimes not too useful (we could encode abstract outcomes in any arbitrary way numerically). While the expectation is not necessarily useful in this concrete example, it is certainly an interesting question to ask "how long should we expect to play our game of shuffling until you get a $A\heartsuit$?"

Random variables allow us to abstract away from the original, potentially complex, sample space Ω and focus on numerical quantities of interest. They provide a bridge from qualitative outcomes to quantitative analysis.

— End Motivation —

1.4. Product spaces, independence, conditional probability

Section goal. Formalize how we can talk about multiple measures (random variables) jointly, what it means for events (random variables) to be independent, and how to "condition" on events (random variables).

Convention 1.62 (assumptions, reminder of the cartesian product). For this section, let $(\Omega, \mathcal{A}, P)$ be a probability space and $\big((\Omega_i, \mathcal{A}_i)\big)_{i \in I}$ a family of measurable spaces for a non-empty index set I. For all $i \in I$, let X_i be a $(\Omega_i, \mathcal{A}_i)$-valued random variable.

The set $\Omega_I := \prod_{i \in I} \Omega_i$ of all maps $(\omega_i)_{i \in I} : I \to \bigcup_{i \in I} \Omega_i$ such that $\omega_i \in \Omega_i$ for all $i \in I$, is called *product space* or *cartesian product*. If all Ω_i are equal, e.g., $\Omega_i = \Omega$ for all $i \in I$, we write $\Omega^I := \Omega_I$. If $n := |I| < \infty$, we use the notation $\Omega^n := \Omega^I$.

Intuitively, the elements of the product space are tuples with one slot for each element of I and each slot can be filled with any element of Ω_i. The definition given above also properly accommodates infinite index sets I.

Remark 1.63 (product σ-algebras, product measures). So far we have only talked about individual probability spaces. There exist natural and unique constructions for a σ-algebra on the product Ω_I, which is called *product σ-algebra* and denoted by $\mathcal{A}_I := \bigotimes_{i\in I} \mathcal{A}_i$. Similarly, there is a natural and unique construction for the so called *product measure*, denoted by $P_I := \bigotimes_{i\in I} P_i$ if one additionally has a probability measure P_i on each $(\Omega_i, \mathcal{A}_i)$, yielding a *product probability space* $(\prod_{i\in I} \Omega_i, \bigotimes_{i\in I} \mathcal{A}_i, \bigotimes_{i\in I} P_i) = (\Omega_I, \mathcal{A}_I, P_I)$.

While this general construction allows us to talk about "joint probability spaces" and "joint distributions" of any (finite) set of random variables, in practice we mostly consider products of either discrete random variables or products of real-valued random variables. (Note how I immediately extended the definition to random variables.) It turns out that $\bigotimes_{i=1}^n \mathcal{B} = \mathcal{B}^n$ and we can talk about the product measure of n real-valued random variables simply in terms of their joint density function $p : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$.

Definition 1.64 (joint/marginal distribution). The induced measure $P_X := P \circ X^{-1}$ under $X := (X_i)_{i\in I}$ on $(\Omega_I, \mathcal{A}_I)$ (this is actually a $\Omega_I, \mathcal{A}_I$)-valued random variable) is called *joint distribution* of the family $(X_i)_{i\in I}$. For each $i \in I$ the induced measure $P_{X_i} := P \circ X_i^{-1}$ is called the *marginal distribution of X_i* with respect to P_X.

Example 1.65 (product pmf/pdf). Let $p_1, \ldots, p_n$ be pmfs on Ω for $n \in \mathbb{N}_{>0}$. Then

$$p(\omega) := \prod_{i\in[n]} p_i(\omega_i) \quad \text{for } \omega = (\omega_i)_{i\in[n]} \in \Omega^n$$

is called the *product pmf* of the $(p_i)_{i\in[n]}$ on Ω^n.

Let $p_1, \ldots, p_n$ be pdfs on $\mathbb{R}$ for $n \in \mathbb{N}_{>0}$. Then

$$p(x) := \prod_{i\in[n]} p_i(x_i) \quad \text{for } x = (x_i)_{i\in[n]} \in \mathbb{R}^n$$

is called the *product pdf* of the $(p_i)_{i\in[n]}$ on $\mathbb{R}^n$.

Intuitively, taking the joint of a finite collection of measures on the same space is represented by the product of the respective pmfs or pdfs.

Theorem 1.66 (computing marginals). *Let $I = [n]$ and $X := (X_i)_{i\in[n]}$. Then the following hold.*

(i) If all X_i are real-valued random variables, with joint cdf

$$F_X(x) = P(X_1 \leq x_1, \ldots, X_n \leq x_n) \quad \textit{for all } x \in \mathbb{R}^n\,,$$

then the marginal cdf for each X_i is

$$F_{X_i}(x_i) = \lim_{x_1\to\infty} \ldots \lim_{x_{i-1}\to\infty} \lim_{x_{i+1}\to\infty} \ldots \lim_{x_n\to\infty} F_X(x_1, \ldots, x_n) \quad \textit{for all } x_i \in \mathbb{R}\,.$$

(ii) *If X is a discrete random variable with joint pmf $p_X : \prod_{i=1}^n \Omega_i \to [0,1]$, then each X_i is discretely distributed with marginal pmf*

$$p_{X_i}(\omega_i) = \sum_{\omega_1 \in \Omega_1} \cdots \sum_{\omega_{i-1} \in \Omega_{i-1}} \sum_{\omega_{i+1} \in \Omega_{i+1}} \cdots \sum_{\omega_n \in \Omega_n} p_X(\omega_1, \ldots, \omega_n) \quad \text{for all } \omega_i \in \Omega_i \,.$$

(iii) *If X is a continuous random variable with joint pdf $p_X : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$, then each X_i is continuously distributed with marginal pdf*

$$p_{X_i}(x_i) = \int_{\mathbb{R}} \cdots \int_{\mathbb{R}} p_X(x_1, \ldots, x_n)\, dx_n \ldots dx_{i+1}\, dx_{i-1} \ldots dx_1 \quad \text{for all } x_i \in \mathbb{R} \,.$$

Definition 1.67 (conditional probability). Let $A, B \in \mathcal{A}$ be events with $P(B) > 0$. Then we denote with

$$P(A \mid B) := \frac{P(A \cap B)}{P(B)}$$

the *conditional probability* of A given B.

Theorem 1.68 (total probability, Bayes formula). *Let $B \in \mathcal{A}$ with $P(B) > 0$ and I a countable index set. Then the following hold.*

(i) *The map*

$$\widetilde{P} : \mathcal{A} \to [0,1] \,, \quad A \mapsto \widetilde{P}(A) := P(A \mid B)$$

is a probability measure on $(\Omega, \mathcal{A})$. The probability space $(B, \mathcal{A}|_B, \widetilde{P}|_B)$ is called trace of $(\Omega, \mathcal{A}, P)$ on B.

(ii) *Let $\{B_i \mid i \in I\} \subseteq \mathcal{A}$ be a measurable* partition *of B, i.e., $\bigcup_{i \in I} B_i = B$ and $B_i \cap B_j = \emptyset$ for all $i, j \in I$ with $i \neq j$. For all $A \in \mathcal{A}$ we then have the* law of total probability

$$P(A \cap B) = \sum_{i \in I} P(A \mid B_i) P(B_i) \,.$$

(iii) *For each $A \in \mathcal{A}$ with $P(A) > 0$ and every measurable partition $\{B_i \mid i \in I\}$ of Ω with $P(B_i) > 0$ for all $i \in I$, we have the* Bayes rule

$$P(B_i \mid A) = \frac{P(A \mid B_i) P(B_i)}{\sum_{j \in I} P(A \mid B_j) P(B_j)} \quad \text{for all } i \in I \,.$$

Lemma 1.69 (trivial factorization). *For $(A_i)_{i \in [n+1]} \subseteq \mathcal{A}$ with $P(\bigcap_{i \in [n]} A_i) > 0$ we have*

$$P\Big(\bigcap_{i \in [n+1]} A_i\Big) = P(A_1) P(A_2 \mid A_1) P(A_3 \mid A_1 \cap A_2) \ldots P\Big(A_{n+1} \,\Big|\, \bigcap_{i \in [n]} A_i\Big) \,.$$

Example 1.70 (Bayes rule example). You and your friend have been jointly working on a research report in a Gdoc. At some point, I realize that the most recent changes include entirely fabricated results. I want to know who is more likely to have made these changes. (We assume only you two have been working on it and the changes were made from a non-logged in, i.e., anonymous, account.)

I recognize that the section in question uses British English. By analyzing the existing text, I also figure out that you have been contributing 70% of the text, whereas your friend only contributed 30%. Moreover, you have used British English in 90% of your code, whereas your friend prefers American English spelling and only uses British English in about 20% of their contributions.

With this information, what is the probability that the fabricated results come from you, given that they are using British English?

Now comes the modeling (and we can argue about it!): Let's say I want to take the spelling into account, then we consider two random variables A (author) taking values in $\{y, f\}$ (you vs your friend) and S (spelling) taking values in $\{\text{uk}, \text{us}\}$. (Ultimately the samples space of interest is thus $\Omega = \{y, f\} \times \{\text{uk}, \text{us}\}$.) Let P be the joint distribution of (A, S) on $(\Omega, \mathcal{P}(\Omega))$ of interest. I obtained information about marginal distributions, namely $P_A(\{y\}) = 0.7$, $P_A(\{f\}) = 0.3$ as well as some conditional probabilities. To write out those, we need the following events

$$\begin{aligned} E_Y &= \{(y, \text{us}), (y, \text{uk})\} && \text{“you wrote it”}\,, \\ E_F &= \{(f, \text{us}), (f, \text{uk})\} && \text{“your friend wrote it”}\,, \\ E_{\text{UK}} &= \{(y, \text{uk}), (f, \text{uk})\} && \text{“British spelling was used”}\,. \end{aligned}$$

Then, I also have the following conditional probabilities

$$\begin{aligned} P(\text{British spelling was used} \mid \text{you wrote it}) &:= P(E_{\text{UK}} \mid E_Y) = 0.9\,, \\ P(\text{British spelling was used} \mid \text{your friend wrote it}) &:= P(E_{\text{UK}} \mid E_F) = 0.2\,. \end{aligned}$$

The marginals can also be written as $P(E_Y) = 0.7$ and $P(E_F) = 0.3$.

The probability I am interested in is $P(E_Y \mid E_{\text{UK}})$, i.e., what's the probability you were the author given the fabricated result section used British spelling. Now we can employ Bayes rule

$$P(E_Y \mid E_{\text{UK}}) = \frac{P(E_{\text{UK}} \mid E_Y)P(E_Y)}{P(E_{\text{UK}})}\,.$$

For the marginal in the denominator, we use the law of total probability

$$P(E_{\text{UK}}) = P(E_{\text{UK}} \mid E_Y)P(E_Y) + P(E_{\text{UK}} \mid E_F)P(E_F) = 0.9 \cdot 0.7 + 0.2 \cdot 0.3 = 0.69\,,$$

where we assert that $\{E_Y, E_F\}$ is a measurable partition of Ω. Hence

$$P(E_Y \mid E_{\text{UK}}) = \frac{P(E_{\text{UK}} \mid E_Y)P(E_Y)}{P(E_{\text{UK}})} = \frac{0.9 \cdot 0.7}{0.69} \approx 91.3\%\,.$$

Now don't fabricate results again!

Remark 1.71 (Bayesian inference).

(i) Bayes rule is often interpreted as a rule of how to update beliefs in the face of new evidence

$$\underbrace{P(A \mid B)}_{\text{posterior}} = \frac{\overbrace{P(B \mid A)}^{\text{likelihood}} \cdot \overbrace{P(A)}^{\text{prior}}}{\underbrace{P(B \mid A)P(A) + P(B \mid A^{\mathsf{c}})P(A^{\mathsf{c}})}_{\text{marginal likelihood / total evidence}}},$$

where A is the event of interest and B is the evidence.

In the example above, we interpret this as follows: My prior belief of you having written the text was $P(E_Y) = 0.7$. After having seen the evidence of British spelling being used (which has a high likelihood for you $P(E_{\text{UK}} \mid E_Y) = 0.9$), my updated belief about you being the author is $P(E_Y \mid E_{\text{UK}}) \approx 0.913$.

(ii) In Bayesian inference, one then also often see variants like

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})},$$

where θ are model parameters to be inferred and $\mathcal{D}$ is data (observed evidence). Given a prior over models $p(\theta)$ (often chosen "uninformatively" or such that "simple models are preferred") and the likelihood $p(\mathcal{D} \mid \theta)$ (how likely is observing data $\mathcal{D}$ if model θ is the ground truth), and the overall probability of observing the data $p(\mathcal{D})$, Bayes rule then gives us a posterior over models given the data, i.e., how probable each model θ is the ground truth given the data. In practice, obtaining $p(\mathcal{D})$ is often the culprit, but a lot can still be done from knowing that $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta)$. In a sense, $p(\mathcal{D})$ is just a "normalizing constant."

Definition 1.72 (independent events).

(i) Two events $A, B \in \mathcal{A}$ are called *independent (under P)*, which we denote by $A \perp\!\!\!\perp B$, if $P(A \cap B) = P(A)P(B)$.

(ii) A family of events $(A_i)_{i \in I}$ in $\mathcal{A}$ is called *independent (under P)*, which we denote by $\perp\!\!\!\perp_{i \in I} A_i$, if for each finite subset $J \subseteq I$ we have $P(\bigcap_{j \in J} A_j) = \prod_{j \in J} P(A_j)$.

Example 1.73 (pairwise independence does not imply independence). Let $\Omega = \{\circ, \triangle, \square, \star\}$ and assume a uniform probability measure P on $(\Omega, \mathcal{P}(\Omega))$. Then for

$$A = \{\circ, \triangle\}, B = \{\circ, \square\}, C = \{\circ, \star\}$$

we have $P(A) = P(B) = P(C) = \frac{1}{2}$ and

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = P(\{\circ\}) = \tfrac{1}{4} = P(A)P(B) = P(A)P(C) = P(B)P(C).$$

So, the events A, B, C are pairwise independent. However,

$$P(A \cap B \cap C) = P(\{\circ\}) = \tfrac{1}{4} \neq \tfrac{1}{8} = P(A)P(B)P(C)\,.$$

So a set of events being pairwise independent does not imply they are jointly independent. The reverse does hold true.

Example 1.74 (independence despite "dependence"). Consider the roll of two distinct fair dice modeled by $\Omega = [6]^2$, σ-algebra $\mathcal{P}(\Omega)$ and a probability measure P following the uniform distribution Lap_Ω. Now consider the events

$$\begin{aligned} A &:= \{\text{dice sum to } 7\} = \{(i,j) \in \Omega \mid i + j = 7\}\,, \\ B &:= \{\text{first die shows } 6\} = \{(i,j) \in \Omega \mid i = 6\}\,. \end{aligned}$$

Then we have $|A| = |B| = 6$ and $|A \cap B| = |\{(6,1)\}| = 1$ and thus $P(A \cap B) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = P(A)P(B)$. Thus the events are independent even though intuitively the sum of the dice depends on what the first one shows. Independence is not equivalent to "things that don't have anything to do with one another."

Definition 1.75 (independence of RVs).

(i) A family $(X_i)_{i \in I}$ of real-valued random variables is *independent* if for each finite subset $J \subseteq I$ and for all $x_j \in \mathbb{R}$ $(j \in J)$, we have

$$P\Big(\bigcap_{j \in J} \{X_j \leq x_j\}\Big) = \prod_{j \in J} P(X_j \leq x_j)\,.$$

For $I = [n]$, $X_1, \ldots, X_n$ are independent if and only if for all $x \in \mathbb{R}^n$ it holds that

$$F_{(X_1,\ldots,X_n)}(x_1, \ldots, x_n) = \prod_{i \in [n]} F_{X_i}(x_i)\,.$$

We then say the *cdf factorizes*.

(ii) For $(\Omega_i, \mathcal{A}_i)$-valued random variables X_i for $i \in [n]$ where are all Ω_i are discrete, we call $X_1, \ldots, X_n$ *independent* if for all $s \in \prod_{i \in [n]} \Omega_i$ we have

$$p_{(X_1,\ldots,X_n)}(s) = \prod_{i \in [n]} p_{X_i}(s_i)\,.$$

We then say the *pmf factorizes*.

(iii) For a continuous random vector $(X_1, \ldots, X_n)$, we call $X_1, \ldots, X_n$ *independent* if

$$p_X(x) = \prod_{i \in [n]} p_{X_i}(x_i)$$

for "Lebesgue-almost-all" $x \in \mathbb{R}^n$, i.e., everywhere on $\mathbb{R}^n$ except for a set of Lebesgue measure zero. We then say the *pdf factorizes*.

Definition 1.76 (conditional cdf, pmf, pdf). Let X, Y be $(\Omega, \mathcal{A})$ and $(\Omega', \mathcal{A}')$-valued random variables and let P be the their joint distribution.

(i) Let Ω, Ω' be discrete and $P(\{X = x\}) > 0$, where

$$\{X = x\} := \{(x', y') \in \Omega \times \Omega' \mid x' = x\} .$$

Then the *conditional pmf of Y given $X = x$* is

$$p_{Y|X=x}(y) = \frac{P(\{X = x\} \cap \{Y = y\})}{P(X = x)} .$$

(ii) Let Ω, Ω' be continuous, X, Y have a density (with respect to the Lebesgue measure) and $x \in \mathbb{R}$ such that $p_X(x) > 0$. Then the *conditional pdf of Y given $X = x$* is

$$p_{Y|X=x}(y) = \frac{p_{X,Y}(x, y)}{p_X(x)} .$$

Definition 1.77 (independent and identically distributed (iid)). A sequence or collection of random variables $X_1, X_2, \ldots, X_n$ (or an infinite sequence $X_1, X_2, \ldots$) is said to be *independent and identically distributed (iid)* if they are independent and identically distributed for some distribution P_X. We then write $X_i \overset{\text{iid}}{\sim} P$. In short, each random variable is a sample from the same underlying probability distribution, and the outcome of one does not influence the outcome of any other.

Remark 1.78 (the centrality of the iid assumption). The *iid assumption* is one of the most fundamental and frequently invoked assumptions in probability, statistics, and machine learning. Its importance stems from several key aspects:

- **Simplification of models:** The iid assumption vastly simplifies mathematical analysis. For instance, the likelihood function for iid data becomes a product of individual marginal likelihoods, $L(\theta \mid x_1, \ldots, x_n) = \prod_{i=1}^n p(x_i \mid \theta)$, which is often much easier to work with (e.g., for Maximum Likelihood Estimation).
- **Foundation for asymptotic theory:** Many cornerstone results, most notably the laws of large numbers (lln) and the central limit theorem (clt), are traditionally stated and proven under the iid assumption. These theorems provide the theoretical basis for why many statistical procedures work well with large sample sizes (e.g., why sample means converge to true means, and why sample means are approximately normally distributed).
- **Benchmark and starting point:** Even when the iid assumption is known to be violated (e.g., in time series data with temporal dependencies, or in clustered data where observations within a cluster are correlated), it often serves as a baseline model.

- **Standard data collection model:** The iid assumption often reflects an idealized model of data collection where samples are drawn randomly and independently from a large, stable population. For example, conducting a simple random sample from a large population to estimate a characteristic (like average height or proportion of voters favoring a candidate) is often modeled as iid.
- **Training data in machine learning:** Following the standard data collection model, in most machine learning settings, the training dataset $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ is assumed to consist of iid samples from an underlying (unknown) joint distribution $P_{X,Y}$. This assumption underpins theoretical guarantees for learning algorithms, such as generalization bounds, which describe how well a model trained on a finite sample is expected to perform on unseen data from the same distribution.

Regarding the data collection model, the sequence of random variables are often used to describe n "observations." When we think of a dataset of n samples, denoted by x_i for $i \in [n]$, these are just fixed values (the ones we have actually observed). For statistical analysis, it is important to think of "all the other possible datasets we could have observed." For example, if I want to estimate the average hight of people from measuring 100 individuals drawn uniformly at random, I might be interested in how close that value may be to the true average. To analyze that, I could repeat the experiment, let's say 1000 times, each time drawing a new random sample of 100 people. This would give me 1000 estimated values for the average hight and by looking at the variance (the spread) of these values, I might get a good idea by how far any single estimate may be away from the true value. To formalize this idea, I can think of each individual measurement still being a random variable, i.e., acting as if I knew there was a measurement for person i to come, but I haven't picked that person yet. At this point, person i's height, denoted by X_i may still be any person's height uniformly at random (that's how I pick persons). Now I can perform computations with these random variables (instead of fixed values) and then also analyze properties of the outcomes like variances. We then typically assume each measurement to be iid, i.e., I pick each observation independently from the others and they all come from the same distribution (I measure the same thing n times).

Note, this example is a bad one, because there is only a finite number of people in the world, so I cannot really draw independently from the same distribution. From my description it sounds like I wouldn't be able to include the same person twice in one sample, so once X_1 is fixed, X_2 is not "independently drawn from the same distribution" anymore.

While powerful and convenient, it is crucial to critically evaluate whether the iid assumption is appropriate for a given real-world problem. Violations of this assumption can lead to incorrect inferences and poorly performing models if not properly addressed. Nevertheless, its role as a foundational concept and a simplifying idealization is undeniable in the landscape of probabilistic modeling and data analysis.

Convention 1.79 (notation). For conciseness, we often use the following notation.

$X, Y, Z, X_1, X_2, \ldots$	random variable (continuous or discrete)
x	value of a random variable X
P	(joint) probability measure
P_X	probability measure induced by X, distribution of X
$P_{Y\|X=x}$	trace of Y given $X = x$
$P_{Y\|X}$	collection of $P_{Y\|X=x}$ for all x
F_X	cumulative distribution function of P_X
$F_{Y\|X=x}$	cumulative distribution function of $P_{Y\|X=x}$
p	(joint) pmf/pdf for discrete/continuous P
p_X	pmf/pdf of P_X
$p(x)$	pmf/pdf of P_X evaluated at point x
$p(x, y)$	joint pmf/pdf of $P_{X,Y}$ evaluated at (x, y)
$p(y \mid x)$	(conditional) pmf/pdf of $P_{Y\|X=x}$ evaluated at y

In the literature, people often use "f" for pdfs instead of p to make it distinct from pmfs. Also, probability measures are often denoted "$\mathbb{P}$" instead of P and (mostly for discrete probability spaces), one often sees "Pr" for "probability," i.e., also to denote a probability measure.

Definition 1.80 (conditional independence of random variables). Let X, Y, Z be random variables over the same probability space. Then X, Y are *conditionally independent given Z* if $F_{X,Y|Z=z}(x, y) = F_{X|Z=z}(x)F_{Y|Z=z}(y)$, which we denote by $X \perp\!\!\!\perp Y \mid Z$. When joint and marginal pmfs/pdfs exist, we also write this as $p(x, y \mid z) = p(x \mid z)p(y \mid z)$ for (Lebesgue almost all) x, y (implicitly assuming all terms are well-defined, i.e., the positivity of the marginals is satisfied).

2. Working with distributions

2.1. Basic properties of distributions

Section goal. Introduce key numerical summaries of distributions, such as their "central tendency" (expectation), "spread" (variance), quantiles, and the "linear relationship" between pairs of variables (covariance and correlation) as well as some characteristics of these properties.

Definition 2.1 (expectation, expected value, mean). Let X be a real-valued random variable with distribution P_X. The *expectation* or *expected value* or *mean* of X, denoted by $\mathbb{E}[X]$ (sometimes also via μ_X) is defined as follows.

(i) If X is discrete with pmf p_X on a countable set $\Omega \subseteq \mathbb{R}$, we have

$$\mathbb{E}[X] = \sum_{x \in \Omega} x\, p_X(x)\,,$$

provided $\sum_{x\in\Omega} |x|\, p_X(x) < \infty$.

(ii) If X is a continuous RVRV (or random vector) with pdf p_X, its expectation is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, p_X(x)\, dx \quad \text{or} \quad \mathbb{E}[X] = \int_{\mathbb{R}} \dots \int_{\mathbb{R}} x\, p_X(x) dx_n \dots dx_1\,,$$

provided $\int_{-\infty}^{\infty} |x|\, p_X(x)\, dx < \infty$.

Generally, the expectation can be defined in a unified way (for all types of distributions and irrespective of the existence of a pdf) via

$$\mathbb{E}[X] = \int_{\mathbb{R}} x\, dP_X(x)\,,$$

provided that $\int_{\mathbb{R}} |x|\, dP_X(x) < \infty$, where "$dP_X$" denotes integration with respect to the measure P_X (which we have not seen rigorously here). We will use this notation to stand for $\sum_{\omega\in\Omega} \dots$ or $\int_{\mathbb{R}} \dots dx$, respectively If the condition on absolute integrability/summability is not met, the expectation is said to "not exist" or be "undefined."

We often use square brackets for the argument of expectation $\mathbb{E}[\bullet]$, because it highlights that the expectation is a *functional*, i.e., a map that takes functions (random variables

are measurable maps) as inputs. Mathematicians often use square bracktes for functionals and parenthesis for functions. However, the literature overwhelmingly does not stick to this convention when it comes to variance and covariances, where they then default back to $\mathrm{var}(X)$ and $\mathrm{cov}(X, Y)$. I'll try to at least also use square brackets there.

Lemma 2.2 (expectations of functions). *Let X be an RVRV with distribution P_X, and let $g : \mathbb{R} \to \mathbb{R}$ be a measurable function. (With respect to what σ-algebras?) Then $Y = g(X)$ is also an RVRV, and its expectation is given by*

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x)\, dP_X(x)\ ,$$

provided that $\int_{\mathbb{R}} |g(x)|\, dP_X(x) < \infty$. For discrete X with pmf p_X this means

$$\mathbb{E}[g(X)] = \sum_{x \in \Omega} g(x)\, p_X(x)\ ,$$

provided $\sum_{x\in\Omega} |g(x)|\, p_X(x) < \infty$ and for continuous X with pdf p_X this means

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, p_X(x)\, dx\ ,$$

provided $\int_{-\infty}^{\infty} |g(x)|\, p_X(x)\, dx < \infty$. This is sometimes also called the "Law of the Unconscious Statistician" (LOTUS) and allows us to calculate the expectation of a function of a random variable, $g(X)$, using the distribution of X directly, without needing to first find the distribution of $Y = g(X)$.

Example 2.3 (expectation examples).

(i) **Discrete case:** Let X be a random variable representing the outcome of rolling a biased three-sided die, with $\Omega = \{1, 2, 3\}$. Suppose the pmf is $p_X(1) = 0.5$, $p_X(2) = 0.3$, and $p_X(3) = 0.2$. The expectation of X is

$$\mathbb{E}[X] = (1 \cdot 0.5) + (2 \cdot 0.3) + (3 \cdot 0.2) = 0.5 + 0.6 + 0.6 = 1.7\ .$$

The condition $\sum |x| p_X(x) = 1.7 < \infty$ is met.

(ii) **Continuous case:** Let $X \sim \mathrm{Unif}(a, b)$ for $a < b$. The pdf is $p_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$ and 0 otherwise. The expectation of X is

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{b-a}\left(\frac{b^2 - a^2}{2}\right) = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}.$$

The condition $\int_a^b |x| \frac{1}{b-a} dx < \infty$ is met since a, b are finite.

(iii) **Why only for real-valued random variables?** Consider a sample space $\Omega = \{\circ, \square, \triangle\}$ with a probability measure P such that $P(\{\circ\}) = 0.5, P(\{\square\}) = 0.3, P(\{\triangle\}) = 0.2$. Let X be a random variable $X(\omega) = \omega$. What would $\mathbb{E}[X]$ be? The sum $\sum_{x\in\Omega} x\, p_X(x)$ would involve terms like $\circ \cdot 0.5 + \square \cdot 0.3 + \triangle \cdot 0.2$. These arithmetic operations (multiplication by a scalar, addition) are not defined for such abstract symbols. Expectation relies on the values of the random variable belonging to a set (like $\mathbb{R}$) where these operations are well-defined and have the usual algebraic properties.

For example, the expectation of a fair die roll makes sense (and turns out to be 3.5), only if we consider the six sides to represent the integers (instead of abstract collections of dots). In the setting above, one could introduce a new real-valued random variable $Y : \Omega \to \mathbb{R}$ on $(\Omega, \mathcal{P}(\Omega), P)$ that maps

$$Y(\circ) = 3\,, \quad Y(\square) = 1\,, \quad Y(\triangle) = 0$$

such that we can compute $\mathbb{E}[Y] = 3 \cdot 0.5 + 1 \cdot 0.3 + 0 \cdot 0.2 = 1.8$.

Proposition 2.4 (properties of expectation). *Let X, Y be RVRVs defined on the same probability space, and let $a, b, c \in \mathbb{R}$ be constants. Assume all relevant expectations exist. Then the expectation satisfies the following properties.*

(i) Linearity: $\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$. *As a special case,* $\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$.

(ii) Expectation of a constant: $\mathbb{E}[c] = c$.

(iii) Non-negativity: *If $X(\omega) \geq 0$ for all $\omega \in \Omega$ (often written as $X \geq 0$), then* $\mathbb{E}[X] \geq 0$.

(iv) Monotonicity: *If $X(\omega) \geq Y(\omega)$ for all $\omega \in \Omega$ (often written as $X \geq Y$), then* $\mathbb{E}[X] \geq \mathbb{E}[Y]$.

(v) Product of independent variables: *If $X \perp\!\!\!\perp Y$, then* $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$.

Proof. Some of these will be shown in the exercises. □

Definition 2.5 (variance, standard deviation). Let X be an RVRV with mean $\mathbb{E}[X]$ (implying that it exists). The *variance* of X, denoted by $\mathrm{var}[X]$ (sometimes also by σ_X^2), is defined as

$$\mathrm{var}[X] = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]\,,$$

provided this expectation exists (for which it suffices that $\mathbb{E}[X^2] < \infty$). The *standard deviation* of X, denoted by $\sigma[X] = \sigma_X$, is the non-negative square root of the variance

$$\sigma[X] = \sigma_X := \sqrt{\mathrm{var}[X]}\,.$$

Variance measures the squared magnitude of fluctuation of X around its mean, while standard deviation provides this measure in the original units of X.

Remark 2.6 (computational formula for variance). A common way to compute variance is using the formula

$$\operatorname{var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \,.$$

Proof. Using $\mu_X := \mathbb{E}[X]$, we compute

$$\begin{aligned}
\operatorname{var}[X] &= \mathbb{E}[(X-\mu_X)^2] = \mathbb{E}[X^2 - 2X\mu_X + \mu_X^2] \\
&= \mathbb{E}[X^2] - \mathbb{E}[2X\mu_X] + \mathbb{E}[\mu_X^2] \quad \text{(by linearity of expectation)} \\
&= \mathbb{E}[X^2] - 2\mu_X\,\mathbb{E}[X] + \mu_X^2 \quad \text{(since } \mu_X \text{ is a constant and linearity)} \\
&= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 = \mathbb{E}[X^2] - \mu_X^2 \,.
\end{aligned}$$

□

Example 2.7 (variance examples). We compute variances and standard deviations for the examples from example 2.3.

(i) **Discrete case (continued):** For the biased three-sided die with $\mathbb{E}[X] = 1.7$, we compute $\mathbb{E}[X^2]$ using lemma 2.2

$$\mathbb{E}[X^2] = (1^2{\cdot}0.5)+(2^2{\cdot}0.3)+(3^2{\cdot}0.2) = (1{\cdot}0.5)+(4{\cdot}0.3)+(9{\cdot}0.2) = 0.5+1.2+1.8 = 3.5.$$

Then, the variance is

$$\operatorname{var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 3.5 - (1.7)^2 = 3.5 - 2.89 = 0.61 \,.$$

The standard deviation is $\sigma_X = \sqrt{0.61} \approx 0.781$.

(ii) **Continuous case (continued):** For $X \sim \operatorname{Unif}(a,b)$ with $\mathbb{E}[X] = \frac{a+b}{2}$, we compute using lemma 2.2

$$\mathbb{E}[X^2] = \int_a^b x^2 \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{1}{b-a}\frac{b^3-a^3}{3} = \frac{a^2+ab+b^2}{3} \,.$$

(Using $b^3 - a^3 = (b-a)(b^2+ab+a^2)$.) Then, the variance is

$$\begin{aligned}
\operatorname{var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 &= \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2 \\
&= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4} \\
&= \frac{4(a^2+ab+b^2) - 3(a^2+2ab+b^2)}{12} \\
&= \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12} = \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12} \,.
\end{aligned}$$

The standard deviation is $\sigma_X = \frac{b-a}{\sqrt{12}} = \frac{b-a}{2\sqrt{3}}$.

Proposition 2.8 (properties of variance). *Let X, Y be RVRVs, and let $a, b \in \mathbb{R}$ be constants. Assume all relevant variances exist.*

(i) Non-negativity: $\operatorname{var}[X] \geq 0$ *and* $\operatorname{var}[X] = 0$ *if and only if* $P(X = \mathbb{E}[X]) = 1$ *(i.e., X is a constant almost surely).*

(ii) Variance of a constant: $\operatorname{var}[b] = 0$.

(iii) Scaling and shifting: $\operatorname{var}[aX + b] = a^2 \operatorname{var}[X]$. *Note that adding a constant does not change the variance.*

(iv) Sum of independent variables: *If $X \perp\!\!\!\perp Y$, then* $\operatorname{var}[X + Y] = \operatorname{var}[X] + \operatorname{var}[Y]$. *More generally, if $X_1, \ldots, X_n$ are pairwise independent, then* $\operatorname{var}[\sum_{i=1}^n X_i] = \sum_{i=1}^n \operatorname{var}[X_i]$.

Definition 2.9 (covariance). Let X and Y be two RVRVs whose means exist. The *covariance* between X and Y, denoted by $\operatorname{cov}[X, Y]$ (or sometimes also by σ_{XY}), is defined as

$$\operatorname{cov}[X, Y] = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right] ,$$

provided this expectation exists. Covariance measures the degree to which X and Y tend to deviate from their respective means in similar directions. A positive covariance indicates that X tends to be large when Y is large. A negative covariance indicates X tends to be small when Y is large.

For a random vector $X = (X_1, \ldots, X_n)$ where each X_i is an RVRV and all $\mathbb{E}[X_i]$ exist, the *covariance (matrix)* of X, denoted by $\operatorname{cov}[X]$ or Σ_X (or simply Σ), is an $n \times n$ matrix whose (i, j)-th entry is $\operatorname{cov}[X_i, X_j]$, i.e.,

$$(\Sigma_X)_{ij} = \operatorname{cov}[X_i, X_j] = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])] .$$

The diagonal entries are the variances $\operatorname{cov}[X_i, X_i] = \operatorname{var}[X_i]$. By definition, the covariance matrix is symmetric, i.e., $\operatorname{cov}[X_i, X_j] = \operatorname{cov}[X_j, X_i]$.

Remark 2.10 (computational formula for covariance). A common way to compute covariance is the formula

$$\operatorname{cov}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] .$$

Proof. Using $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$, this can be shown via

$$\begin{aligned}
\operatorname{cov}[X, Y] &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y] \\
&= \mathbb{E}[XY] - \mathbb{E}[X\mu_Y] - \mathbb{E}[Y\mu_X] + \mathbb{E}[\mu_X\mu_Y] \quad \text{(by linearity)} \\
&= \mathbb{E}[XY] - \mu_Y\,\mathbb{E}[X] - \mu_X\,\mathbb{E}[Y] + \mu_X\mu_Y \quad \text{(constants factor out)} \\
&= \mathbb{E}[XY] - \mu_Y\mu_X - \mu_X\mu_Y + \mu_X\mu_Y = \mathbb{E}[XY] - \mu_X\mu_Y .
\end{aligned}$$

□

Example 2.11 (covariance of two random variables). Let (X, Y) be a discrete random vector with the following joint pmf

$$p(x,y) = \begin{cases} 0.4 & \text{if } x = 0 \text{ and } y = 0\,, \\ 0.1 & \text{if } x = 0 \text{ and } y = 1\,, \\ 0.1 & \text{if } x = 1 \text{ and } y = 0\,, \\ 0.4 & \text{if } x = 1 \text{ and } y = 1\,. \end{cases}$$

We first, compute the marginals

$$\begin{aligned} p_X(0) &= p(0,0) + p(0,1) = 0.4 + 0.1 = 0.5\,, \\ p_X(1) &= p(1,0) + p(1,1) = 0.1 + 0.4 = 0.5\,, \\ p_Y(0) &= p(0,0) + p(1,0) = 0.4 + 0.1 = 0.5\,, \\ p_Y(1) &= p(0,1) + p(1,1) = 0.1 + 0.4 = 0.5\,, \end{aligned}$$

and means

$$\begin{aligned} \mu_X &= \mathbb{E}[X] = (0 \cdot 0.5) + (1 \cdot 0.5) = 0.5\,, \\ \mu_Y &= \mathbb{E}[Y] = (0 \cdot 0.5) + (1 \cdot 0.5) = 0.5\,. \end{aligned}$$

Next, we compute $\mathbb{E}[XY]$. We note that XY is an RVRV again, that takes the value 1 if $X = 1, Y = 1$, and 0 otherwise, i.e.,

$$\mathbb{E}[XY] = (0 \cdot 0 \cdot 0.4) + (0 \cdot 1 \cdot 0.1) + (1 \cdot 0 \cdot 0.1) + (1 \cdot 1 \cdot 0.4) = 0 + 0 + 0 + 0.4 = 0.4\,.$$

Finally, the covariance is

$$\operatorname{cov}[X,Y] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0.4 - 0.5 \cdot 0.5 = 0.4 - 0.25 = 0.15\,.$$

Since $\operatorname{cov}[X,Y] = 0.15 > 0$, X and Y tend to vary in the same direction. For instance, if $X = 1$ (above its mean), Y is more likely to be 1 (above its mean), as $P(Y = 1 \mid X = 1) = \frac{p(1,1)}{p_X(1)} = \frac{0.4}{0.5} = 0.8$.

Proposition 2.12 (properties of covariance). *Let W, X, Y, Z be RVRVs and $a, b, c, d \in \mathbb{R}$ be constants. Assume all relevant covariances exist.*

(i) Symmetry: $\operatorname{cov}[X,Y] = \operatorname{cov}[Y,X]$.

(ii) Relation to variance: $\operatorname{cov}[X,X] = \operatorname{var}[X]$.

(iii) Bilinearity:

$$\begin{aligned} \operatorname{cov}[aX + bY, Z] &= a \operatorname{cov}[X,Z] + b \operatorname{cov}[Y,Z]\,, \\ \operatorname{cov}[X, cY + dZ] &= c \operatorname{cov}[X,Y] + d \operatorname{cov}[X,Z]\,. \end{aligned}$$

(iv) Covariance with a constant: $\operatorname{cov}[X,c] = 0$.

(v) Independence implies zero covariance: *If $X \perp\!\!\!\perp Y$, then* $\operatorname{cov}[X,Y]=0$. *(The converse is not generally true:* $\operatorname{cov}[X,Y]=0$ *does not imply* $X \perp\!\!\!\perp Y$*.)*

(vi) Variance of a sum:

$$\operatorname{var}[X+Y]=\operatorname{var}[X]+\operatorname{var}[Y]+2\operatorname{cov}[X,Y]\,.$$

More generally, for RVRVs $X_1,\dots,X_n$ and constants $a_1,\dots,a_n$:

$$\operatorname{var}\left[\sum_{i=1}^n a_iX_i\right]=\sum_{i=1}^n a_i^2\operatorname{var}[X_i]+2\sum_{1\le i<j\le n}a_ia_j\operatorname{cov}[X_i,X_j]\,.$$

If $X_1,\dots,X_n$ are pairwise independent, then all $\operatorname{cov}[X_i,X_j]=0$ *for $i\neq j$, and the formula simplifies to* $\operatorname{var}[\sum_{i=1}^n a_iX_i]=\sum_{i=1}^n a_i^2\operatorname{var}[X_i]$.

Definition 2.13 (correlation coefficient). The *correlation coefficient* (or *Pearson correlation coefficient*) between two RVRVs X and Y with standard deviations $\sigma_X>0$ and $\sigma_Y>0$ is defined as

$$\rho_{X,Y}=\rho[X,Y]=\operatorname{corr}[X,Y]:=\frac{\operatorname{cov}[X,Y]}{\sigma_X\sigma_Y}=\frac{\operatorname{cov}[X,Y]}{\sqrt{\operatorname{var}[X]\operatorname{var}[Y]}}\in[-1,1]\,.$$

The correlation coefficient is a dimensionless, standardized measure of the strength and direction of the *linear* relationship between X and Y. If $\rho[X,Y]=0$, we call X and Y *uncorrelated*.

Proposition 2.14 (properties of correlation coefficient). *Let X,Y be RVRVs with $\sigma_X>0,\sigma_Y>0$.*

(i) Range: $-1\le\rho_{X,Y}\le 1$.

(ii) Perfect linear relationship: $\rho_{X,Y}=1$ *if and only if* $Y=aX+b$ *for some constants $a,b\in\mathbb{R}$ with $a>0$ (almost surely).* $\rho_{X,Y}=-1$ *if and only if* $Y=aX+b$ *for some constants $a,b\in\mathbb{R}$ with $a<0$ (almost surely).*

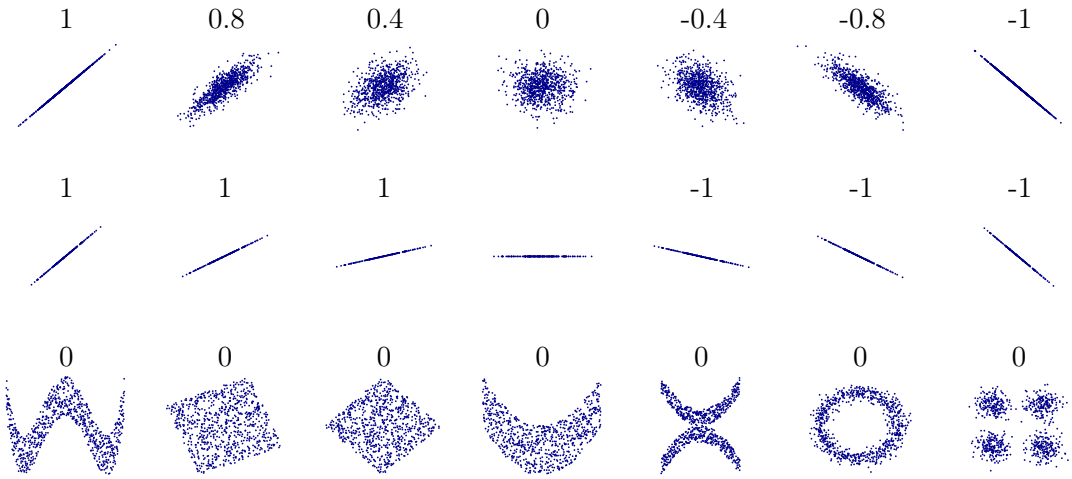
(iii) Invariance to linear transformations: *For constants $a,b,c,d\in\mathbb{R}$ with $a\neq 0,c\neq 0$:*

$$\rho_{aX+b,cY+d}=\operatorname{sign}(ac)\rho_{X,Y}\,.$$

Thus, correlation is invariant to scaling (up to the sign) and shifting of X and Y.

(iv) Independence implies zero correlation: *If $X \perp\!\!\!\perp Y$, then* $\operatorname{cov}[X,Y]=0$, *so* $\rho_{X,Y}=0$. *(The converse is not generally true:* $\rho_{X,Y}=0$ *only means no* linear *relationship, but there could be a non-linear relationship, so X and Y might not be independent.)*

Example 2.15 (correlation coefficients). This illustration shows a series of 2-dimensional continuous distributions. These are typically depicted via contour or surface plots of their joint pdfs or via a scatter plots of iid samples of the joint distribution, which we see here (image taken from Wikipedia). The scatter plots of those points provides a good intuition for "what the distribution looks like" as regions of high point density are regions of high joint pdf. Above each distribution, we show the (Pearson) correlation coefficient. The last row provides various examples of distributions where $X \not\perp\!\!\!\perp Y$ but $\rho[X, Y] = 0$. These pictures also illustrate how the correlation coefficient captures the "linear dependence," and how the intuition that the correlation coefficient is the "slope of a linear regression line" breaks down (middle row) as it is also normalized/scaled.



Definition 2.16 (partial correlation coefficient). Let X, Y and $Z = (Z_1, \ldots, Z_k)$ be RVRVs defined on the same probability space for $k \in \mathbb{N}_{>0}$. The *partial correlation coefficient* between X and Y given Z, denoted by $\rho_{X,Y|Z} = \rho[X, Y \mid Z]$, measures the linear relationship between X and Y after accounting for (or "removing") the linear effects of the variables in Z.

One way to define it is via residuals from linear regression. Let $\hat{X}$ be the best linear predictor of X using Z, and $\hat{Y}$ be the best linear predictor of Y using Z. The residuals are $X_{\text{res}} = X - \hat{X}$ and $Y_{\text{res}} = Y - \hat{Y}$. Then,

$$\rho_{X,Y|Z} = \rho_{X_{\text{res}},Y_{\text{res}}} \,.$$

For the case where Z consists of a single variable, the partial correlation can be computed using the (pairwise) correlations via

$$\rho[X, Y \mid Z] = \rho_{X,Y|Z} = \frac{\rho_{X,Y} - \rho_{X,Z}\rho_{Y,Z}}{\sqrt{(1 - \rho_{X,Z}^2)(1 - \rho_{Y,Z}^2)}} \,,$$

provided that $1 - \rho_{X,Z}^2 > 0$ and $1 - \rho_{Y,Z}^2 > 0$. For more than one variable in Z, there exist recursive formulas. If $\rho[X, Y \mid Z] = 0$, we call X and Y *partially uncorrelated given Z*.

Remark 2.17 (interpretation of partial correlation). Partial correlation is a useful concept for understanding complex relationships among multiple variables. For example, if X and Y are correlated, this correlation might be due to both X and Y being influenced by a third variable Z (a confounder). The partial correlation $\rho_{X,Y|Z}$ gives the correlation between X and Y if the influence of Z were held constant or removed. If $\rho_{X,Y|Z}$ is close to zero while $\rho_{X,Y}$ is large, it suggests that the relationship between X and Y might be largely explained by Z.

Overview 2.18 (notation overview). In summary, for X, Y, Z RVRVs we have

$$\begin{aligned}
&\textit{expectation of } X && \mathbb{E}[X] := \int x\, dP_X(x) = \int x p(x)\, dx\,, \\
&\textit{variance of } X && \operatorname{var}[X] := \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2\,, \\
&\textit{standard deviation of } X && \sigma[X] := \sqrt{\operatorname{var}[X]}\,, \\
&\textit{covariance of } X, Y && \operatorname{cov}[X, Y] := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]\,, \\
&\textit{corr. coeff. of } X, Y && \rho[X, Y] := \frac{\operatorname{cov}[X, Y]}{\sigma[X] \cdot \sigma[Y]}\,, \\
&\textit{par. corr. coeff. of } X, Y \mid Z && \rho[X, Y \mid Z] := \frac{\rho[X, Y] - \rho[X, Z]\rho[Y, Z]}{\sqrt{(1 - \rho[X, Z]^2)(1 - \rho[Y, Z]^2)}}\,.
\end{aligned}$$

In the literature, notation varies wildly. Some people also use $\mathbb{P}$ for probability measures, f for pdfs, $\mathbb{F}$ for cdfs, and $\mathbb{V}$ for variance—however, usually not in a systematic/consistent way. They typically use the σ_X and $\rho_{X,Y}$ notation for standard deviations and correlation coefficients (unlike our functional notation), round parenthesis for variances and covariances $\operatorname{var}(\bullet), \operatorname{cov}(\bullet)$, but square brackets for expectations $\mathbb{E}[X]$. Some people advise to use no brackets at all for expectations "whenever unambiguous," i.e., to simply write $\mathbb{E}\,X$, or even just EX. The mantra is very much "anything goes."

Convention 2.19 (integrability). Unless stated otherwise, we typically assume implicitly that RVRVs are *integrable*, i.e., $\mathbb{E}[|X|] < \infty$ and *square-integrable*, i.e., $\mathbb{E}[X^2] < \infty$. Hence, the expectation and the variance always exist if we do not explicitly mention to watch out for it. Similarly, when we do calculations with continuous distributions, please feel free to assume that (joint) pdfs exist whenever convenient unless we point out explicitly that pdfs may not exist.

Definition 2.20 (quantile, median). Let X be an RVRV. For any $q \in (0, 1)$, a *q-quantile* (or *percentile* if q is expressed as a percentage) of (the distribution of) X is a value x_q such that

$$P(X \le x_q) \ge q \quad \text{and} \quad P(X \ge x_q) \ge 1 - q\,.$$

Equivalently, $F_X(x_q) \ge q$ and $F_X(x_q^-) \le q$, where $F_X(x_q^-) = \lim_{x \uparrow x_q} F_X(x) = P(X < x_q)$.

If F_X is continuous and strictly increasing, the q-quantile x_q is unique and is given by $x_q = F_X^{-1}(q)$, i.e., the value x_q such that $F_X(x_q) = q$.

If F_X has flat regions or jumps, a q-quantile may not be unique. A common convention is to define the q-quantile as

$$x_q = \inf\{x \in \mathbb{R} \mid F_X(x) \geq q\} \,.$$

This is also known as the *generalized inverse distribution function* or *quantile function*.

There are a few quantiles that got special names.

- The *median* is the 0.5-quantile, denoted m or $x_{0.5}$. Intuitively speaking, it splits the distribution into two "equally heavy" (i.e., equally probable) halves.
- *Quartiles*:
  - First quartile (Q_1): $x_{0.25}$
  - Second quartile (Q_2): $x_{0.50}$ (the median)
  - Third quartile (Q_3): $x_{0.75}$
- *Deciles* divide the distribution into 10 equal parts, and *percentiles* into 100 equal parts. In statistics, health, sports, or standardized tests we often talk about the *5th and 95th percentile* (or the *1st and 99th percentile*) as well.

Example 2.21 (quantiles of an exponential distribution). Let $X \sim \mathrm{Exp}(\lambda)$ with $\lambda > 0$. Its cdf is $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$, and $F_X(x) = 0$ for $x < 0$. This cdf is continuous and strictly increasing for $x \geq 0$. To find the q-quantile x_q, we solve $F_X(x_q) = q$ for x_q

$$1 - e^{-\lambda x_q} = q \;\Leftrightarrow\; e^{-\lambda x_q} = 1 - q \;\Leftrightarrow\; -\lambda x_q = \ln(1-q) \;\Leftrightarrow\; x_q = -\frac{\ln(1-q)}{\lambda} \,.$$

For the median ($q = 0.5$) we thus find $m = \frac{\ln(2)}{\lambda}$. For an exponential distribution, the mean is $\mathbb{E}[X] = 1/\lambda$ while the median is $(\ln 2)/\lambda \approx 0.693/\lambda < \mathbb{E}[X]$. So the median is less than the mean and we call the exponential distribution "right-skewed."

Remark 2.22 (meaning and robustness of the median vs. mean). The median and the mean ($\mathbb{E}[X]$) are both measures of the central tendency of a distribution, but they capture different aspects.

(i) The mean is the "center of mass" or the average value of the random variable. It is influenced by all values of the distribution, including extreme ones. Having very high outliers (even if they comprise little probability) will affect the mean.

(ii) The median is the "middle value" that divides the probability mass in half with 50% of the probability mass at or below the median, and 50% at or above it. The median is robust to outliers. Extreme values have little to no effect on the median. For skewed distributions, e.g., the distribution of wealth in a country, the median is often considered a more representative measure of central tendency than the mean.

(iii) The mean $\mathbb{E}[X]$ exists only if $\mathbb{E}[|X|] < \infty$. For some distributions (e.g., Cauchy distribution), the mean is undefined. Quantiles (and thus also the median) always exist for any probability distribution.

(iv) The mean has convenient mathematical properties, particularly linearity, whereas medians are not linear and often harder to handle. The mean is the typical target of prediction (regression) tasks, but there are also techniques for "quantile regression," which often come with added challenges.

(v) The mean $\mu = \mathbb{E}[X]$ is the value c that minimizes $\mathbb{E}[(X-c)^2]$ (mean squared error). The median m is a value c that minimizes $\mathbb{E}[|X-c|]$ (mean absolute error).

Assuming $\mathbb{E}[X^2]$ exists, let $f(c) := \mathbb{E}[(X-c)^2]$. To find the minimum of

$$f(c) = \mathbb{E}[X^2 - 2Xc + c^2] = \mathbb{E}[X^2] - 2c\,\mathbb{E}[X] + c^2\,,$$

a quadratic function of c, we set the derivative with respect to c to zero to obtain

$$\frac{df}{dc} = -2\,\mathbb{E}[X] + 2c \overset{!}{=} 0\,.$$

So $c = \mathbb{E}[X]$ is indeed an extremum and since $\frac{d^2 f}{dc^2} = 2 > 0$, we have shown that $\mathbb{E}[X]$ is the unique minimizer of the mean squared error.

Let $g(c) = \mathbb{E}[|X-c|]$ and assume for simplicity that the pdf $p_X(x)$ of X exists and that we can interchange differentiation and expectation. Then $g(c)$ can be written as

$$g(c) = \int_{-\infty}^{\infty} |x-c|\,p_X(x)\,dx = \int_{-\infty}^{c} (c-x)p_X(x)\,dx + \int_{c}^{\infty} (x-c)p_X(x)\,dx\,.$$

Differentiating with respect to c (using Leibniz integral rule, assuming $p_X(x)$ is well-behaved): The derivative of the first term is $\int_{-\infty}^{c} p_X(x)\,dx = P(X \le c)$ and of the second term is $-\int_{c}^{\infty} p_X(x)\,dx = -P(X > c)$. Thus,

$$\frac{dg}{dc} = P(X \le c) - P(X > c) \overset{!}{=} 0$$

implies $P(X \le c) = P(X > c)$, which means $2P(X \le c) = 1$ and we find $P(X \le c) = F_X(c) = 1/2$, i.e., c is the median m.

(vi) In symmetric distributions (like the Normal distribution), the mean and median coincide. For skewed distributions, they will typically differ, with the mean being pulled towards the longer tail. A full set of quantiles (e.g., quartiles or deciles), are often used as a fairly complete and informative picture of a distribution (typically more informative than mean and variance).

Proposition 2.23 (quantile transformation). *Let $F : \mathbb{R} \to [0,1]$ be a monotonically increasing, right-continuous function such that*

$$\lim_{c\to-\infty} F(c) = 0 \quad \textit{and} \quad \lim_{c\to\infty} F(c) = 1.$$

That is, F is a valid cdf for some probability measure λ_F on $(\mathbb{R}, \mathcal{B})$.

Then there is a random variable $X : (0,1) \to \mathbb{R}$ on the probability space $((0,1), \mathcal{B}|_{(0,1)}, \mathrm{Unif}(0,1))$ with $F_X = F$ given by

$$X : (0,1) \to \mathbb{R} : \; u \mapsto X(u) := \inf\{c \in \mathbb{R} \mid F(c) \geq u\}$$

called the quantile transformation.

Proof. We proceed in steps.

(i) First, X is well-defined, i.e., $-\infty < X(u) < \infty$ for all $u \in (0,1)$. Assume there exists $u \in (0,1)$ such that $X(u) = -\infty$ (the case $+\infty$ is similar). Because

$$X(u) := \inf\{c \in \mathbb{R} \,|\, u \leq F(c)\},$$

this implies that the set has no lower bound, so there exists a sequence $(c_n)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}}$ with $c_n \to -\infty$ and $F(c_n) \geq u > 0$ for all $n \in \mathbb{N}$. This contradicts $\lim_{x\to-\infty} F(x) = 0$.

(ii) Next, we show that X is increasing and left-continuous, and that for all $c \in \mathbb{R}$, $u \in (0,1)$ we have

$$X(u) \leq c \iff u \leq F(c).$$

This means that X is the *left-continuous inverse of F*. Since $X(u) = \inf\{c \in \mathbb{R} \mid u \leq F(c)\} \leq c$, there exists $c' \leq c$ such that $u \leq F(c') \leq F(c)$ (monotonicity). We then have $X \circ F(c) = \inf\{c' \in \mathbb{R} \mid F(c) \leq F(c')\} = c$. Also, for $u \in (0,1)$, $F \circ X(u) = F(\inf\{c \in \mathbb{R} \mid F(c) \geq u\})$.

(iii) Now, we show that X is a random variable, i.e., that $\{X \leq c\} \in \mathcal{B}|_{(0,1)}$. As F is right-continuous and increasing, the infimum is attained, i.e., there exists $c' \in \mathbb{R}$ such that

$$c' = \inf\{c \in \mathbb{R} \mid u \leq F(c)\} \quad \text{with} \quad F(c') \geq u.$$

In particular,

$$X^{-1}((-\infty, c]) = \{u \in (0,1) \mid X(u) \leq c\} = (0, F(c)] \in \mathcal{B}|_{(0,1)},$$

and since $\mathcal{B} = \sigma(\{(-\infty, c] \mid c \in \mathbb{R}\})$, X is $\mathcal{B}|_{(0,1)}$-$\mathcal{B}$-measurable.

(iv) Finally, we show that for all $c \in \mathbb{R}$ we have that $F_X(c) = F(c)$. For this, we first recognize that $\mathrm{Unif}(0,1) = \lambda|_{(0,1)} =: P$ for which we compute

$$F_X(c) = P_X((-\infty, c]) = \lambda_{(0,1)}(X \leq c) = \lambda_{(0,1)}((0, F(c)]) = F(c)\,.$$

Taken together, this proves the statement. □

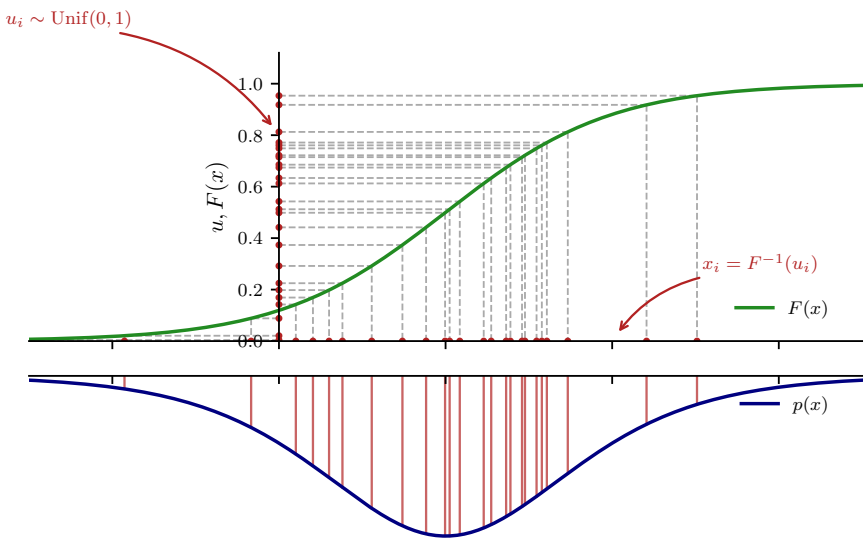
Remark 2.24 (inverse transform sampling). By proposition 2.23, every probability measure P on $(\mathbb{R}, \mathcal{B})$ is the distribution of a random variable on the probability space $((0,1), \mathcal{B}|_{(0,1)}, \mathrm{Unif}(0,1))$. *Inverse transform sampling* is the procedure to generate a sample at random from an unknown/not-implemented probability distribution given its cumulative distribution function:

(i) Sample $U \sim \mathrm{Unif}(0,1)$, i.e., generate $\{u_1, \ldots, u_m\}$.

(ii) Find the generalized inverse $X = F^{-1}$ of F.

(iii) Obtain the sample $\{X(u_1), \ldots, X(u_m)\}$.

The process is visualized below. Samples u_i are drawn from a uniform distribution on $(0,1)$ (shown on the y-axis). These are mapped through the inverse CDF F^{-1} to produce samples x_i (shown on the x-axis). The density of the resulting x_i values will correspond to the probability density function $p(x)$ associated with the CDF $F(x)$.



Example 2.25 (inverse transform sampling for $\mathrm{Exp}(\lambda)$). Let $X \sim \mathrm{Exp}(\lambda)$. Its cdf is $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ and we have found the quantile function $X(u) = -\frac{\ln(1-u)}{\lambda}$. Then $X(U)$ is $\mathrm{Exp}(\lambda)$ distributed for $U \sim \mathrm{Unif}(0,1)$. Since $1 - U \sim \mathrm{Unif}(0,1)$ as well, this is often implemented as $x = -\frac{\ln(U)}{\lambda}$.

Lemma 2.26 (expected value and variance of common discrete distributions). *The following formulas give the expectation* $\mathbb{E}[\cdot]$ *and variance* $\mathrm{var}[\cdot]$ *for each distribution. Verifying all of these by manually computing the respective sums will be an excellent exercise to test and hone your "computing infinite series" skill. However, I am not sure how much valuable insight one gains from these computations so we will skip them.*

(i) Uniform distribution: if $X \sim \mathrm{Lap}_\Omega$ *on a finite* $\Omega \subseteq \mathbb{R}$*, then*

$$\mathbb{E}[X] = \frac{1}{|\Omega|}\sum_{\omega\in\Omega}\omega, \quad \mathrm{var}[X] = \frac{1}{|\Omega|}\sum_{\omega\in\Omega}\big(\omega - \mathbb{E}[X]\big)^2.$$

(ii) Hypergeometric distribution: if $X \sim \mathrm{Hyp}(N, K, n)$*, then*

$$\mathbb{E}[X] = n\,\frac{K}{N}, \quad \mathrm{var}[X] = n\,\frac{K}{N}\Big(1 - \frac{K}{N}\Big)\frac{N-n}{N-1}.$$

(iii) Bernoulli process: if $(X_1, \dots, X_n) \sim \mathrm{Ber}(n, p)$*, then for each* $i \in [n]$

$$\mathbb{E}[X_i] = p, \quad \mathrm{var}[X_i] = p(1-p), \quad \mathrm{cov}[X_i, X_j] = 0 \quad (i \neq j).$$

(iv) Binomial distribution: if $X \sim \mathrm{Bin}(n, p)$*, then*

$$\mathbb{E}[X] = n\,p, \quad \mathrm{var}[X] = n\,p(1-p).$$

(v) Geometric distribution: if $X \sim \mathrm{Geo}(p)$*, then*

$$\mathbb{E}[X] = \frac{1}{p}, \quad \mathrm{var}[X] = \frac{1-p}{p^2}.$$

(vi) Negative binomial distribution: if $X \sim \mathrm{NegBin}(r, p)$*, then*

$$\mathbb{E}[X] = \frac{r}{p}, \quad \mathrm{var}[X] = \frac{r(1-p)}{p^2}.$$

(vii) Poisson distribution: if $X \sim \mathrm{Poi}(\lambda)$*, then*

$$\mathbb{E}[X] = \lambda, \quad \mathrm{var}[X] = \lambda.$$

Lemma 2.27 (expected value and variance of common continuous distributions). *The following formulas give the expectation* $\mathbb{E}[X]$ *and variance* $\mathrm{var}[X]$ *for each distribution. Verifying all of these by manually computing the respective integrals will be an excellent exercise to test and hone your integration skills (some might require techniques like integration by parts or using properties of the Gamma function). However, I am not sure how much valuable insight one gains from these computations so we will skip them.*

(i) Uniform distribution: if $X \sim \mathrm{Unif}(a, b)$, then

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \mathrm{var}[X] = \frac{(b-a)^2}{12}.$$

(ii) Normal distribution: if $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{E}[X] = \mu, \quad \mathrm{var}[X] = \sigma^2.$$

(The parameters μ and σ^2 are precisely the mean and variance.)

(iii) Gamma distribution: if $X \sim \Gamma(\alpha, \beta)$ with $\alpha > 0, \beta > 0$, then

$$\mathbb{E}[X] = \frac{\alpha}{\beta}, \quad \mathrm{var}[X] = \frac{\alpha}{\beta^2}.$$

(iv) Exponential distribution: if $X \sim \mathrm{Exp}(\lambda)$ with $\lambda > 0$, then

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \mathrm{var}[X] = \frac{1}{\lambda^2}.$$

(This is a special case of the Gamma distribution with $\alpha = 1, \beta = \lambda$.)

(v) Chi-squared distribution: if $X \sim \chi^2_k$ with $k > 0$ degrees of freedom, then

$$\mathbb{E}[X] = k, \quad \mathrm{var}[X] = 2k.$$

(This is a special case of the Gamma distribution with $\alpha = k/2, \beta = 1/2$.)

(vi) Student's t distribution: if $X \sim t_\nu$ with $\nu > 0$ degrees of freedom, then

$$\mathbb{E}[X] = \begin{cases} 0 & \text{if } \nu > 1\,, \\ \text{undefined} & \text{if } \nu \le 1\,. \end{cases}, \quad \mathrm{var}[X] = \begin{cases} \frac{\nu}{\nu-2} & \text{if } \nu > 2\,, \\ \infty & \text{if } 1 < \nu \le 2\,, \\ \text{undefined} & \text{if } \nu \le 1\,. \end{cases}$$

(vii) Beta distribution: if $X \sim \mathrm{Beta}(\alpha, \beta)$ with $\alpha > 0, \beta > 0$, then

$$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}, \quad \mathrm{var}[X] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Definition 2.28 (conditional expectation). Let X and Y be two RVRVs. The *conditional expectation of Y given that $X = x$*, denoted $\mathbb{E}[Y \mid X = x]$, is the expected value of Y computed using the conditional distribution of Y given $X = x$.

- If X, Y are discrete with joint pmf $p_{X,Y}(x, y)$ and marginal pmf $p_X(x) > 0$, then

$$\mathbb{E}[Y \mid X = x] = \sum_y y\, p_{Y|X=x}(y) = \sum_y y\, \frac{p_{X,Y}(x, y)}{p_X(x)}\,.$$

- If X, Y are continuous with joint pdf $p_{X,Y}(x, y)$ and marginal pdf $p_X(x) > 0$, then

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y\, p_{Y|X=x}(y)\, dy = \int_{-\infty}^{\infty} y\, \frac{p_{X,Y}(x, y)}{p_X(x)}\, dy\,.$$

$\mathbb{E}[Y \mid X = x]$ is a function of x.

Example 2.29 (conditional expectation). Consider the discrete random vector (X, Y) from example 2.11 with joint pmf $p(0, 0) = 0.4, p(0, 1) = 0.1, p(1, 0) = 0.1, p(1, 1) = 0.4$. We found $p_X(0) = 0.5$ and $p_X(1) = 0.5$.

- The conditional expectation of Y given $X = 0$ is $p_{Y|X=0}(0) = \frac{p(0,0)}{p_X(0)} = \frac{0.4}{0.5} = 0.8$, $p_{Y|X=0}(1) = \frac{p(0,1)}{p_X(0)} = \frac{0.1}{0.5} = 0.2$. So, $\mathbb{E}[Y \mid X = 0] = (0 \cdot 0.8) + (1 \cdot 0.2) = 0.2$.
- The conditional expectation of Y given $X = 1$ is $p_{Y|X=1}(0) = \frac{p(1,0)}{p_X(1)} = \frac{0.1}{0.5} = 0.2$, $p_{Y|X=1}(1) = \frac{p(1,1)}{p_X(1)} = \frac{0.4}{0.5} = 0.8$. So, $\mathbb{E}[Y \mid X = 1] = (0 \cdot 0.2) + (1 \cdot 0.8) = 0.8$.

Definition 2.30 (conditional expectation as RV). The *conditional expectation of Y given X*, denoted $\mathbb{E}[Y \mid X]$, is a random variable. Its value depends on the value taken by X. If X takes the value x, then $\mathbb{E}[Y \mid X]$ takes the value $\mathbb{E}[Y \mid X = x]$. More formally, let $h(x) = \mathbb{E}[Y \mid X = x]$. Then $\mathbb{E}[Y \mid X]$ is the random variable $h(X)$.

Remark 2.31. The random variable $\mathbb{E}[Y \mid X]$ can be thought of as the "best guess" or predictor of Y based on the information provided by X, in the sense of minimizing mean squared error. It has the property that $\mathbb{E}[(Y - \mathbb{E}[Y \mid X])g(X)] = 0$ for any suitable function g.

In machine learning, whenever you see $\mathbb{E}[Y \mid X]$ you should think of any regression technique that aims at predicting Y from X (as a point estimator). Depending on what X, Y represent, this could be a linear regression, a random forest, a (convolutional) neural network, etc.

Theorem 2.32 (law of total expectation / tower property / iterated expectation). *For any two RVRVs X and Y (assuming relevant expectations exist),*

$$\mathbb{E}[Y] = \mathbb{E}\left[\mathbb{E}[Y \mid X]\right]\,.$$

Proof. We show this for the continuous case (discrete case is analogous with sums). Let

$h(X) = \mathbb{E}[Y \mid X]$. So $h(x) = \mathbb{E}[Y \mid X = x] = \int y \, \frac{p_{X,Y}(x,y)}{p_X(x)} \, dy$. Then,

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[Y \mid X]] &= \mathbb{E}[h(X)] \\
&= \int h(x) \, p_X(x) \, dx \quad \text{(by LOTUS)} \\
&= \int \left(\int y \, \frac{p_{X,Y}(x,y)}{p_X(x)} \, dy \right) p_X(x) \, dx \\
&= \iint y \, p_{X,Y}(x,y) \, dy \, dx \quad \text{(assuming } p_X(x) > 0 \text{ where } p_{X,Y}(x,y) \text{ is non-zero)} \\
&= \int y \left(\int p_{X,Y}(x,y) \, dx \right) dy \\
&= \int y \, p_Y(y) \, dy \quad \text{(since } \int p_{X,Y}(x,y) \, dx = p_Y(y)\text{)} \\
&= \mathbb{E}[Y] \, .
\end{aligned}$$

□

Example 2.33 (law of total expectation). Suppose the number of customers N arriving at a store in a day is a Poisson random variable with mean λ, $N \sim \text{Poi}(\lambda)$. Each customer i, independently, makes a purchase with probability q. Let K be the total number of purchases. We want to compute the expected number of purchases $\mathbb{E}[K]$. Given $N = n$ (a fixed number of customers), K is the number of successes in n Bernoulli trials, so $K \mid N = n \sim \text{Bin}(n, q)$. (This is extremely bad notation that is still pretty clear and compelling. We should rather write $p_{K|N=n}(k) = p_{\text{Bin}(n,q)}(k)$.) Thus, $\mathbb{E}[K \mid N = n] = nq$. So, the random variable $\mathbb{E}[K \mid N]$ is Nq. By the law of total expectation, we find

$$\mathbb{E}[K] = \mathbb{E}[\mathbb{E}[K \mid N]] = \mathbb{E}[Nq] = q \, \mathbb{E}[N] = q\lambda \, .$$

Definition 2.34 (conditional variance). The *conditional variance of Y given $X = x$*, denoted $\text{var}[Y \mid X = x]$, is

$$\text{var}[Y \mid X = x] = \mathbb{E}\left[(Y - \mathbb{E}[Y \mid X = x])^2 \mid X = x \right] \, .$$

It can also be computed as $\mathbb{E}[Y^2 \mid X = x] - (\mathbb{E}[Y \mid X = x])^2$. The *conditional variance of Y given X*, $\text{var}[Y \mid X]$, is a random variable whose value is $\text{var}[Y \mid X = x]$ when $X = x$.

Theorem 2.35 (law of total variance). *For any two RVRVs X and Y (assuming relevant expectations and variances exist),*

$$\text{var}[Y] = \mathbb{E}\left[\text{var}[Y \mid X]\right] + \text{var}\left[\mathbb{E}[Y \mid X]\right] \, .$$

In words, the total variance is the expected conditional variance plus the variance of the conditional expectation.

Proof. Let $\mu_Y = \mathbb{E}[Y]$ and $h(X) = \mathbb{E}[Y \mid X]$.

$$\begin{aligned}\operatorname{var}[Y] &= \mathbb{E}[(Y-\mu_Y)^2] \\ &= \mathbb{E}[(Y-h(X)+h(X)-\mu_Y)^2] \\ &= \mathbb{E}[(Y-h(X))^2] + \mathbb{E}[(h(X)-\mu_Y)^2] + 2\,\mathbb{E}[(Y-h(X))(h(X)-\mu_Y)]\,.\end{aligned}$$

The second term is $\mathbb{E}[(\mathbb{E}[Y \mid X] - \mathbb{E}[\mathbb{E}[Y \mid X]])^2] = \operatorname{var}[\mathbb{E}[Y \mid X]]$ by theorem 2.32. The first term, $\mathbb{E}[(Y-h(X))^2] = \mathbb{E}[(Y-\mathbb{E}[Y \mid X])^2]$. Using iterated expectation: $\mathbb{E}[(Y-\mathbb{E}[Y \mid X])^2] = \mathbb{E}\left[\mathbb{E}[(Y-\mathbb{E}[Y \mid X])^2 \mid X]\right] = \mathbb{E}[\operatorname{var}[Y \mid X]]$. The cross-term $2\,\mathbb{E}[(Y-h(X))(h(X)-\mu_Y)]$: Let $g(X) = h(X)-\mu_Y$. Then the cross-term is $2\,\mathbb{E}[(Y-\mathbb{E}[Y \mid X])g(X)]$. Using iterated expectation: $\mathbb{E}[\mathbb{E}[(Y-\mathbb{E}[Y \mid X])g(X) \mid X]]$. Since $g(X)$ is a function of X, it's constant when conditioning on X: $\mathbb{E}[g(X)\,\mathbb{E}[Y-\mathbb{E}[Y \mid X] \mid X]] = \mathbb{E}[g(X)(\mathbb{E}[Y \mid X] - \mathbb{E}[\mathbb{E}[Y \mid X] \mid X])] = \mathbb{E}[g(X)(\mathbb{E}[Y \mid X] - \mathbb{E}[Y \mid X])] = \mathbb{E}[g(X)\cdot 0] = 0$. So the cross-term vanishes. Combining the pieces gives the result. □

Example 2.36 (law of total variance). Continuing the store example from example 2.33, we have $N \sim \operatorname{Poi}(\lambda)$ and $K \mid N = n \sim \operatorname{Bin}(n, q)$. We found $\mathbb{E}[K \mid N] = Nq$. We also need $\operatorname{var}[K \mid N = n] = nq(1-q)$. So $\operatorname{var}[K \mid N] = Nq(1-q)$.

- Expected Conditional Variance: $\mathbb{E}[\operatorname{var}[K \mid N]] = \mathbb{E}[Nq(1-q)] = q(1-q)\,\mathbb{E}[N] = q(1-q)\lambda$.
- Variance of Conditional Expectation: $\operatorname{var}[\mathbb{E}[K \mid N]] = \operatorname{var}[Nq] = q^2 \operatorname{var}[N] = q^2\lambda$ (since for Poisson, $\operatorname{var}[N] = \lambda$).

By the law of total variance we find

$$\operatorname{var}[K] = \mathbb{E}[\operatorname{var}[K \mid N]] + \operatorname{var}[\mathbb{E}[K \mid N]] = q(1-q)\lambda + q^2\lambda = \lambda q(1-q+q) = \lambda q\,.$$

If $q = 1$ (every customer buys), then $K = N$, so $\operatorname{var}[K] = \operatorname{var}[N] = \lambda$ and if $q = 0$ (no customer buys), then $K = 0$, so $\operatorname{var}[K] = 0$. Our formula gives $\lambda(0) = 0$.

Remark 2.37 (Bayesian interpretation of total variance). In Bayesian modeling, if Y is a predicted outcome and θ represent model parameters, then the law of total variance $\operatorname{var}[Y] = \mathbb{E}[\operatorname{var}[Y \mid \theta]] + \operatorname{var}[\mathbb{E}[Y \mid \theta]]$ is interpreted as follows: $\mathbb{E}[\operatorname{var}[Y \mid \theta]]$ is termed *aleatoric uncertainty*—the inherent randomness in Y for fixed parameters (within-model variance). The second term, $\operatorname{var}[\mathbb{E}[Y \mid \theta]]$, is termed *epistemic uncertainty*—our uncertainty about the true parameters, reflecting how much the model's mean prediction varies across different parameter settings (between-model variance). This decomposition helps in understanding sources of prediction uncertainty.

2.2. Basic probability inequalities

Section goal. Introduce fundamental inequalities that provide bounds on probabilities without requiring full knowledge of the distribution but only relying on properties like the mean or variance.

Theorem 2.38 (Markov's inequality). *Let $(\Omega, \mathcal{A}, P)$ be a probability space, $X : \Omega \to \mathbb{R}$ be an RVRV (whose density exists and is integrable) and let $f : S_X \to \mathbb{R}_{\geq 0}$ be monotonically increasing, where $S_X = \operatorname{im}(X) := X(\Omega) \subseteq \mathbb{R}$. Then for each $\epsilon \in \mathbb{R}$ with $f(\epsilon) > 0$ we have*

$$P(X \geq \epsilon) \leq \frac{1}{f(\epsilon)} \mathbb{E}[f(X)] \,.$$

In the literature, the Markov inequality is often shown only as a special case: Let X be a non-negative *RVRV (i.e., $P(X \geq 0) = 1$). Then for any constant $\epsilon > 0$,*

$$P(X \geq \epsilon) \leq \frac{\mathbb{E}[X]}{\epsilon} \,.$$

Proof. First note that f is $\mathcal{B}|_{S_X}$-$\mathcal{B}|_{\mathbb{R}_{\geq 0}}$-measurable by Exercise 2.1, as for $a \in \mathbb{R}_{\geq 0}$

$$f^{-1}([a, \infty)) = \{x \in \mathbb{R}_{\geq 0} \mid f(x) > a\} = [A, \infty), \quad \text{where } A := \inf\{x \in \mathbb{R}_{\geq 0} \mid f(x) > a\} \,.$$

Moreover, by monotonicity we have

$$\{x \in \mathbb{R} \mid f(x) \geq f(\varepsilon)\} = \{x \in \mathbb{R} \mid x \geq \varepsilon\}.$$

Hence, we compute

$$P(X \geq \epsilon) = \mathbb{E}[\chi_{\{X \geq \epsilon\}}] = \mathbb{E}\Big[\frac{f(\epsilon)}{f(\epsilon)} \chi_{\{X \geq \epsilon\}}\Big] \leq \mathbb{E}\Big[\frac{f(X)}{f(\epsilon)} \chi_{\{X \geq \epsilon\}}\Big] \leq \frac{\mathbb{E}[f(X)]}{f(\epsilon)} \,.$$

More explicitly, the "simpler" statement for a non-negative RVRV X with pdf $p_X(x)$ (the discrete case is analogous with sums instead of integrals) can be computed like follows:

$$\begin{aligned}
\mathbb{E}[X] &= \int_0^\infty x\, p_X(x)\, dx \\
&= \int_0^\epsilon x\, p_X(x)\, dx + \int_\epsilon^\infty x\, p_X(x)\, dx \\
&\geq \int_\epsilon^\infty x\, p_X(x)\, dx \quad (\text{since } x\, p_X(x) \geq 0 \text{ for } x \in [0, \epsilon]) \\
&\geq \int_\epsilon^\infty \epsilon\, p_X(x)\, dx \quad (\text{since } x \geq \epsilon \text{ in the integral's range}) \\
&= \epsilon \int_\epsilon^\infty p_X(x)\, dx \\
&= \epsilon\, P(X \geq \epsilon) \,.
\end{aligned}$$

Since $\epsilon > 0$, dividing by ϵ gives the desired result.

Note that the Markov inequality is very crude/lose, but requires hardly any assumptions on the distribution of X and we only use the mean of X for the bound. □

Example 2.39 (Markov's inequality application). Suppose the average daily number of hits on a website is $\mathbb{E}[X] = 50$. What can we say about the probability that the website receives at least 150 hits on a given day? Since the number of hits X must be non-negative, we can use Markov's inequality with $a = 150$

$$P(X \geq 150) \leq \frac{\mathbb{E}[X]}{150} = \frac{50}{150} = \frac{1}{3}\,.$$

So, the probability of having at least 150 hits is at most $1/3$. This bound is quite general and might not be very tight.

Theorem 2.40 (Chebyshev's inequality). *Let X be an RVRV with finite mean $\mathbb{E}[X]$ and finite variance $\operatorname{var}[X] > 0$. Then for any $\epsilon > 0$,*

$$P(|X - \mathbb{E}[X]| \geq \epsilon) \leq \frac{\operatorname{var}[X]}{\epsilon^2}\,.$$

Proof. In theorem 2.38, let $f(x) = x^2$, which gives $P(|X| \geq \epsilon) \leq \epsilon^{-2}\,\mathbb{E}[X^2]$. When then substituting $Y := X - \mathbb{E}[X]$ we get

$$P(|X - \mathbb{E}[X]| \geq \epsilon) \leq \epsilon^{-2}\,\mathbb{E}[(X - \mathbb{E}[X])^2] = \epsilon^{-2}\operatorname{var}[X]\,.$$

Chebyshev's inequality requires more information (mean and variance), but can give potentially better (tighter) bounds. □

Example 2.41 (Chebyshev's inequality application). Suppose a random variable X has mean $\mu_X = 10$ and variance $\sigma_X^2 = 4$ (so $\sigma_X = 2$). What is the probability that X deviates from its mean by more than 3 standard deviations (i.e., by more than $3\sigma_X = 6$)? Using Chebyshev's inequality with $k = 3$ we get

$$P(|X - 10| \geq 3 \cdot 2) = P(|X - 10| \geq 6) \leq \frac{1}{3^2} = \frac{1}{9}\,.$$

This means the probability that X is outside the interval $(10 - 6, 10 + 6) = (4, 16)$ is at most $1/9$. Chebyshev's inequality provides a universal bound, applicable regardless of the specific shape of the distribution of X, as long as its mean and variance are finite.

Theorem 2.42 (Jensen's Inequality). *Let X be an RVRV such that $\mathbb{E}[X]$ is finite. Let $g : \mathbb{R} \to \mathbb{R}$ be a convex function, that is for any $x, y \in \mathbb{R}$ and $t \in [0, 1]$ we have $f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2)$. Then*

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])\,.$$

If g is strictly convex, then equality holds if and only if $X = \mathbb{E}[X]$ with probability 1 (i.e., X is a constant). If g is concave, the inequality is reversed, $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$.

Proof. We sketch the proof for differentiable g. A convex differentiable function g has the property that for any point x_0, its graph lies above its tangent line at x_0. The tangent line at $x_0 = \mathbb{E}[X]$ is given by $L(x) = g(\mathbb{E}[X]) + g'(\mathbb{E}[X])(x - \mathbb{E}[X])$ (Taylor approximation). Since g is convex, $g(x) \geq L(x)$ for all x. Therefore, $g(X) \geq g(\mathbb{E}[X]) + g'(\mathbb{E}[X])(X - \mathbb{E}[X])$. Taking expectations of both sides yields

$$\begin{aligned}
\mathbb{E}[g(X)] &\geq \mathbb{E}[g(\mathbb{E}[X]) + g'(\mathbb{E}[X])(X - \mathbb{E}[X])] \\
&= \mathbb{E}[g(\mathbb{E}[X])] + \mathbb{E}[g'(\mathbb{E}[X])(X - \mathbb{E}[X])] \\
&= g(\mathbb{E}[X]) + g'(\mathbb{E}[X])\,\mathbb{E}[X - \mathbb{E}[X]] \quad \text{(since } g(\mathbb{E}[X]) \text{ and } g'(\mathbb{E}[X]) \text{ are constants)} \\
&= g(\mathbb{E}[X]) + g'(\mathbb{E}[X])(\mathbb{E}[X] - \mathbb{E}[\mathbb{E}[X]]) \\
&= g(\mathbb{E}[X]) + g'(\mathbb{E}[X])(\mathbb{E}[X] - \mathbb{E}[X]) \\
&= g(\mathbb{E}[X]) + g'(\mathbb{E}[X]) \cdot 0 \\
&= g(\mathbb{E}[X])\,.
\end{aligned}$$

A more rigorous proof for non-differentiable g uses the property that for a convex function, there exists a supporting line at every point. □

Example 2.43 (Jensen's inequality application).

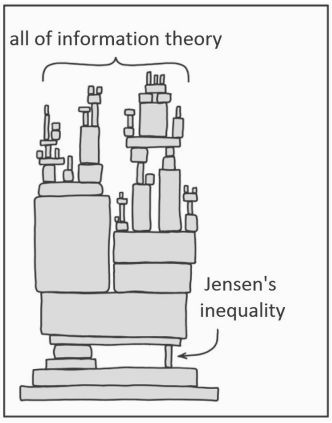
(i) Let $g(x) = -\ln(x)$ for $x > 0$. This function is convex. If X is a positive RVRV, then $\mathbb{E}[-\ln(X)] \geq -\ln(\mathbb{E}[X])$, which means $\mathbb{E}[\ln(X)] \leq \ln(\mathbb{E}[X])$. This shows that the logarithm of the expectation is greater than or equal to the expectation of the logarithm. This is related to the geometric mean being less than or equal to the arithmetic mean.

(ii) Let X be a RVRV with finite variance. Let m be the median, μ the mean, and σ the standard deviation of X. Applying Jensen's inequality twice, we get

$$\begin{aligned}
|\mu - m| &= |\,\mathbb{E}[X - m]| \\
&\leq \mathbb{E}[|X - m|] \qquad \text{Jensen} \\
&\leq \mathbb{E}[|X - \mu|] \qquad \text{“median minimizes expected absolute deviation”} \\
&\leq \sqrt{\mathbb{E}[(X - \mu)^2]} \qquad \text{Jensen} \\
&= \sigma\,.
\end{aligned}$$

The absolute difference between the mean and the median is bounded by the standard deviation.

Remark 2.44 (Jensen oh Jensen). Jensen's inequality is extremely important, but also infamous among researchers, because it's always "the wrong way round." Random memes from the internet:



Big calibre researchers like Francis Bach describe this in blog post: "...let me be very clear. Jensen's inequality is often not in the direction that you would hope it to be."

Remark 2.45 (concentration inequalities). The Markov and Chebyshev inequalities bound the probability that "X becomes large or far from its mean." In that sense, they describe how "concentrated" the probability mass is. More generally, when a function (e.g., the sum or average) of multiple, typically *independent*, random variables "does not depend too much on any single one of them," there is a plethora of theory showing that the probability of that function concentrates closely around its expected value. This is captured by so-called *concentration inequalities*, which form an important cornerstone of statistics and machine learning and have fancy names like Chernoff, Hoeffding, Azuma, McDiarmid, Bennett, or Bernstein bounds (quick overview). While Markov and Chebyshev provide "polynomial" bounds (polynomial in ϵ), we can get much better "exponential" bounds for the concentration of sums of independent bounded RVs. A glance into the "standard reference of concentration inequalities" gives a good feeling for how the different variants look like.

2.3. Computations with distributions

Section goal. We introduce and get used to a set of tools that are useful for performing computations with distributions, including characteristic functions (cf), transformations of distributions, and explicit computation rules for special distributions (e.g., Gaussians).

Definition 2.46 (characteristic function (cf)). Let X be an RVRV. The *characteristic function (cf)* of X, denoted $\phi_X(t)$ or $\varphi_X(t)$, is defined as

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\cos(tX) + i\sin(tX)]\,,$$

where $i = \sqrt{-1}$ is the imaginary unit.

(i) If X is discrete with pmf $p_X(x)$, $\phi_X(t) = \sum_x e^{itx} p_X(x)$.

(ii) If X is continuous with pdf $p_X(x)$, $\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} p_X(x)\, dx$.

The characteristic function always exists for any random variable X and any $t \in \mathbb{R}$, because $|e^{itX}| = |\cos(tX) + i\sin(tX)| = \sqrt{\cos^2(tX) + \sin^2(tX)} = 1$, so the expectation is of a bounded function.

Remark 2.47 (properties of cfs). Let X, Y be RVRVs.

(i) $\phi_X(0) = 1$ and $|\phi_X(t)| \leq 1$.

(ii) $\phi_{aX+b}(t) = e^{itb}\phi_X(at)$.

(iii) We can compute *moments* from the derivatives of $\phi(t)$ via $\mathbb{E}[X^n] = i^{-n}\phi_X^{(n)}(0)$ (if the moments exist).

(iv) If $X \perp\!\!\!\perp Y$, then $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$.

(v) **Uniqueness:** The cf uniquely determines the distribution. There are explicit inversion formulas to recover the cdf or pdf from the cf (e.g., Lévy's inversion theorem).

(vi) **(Lévy's) continuity theorem:** A sequence of RVRVs X_n converges in distribution to X if and only if their characteristic functions $\phi_{X_n}(t)$ converge pointwise to $\phi_X(t)$ for all t, and $\phi_X(t)$ is continuous at $t = 0$. This is crucial for proving the central limit theorem (clt). For more concrete statements see the wikipedia article.

Example 2.48 (cf of normal distribution). The characteristic function of $X \sim \mathcal{N}(\mu, \sigma^2)$ is

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \exp\left(i\mu t - \frac{1}{2}\sigma^2 t^2\right).$$

Remark 2.49 (moment generating functions). Another tool often introduced alongside or instead of characteristic functions is the *moment generating function (MGF)*, defined as $M_X(t) = \mathbb{E}[e^{tX}]$ for real t where this expectation exists (typically in a neighborhood of $t = 0$). If it exists and is differentiable, moments can be derived as $\mathbb{E}[X^n] = M_X^{(n)}(0)$, i.e., the n-th derivative of the MGF evaluated at $t = 0$. Like cfs, MGFs have the property that $M_{X+Y}(t) = M_X(t)M_Y(t)$ if $X \perp\!\!\!\perp Y$, and they also uniquely determine the distribution if they exist in a neighborhood of zero. However, MGFs do not always exist (e.g., for the Cauchy distribution), whereas the characteristic function $\phi_X(t) = \mathbb{E}[e^{itX}]$ always exists for all real t.

Together with the uniqueness and continuity results, characteristic functions are extremely powerful "calculation tools." They are often used to prove results about distributions by employing tools from (complex) analysis (think derivatives, Taylor series, convergence of functions, etc).

Example 2.50 (sums of independent RVs). Characteristic functions are particularly useful for determining the distribution of sums of independent RVs.

(i) **Sum of independent Poissons is Poisson:** Let $X_1 \sim \mathrm{Poi}(\lambda_1)$ and $X_2 \sim \mathrm{Poi}(\lambda_2)$ be independent RVs. Then $X_1 + X_2 \sim \mathrm{Poi}(\lambda_1 + \lambda_2)$.

(ii) **Sum of independent Gammas (with same rate) is Gamma:** Let $X_1 \sim \Gamma(\alpha_1, \beta)$ and $X_2 \sim \Gamma(\alpha_2, \beta)$ be independent RVs. The characteristic function of $X \sim \Gamma(\alpha, \beta)$ is $\phi_X(t) = (1 - it/\beta)^{-\alpha}$.
To see this:

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \int_0^\infty e^{itx} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}\, dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\beta-it)x}\, dx\,.$$

Let $u = (\beta - it)x$, so $x = u/(\beta - it)$ and $dx = du/(\beta - it)$. The integral becomes

$$\int_0^\infty \left(\frac{u}{\beta - it}\right)^{\alpha-1} e^{-u} \frac{du}{\beta - it} = \frac{1}{(\beta - it)^\alpha} \int_0^\infty u^{\alpha-1} e^{-u}\, du = \frac{\Gamma(\alpha)}{(\beta - it)^\alpha}\,.$$

Thus, $\phi_X(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha)}{(\beta-it)^\alpha} = \left(\frac{\beta}{\beta-it}\right)^\alpha = \left(1 - \frac{it}{\beta}\right)^{-\alpha}$.

For independent X_1 and X_2 we compute for $S = X_1 + X_2$

$$\phi_S(t) = \phi_{X_1}(t)\phi_{X_2}(t) = (1 - it/\beta)^{-\alpha_1}(1 - it/\beta)^{-\alpha_2} = (1 - it/\beta)^{-(\alpha_1+\alpha_2)}\,.$$

This is the characteristic function of a Gamma distribution with parameters $(\alpha_1 + \alpha_2, \beta)$ and by uniqueness $S \sim \Gamma(\alpha_1 + \alpha_2, \beta)$. Since $\mathrm{Exp}(\lambda) = \Gamma(1, \lambda)$ and $\chi^2_k = \Gamma(k/2, 1/2)$ this additivity extends to exponential and χ^2 distributions.

Proposition 2.51 (transformation of densities / change of variables). *Let X be a continuous RVRV with density p_X, $g : \mathbb{R} \to \mathbb{R}$ a continuously differentiable, strictly monotonic function on the support of X, $S_X := \{x \in \mathbb{R} \mid p_X(x) > 0\}$, and set $Y := g(X)$. Then the pdf of Y is given by*

$$p_Y(y) = p_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| \quad \text{for } y \in g(S_X)\,.$$

For $y \notin g(S_X)$, $p_Y(y) = 0$.

For a random vector X in $\mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^n$ a continuously differentiable bijective function for which the determinant of the Jacobian J_g is non-zero for all $x \in S_X$ we have

$$p_Y(y) = p_X(g^{-1}(y)) \left|\det(J_{g^{-1}}(y))\right| \quad \text{for } y \in g(S_X)\,,$$

where $J_{g^{-1}}(y) = \left[\frac{\partial g_i^{-1}}{\partial y_j}\right]_{i,j=1}^n$ is the Jacobian matrix of the inverse transformation g^{-1} at y. Using $J_{g^{-1}}(y) = [J_g(g^{-1}(y))]^{-1}$, we have equivalently

$$p_Y(y) = p_X(g^{-1}(y)) \frac{1}{|\det(J_g(g^{-1}(y)))|} = p_X(g^{-1}(y)) \left|\det(J_g(x))\right|^{-1}_{x=g^{-1}(y)}\,.$$

Proof sketch for the scalar case. We first compute the cdf of Y and differentiate it to get the pdf. Assume g is strictly increasing. Then

$$F_Y(y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y)) \,.$$

Differentiating with respect to y using the chain rule yields

$$p_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(g^{-1}(y)) = p_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) \,.$$

Since g is strictly increasing, g^{-1} is also strictly increasing, so its derivative $\frac{d}{dy} g^{-1}(y)$ is positive. Thus, $\frac{d}{dy} g^{-1}(y) = \left| \frac{d}{dy} g^{-1}(y) \right|$. If g is strictly decreasing,

$$F_Y(y) = P(g(X) \leq y) = P(X \geq g^{-1}(y)) = 1 - P(X < g^{-1}(y)) = 1 - F_X(g^{-1}(y)) \,.$$

Differentiating with respect to y yields

$$p_Y(y) = \frac{d}{dy}(1 - F_X(g^{-1}(y))) = -p_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) \,.$$

Since now $\frac{d}{dy} g^{-1}(y)$ is negative we obtain the result. If g is not strictly monotonic over the entire support, the support of X may need to be partitioned into regions where g is strictly monotonic, and the contributions summed up. □

Example 2.52 (affine transformations).

(i) Let X be a RVRV with density p_X. Then $Y = aX + b$ for $a \in \mathbb{R} \setminus \{0\}$ and $b \in \mathbb{R}$ has density $p_Y(y) = \frac{1}{|a|} p_X((y-b)/a)$.

(ii) If X is a random vector in $\mathbb{R}^n$ with density p_X, then $Y = AX + b$ for invertible $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$ has density $p_Y(y) = \frac{p_X(A^{-1}(y-b))}{|\det(A)|}$.

(iii) If $X \sim \mathcal{N}(0,1)$, i.e., X is a *standard Gaussian*, then $Z := \sigma X + \mu$ is $\mathcal{N}(\mu, \sigma^2)$ distributed (for $\mu, \sigma \in \mathbb{R}$ with $\sigma \neq 0$). Conversely, starting from a $Z \sim \mathcal{N}(\mu, \sigma^2)$ distributed RV, we get a standard normal via $(Z - \mu)/\sigma \sim \mathcal{N}(0,1)$.

Lemma 2.53 (computation rules for Gaussians). *The pdf of a multi-variate Gaussian with mean $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ is given by*

$$p(x) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} \exp\Big(-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu) \Big) \,,$$

where $|\Sigma|$ denotes the determinant of Σ. We then write $X \sim \mathcal{N}(\mu, \Sigma)$. Any valid covariance matrix has to be symmetric ($\Sigma^T = \Sigma$) and positive semi-definite, meaning that $x^T \Sigma x \geq 0$ for all $x \in \mathbb{R}^d$. To obtain non-degenerate densities, we often make the stronger assumption of Σ being positive definite, meaning that $x^T \Sigma x > 0$ for all $x \in \mathbb{R}^d \setminus \{\mathbf{0}\}$. We also denote positive semi-definiteness by $\Sigma \succeq 0$ and positive definiteness by $\Sigma \succ 0$.

Let $X \in \mathbb{R}^{d_X}, Y \in \mathbb{R}^{d_Y}$ be random vectors that are jointly distributed according to a multi-variate Gaussian with mean $\mu = (\mu_X, \mu_Y) \in \mathbb{R}^d$ for $d := d_X + d_Y$ and covariance matrix

$$\Sigma = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_{YY} \end{pmatrix} \in \mathbb{R}^{d \times d} ,$$

where $\Sigma_{XX} \in \mathbb{R}^{d_X \times d_X}$, $\Sigma_{XY} \in \mathbb{R}^{d_X \times d_Y}$, and $\Sigma_{YY} \in \mathbb{R}^{d_Y \times d_Y}$. Calling $Z := (X, Y)$ we thus have $Z \sim \mathcal{N}(\mu, \Sigma)$. Then the following hold.

(i) The random variables X and Y are independent if and only if $\Sigma_{XY} = 0$.

(ii) The random variable $AZ + b$ for $A \in \mathbb{R}^{d \times d}$ and $b \in \mathbb{R}^d$ follows the multi-variate normal $\mathcal{N}(A\mu + b, A\Sigma A^T)$. (You can assume that $A\Sigma A^T \succ 0$.)

(iii) The marginal distribution of X is given by $\mathcal{N}(\mu_X, \Sigma_{XX})$ (and the marginal distribution of Y is given by $\mathcal{N}(\mu_Y, \Sigma_{YY})$).

(iv) The conditional distribution of X given $Y = y$ is given by $\mathcal{N}(\mu_{X|y}, \Sigma_{X|y})$ where

$$\mu_{X|y} = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y) , \qquad \Sigma_{X|y} = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{XY}^T .$$

(v) If we assume $d_X = d_Y$ and $\Sigma_{XY} = 0$, i.e., the random vectors X and Y are over the same space and are uncorrelated and thus independent. Then the random variable $aX + bY$ is distributed according to $\mathcal{N}(a\mu_X + b\mu_Y, a^2\Sigma_{XX} + b^2\Sigma_{YY})$ for $a, b \in \mathbb{R}$.

(vi) The previous statement can be extended: Even in the original case ($d_X \neq d_Y$ and $\Sigma_{XY} \neq 0$), we have that for all matrices $A \in \mathbb{R}^{k \times d_X}, B \in \mathbb{R}^{k \times d_Y}$ and vectors $c \in \mathbb{R}^k$, the random vector $AX + BY + c$ is distributed according to multi-variate normal in k dimensions given by

$$\mathcal{N}(A\mu_X + B\mu_Y + c, A\Sigma_{XX}A^T + B\Sigma_{XY}^T A^T + A\Sigma_{XY}B^T + B\Sigma_{YY}B^T) .$$

(Again, the resulting normal may be "degenerate," meaning that the covariance matrix may not be positive definite depending on A and B.)

Remark 2.54 (the importance/usefulness of Gaussians). In words, the statements in lemma 2.53 mean that (i) for Gaussians zero correlation and independence are equivalent and (ii)-(vi) Gaussians are closed under affine transformations, marginalization, conditioning, and linear combinations.

Since most computations and manipulations involving only Gaussians result again in Gaussians, and Gaussians are fully determined by the mean vector and covariance matrix, "every computation with Gaussian distributions just become linear algebra." This is one of the many reasons why Gaussians are so commonly used and play such a big role in statistics and machine learning (others being their maximum entropy property, the central limit theorem, etc).

For example, Gaussian processes are a much used probabilistic regression techniques heavily relying on computations with Gaussians. Gaussians are typically the base distribution

for normalizing flows or diffusion models and Gaussian splatting (heavily leveraging fast computations with Gaussians) have recently shown strong performance in 3D rendering.

2.4. Convergence of random variables and limit theorems

Section goal. Introduce different ways in which a sequence of RVs can converge to another RVs or a constant, and then state and discuss the foundational limit theorems of probability: the laws of large numbers (lln) and the central limit theorem (clt).

Remark 2.55 (why sequences). Understanding how sequences of random variables behave as their length (number) increases is crucial in probability and statistics. We have already discussed "concentration inequalities" specifically for sums/means of RVs, where we are interested in how the average of many observations gets closer to the true mean of the underlying process. In these settings, RVs in the sequence are often thought of as independent observations, i.e., represent observations in a dataset, as described in remark 1.78. The lln and clt feed into that theme of results, but first we need to formalize different notions of such "getting closer," i.e., convergence of sequences of RVs.

After that, the laws of large numbers and the central limit theorem will state that (and how) means of sequences of iid random variables get closer and closer to their mean (in probability or almost surely).

Definition 2.56 (modes of convergence). Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of RVRV and let X be another RVRV (which can also be a constant), all defined on the same probability space $(\Omega, \mathcal{A}, P)$. The sequence $(X_n)_{n\in\mathbb{N}}$

(i) *converges in probability* to X, denoted $X_n \xrightarrow{P} X$, if for every $\epsilon > 0$,

$$\lim_{n\to\infty} P(|X_n - X| \geq \epsilon) = 0\,.$$

Intuitively, this means that for large n, it is very unlikely that X_n is far from X.

(ii) *converges almost surely* to X, denoted $X_n \xrightarrow{a.s.} X$ or $X_n \to X$ a.s., if

$$P\left(\left\{\omega \in \Omega \,\middle|\, \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1\,.$$

Intuitively, this means that for almost all outcomes ω, the sequence of numbers $X_n(\omega)$ converges to $X(\omega)$.

(iii) *converges in distribution* (or *converges weakly*) to X, denoted $X_n \xrightarrow{d} X$ or $X_n \Rightarrow X$, if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

for every $x \in \mathbb{R}$ at which F_X is continuous. This means that the overall shape of the distribution of X_n approaches the shape of the distribution of X. Equivalently, $\lim_{n\to\infty} \phi_{X_n}(t) = \phi_X(t)$ for all $t \in \mathbb{R}$.

Remark 2.57 (relations between modes of convergence). The different modes of convergence are related via

$$X_n \xrightarrow{a.s.} X \quad \Longrightarrow \quad X_n \xrightarrow{P} X \quad \Longrightarrow \quad X_n \xrightarrow{d} X\,.$$

In words,

$$\text{almost sure conv.} \Longrightarrow \text{conv. in probability} \Longrightarrow \text{conv. in distribution}\,.$$

While the opposite implications generally do not hold, whenever $X_n \xrightarrow{P} X$, there exists a subsequence of (X_n) that converges almost surely to X. If $X_n \xrightarrow{d} c$ for $c \in \mathbb{R}$ then we also have $X_n \xrightarrow{P} c$.

Example 2.58 (convergence in P $\not\Rightarrow$ a.s. convergence). For $([0,1], \mathcal{B}|_{[0,1]}, \lambda|_{[0,1]})$, we define the "traveling bump" as follows. For $n = 2^k + j$ where $0 \leq j < 2^k$ $(j, k \in \mathbb{N})$, define

$$X_n(\omega) = \chi_{[j/2^k,(j+1)/2^k]}(\omega)\,.$$

For any $\epsilon \in (0,1)$, $P(|X_n - 0| \geq \epsilon) = P(X_n = 1)$. For $n = 2^k + j$ $P(X_n = 1)$ is the length of the indicator interval, which is $1/2^k$. As $n \to \infty$, $k \to \infty$, so $P(X_n = 1) \to 0$, i.e., $X_n \xrightarrow{P} 0$.

However, for $\omega \in [0,1]$, for each k, there is exactly one $j \in \{0, \ldots, 2^k - 1\}$ such that $\omega \in [j/2^k, (j+1)/2^k]$. So, for any ω, $X_n(\omega) = 1$ for infinitely many n and $\lim_{n\to\infty} X_n(\omega)$ does not exist (it certainly does not converge to 0). Thus, $P(\lim_{n\to\infty} X_n(\omega) = 0) = 0 \neq 1$.

Example 2.59 (conv. in d $\not\Rightarrow$ conv. in P). Let $U \sim \mathrm{Ber}(0.5)$ and define $X_n := U$ for all $n \in \mathbb{N}$ and $Y := 1 - U$. Then $X_n \xrightarrow{d} Y$, as $F_{X_n} = F_Y = F_{\mathrm{Ber}(0.5)}$. However, $|X_n - Y| = |U - (1-U)| = |2U - 1| = 1$ (note that $U \in \{0,1\}$). So, $P(|X_n - Y| = 1) = 1$ for all n.

Remark 2.60 (interpreting "closeness" in convergence). The different modes of convergence imply different types of "closeness."

- Convergence in probability ($X_n \xrightarrow{P} X$) and almost sure convergence ($X_n \xrightarrow{a.s.} X$) are strong notions that relate the random variables X_n and X on the same underlying probability space $(\Omega, \mathcal{A}, P)$, i.e., they relate the random variables on the "outcome label of the actual random experiments." The random variables actually end up "taking the same values" for any given instantiation. This only works, when they "measure the exact same processes." Almost sure convergence means that for almost every specific outcome $\omega \in \Omega$, the sequence of real numbers $X_n(\omega)$ converges to the real number $X(\omega)$. This is a very direct, pointwise relationship on the sample space. Convergence in probability is slightly weaker but still implies that for large n, $X_n(\omega)$ is very likely to be close to $X(\omega)$ for any given ω. Both imply that X_n and X are "coupled" or "live on the same space" and their values become similar.

- Convergence in distribution ($X_n \xrightarrow{d} X$) is a much weaker notion. It only means that the cdfs F_{X_n} converge to F_X (at continuity points). The random variables X_n and X do not even need to be defined on the same probability space. Their individual values $X_n(\omega)$ and $X(\omega)$ for some ω might be entirely unrelated. One X_i may be measurements in a physics laboratory, another a measurement of microbes in a river, and X may be measuring some properties of galaxies. All that matters is that their overall "probabilistic shape" (their distributions) ultimately become similar.

This distinction is particularly important for the Central Limit Theorem, which states convergence in distribution. The standardized sum Z_n doesn't become "equal" to a standard normal Z in a pointwise sense; rather, its distribution just increasingly resembles that of Z.

Theorem 2.61 (weak law of large numbers (wlln)). *Let $X_1, X_2, \ldots$ be a sequence of identically distributed RVRVs with finite mean $\mathbb{E}[X_i] = \mu < \infty$ that are pairwise uncorrelated. Let $\bar{X}_n := \frac{1}{n}\sum_{i=1}^n X_i$ be the sample mean. Then,*

$$\bar{X}_n \xrightarrow{P} \mu\,.$$

Proof sketch for finite variance. If we additionally assume that the X_i have finite variance $\mathrm{var}[X_i] = \sigma^2 < \infty$, the proof is straightforward using Chebyshev's inequality. We have $\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n}(n\mu) = \mu$. Since the X_i are iid,

$$\mathrm{var}[\bar{X}_n] = \mathrm{var}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n^2}\sum_{i=1}^n \mathrm{var}[X_i] = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}\,.$$

By Chebyshev's inequality theorem 2.40, for any $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\mathrm{var}[\bar{X}_n]}{\epsilon^2} = \frac{\sigma^2/n}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \xrightarrow{n\to\infty} 0\,.$$

The wlln holds even if only the mean is finite (i.e., variance can be infinite), but the proof is more involved than. The rate of convergence is slower when variances can vary, especially when they become large or infinite. Note that it also holds when the $X_1, X_2, \ldots$ are iid, since that implies that they are pairwise uncorrelated. □

Theorem 2.62 (strong law of large numbers (slln)). *Let $X_1, X_2, \ldots$ be a sequence of iid RVRVs with finite mean $\mathbb{E}[X_i] = \mu < \infty$. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ be the sample mean. Then,*

$$\bar{X}_n \xrightarrow{a.s.} \mu\,.$$

Remark 2.63 (wlln vs. slln).

- The slln implies the wlln and states that the sample mean $\bar{X}_n$ converges to the true mean μ for *almost every sequence of outcomes*, not just that the probability of large deviations becomes small.

- The proof of the slln is more complex (especially the version not assuming finite variance) and typically requires the Borel-Cantelli lemmas to show that deviations from the mean do not occur infinitely often.
- The wikipedia article gives good structured overviews of how the proofs work in case you are interested. Note specifically that the wlln can be proven elegantly using characteristic functions and Lévy's continuity theorem.
- The llns are fundamental in statistics, providing justification for using sample averages to estimate population means. This can hardly be overstated: Our goal is to reason about properties of an entire distribution (e.g., "all humans," or "any conditions I could ever encounter"), but all we have is a fixed number of concrete observations (data). Without llns (or the clt or concentration inequalities), a priori, we would have no reason to believe that the average of the observed data should have anything to do with the true mean of the population/distribution of interest.
- For both the wlln and slln, there are many variants trying to relax some of the assumptions a bit, while adding others.
- What the llns do not yet tell us, is *how* the sample mean is distributed around the true mean for large (but finite) n and what the rate of convergence is. This is addressed by the clt.

Theorem 2.64 (central limit theorem (clt)). *Let $X_1, X_2, \ldots$ be a sequence of iid RVRVs with finite mean $\mathbb{E}[X_i] = \mu$ and finite, non-zero variance $\operatorname{var}[X_i] = \sigma^2 \in \mathbb{R}_{>0}$. Let $S_n := \sum_{i=1}^n X_i$ be the sum, and $\bar{X}_n = S_n/n$ be the sample mean. Consider the standardized sum (or standardized mean):*

$$Z_n = \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\operatorname{var}[S_n]}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \,.$$

Then

$$Z_n \xrightarrow{d} \mathcal{N}(0,1) \,.$$

The quantity $\sigma/\sqrt{n}$ that dictates the "asymptotic standard deviation" of the normal distribution as n goes to infinity is called standard error.

Proof idea using characteristic functions. The proof typically involves characteristic functions and Lévy's continuity theorem.

(i) Let $Y_i = (X_i - \mu)/\sigma$. Then $\mathbb{E}[Y_i] = 0$ and $\operatorname{var}[Y_i] = 1$. The standardized sum can be written as $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$.

(ii) The characteristic function of Z_n is $\phi_{Z_n}(t) = \mathbb{E}[e^{itZ_n}] = \mathbb{E}\left[\exp\left(\frac{it}{\sqrt{n}} \sum Y_i\right)\right]$. Since Y_i are iid, this becomes $\phi_{Z_n}(t) = \left(\phi_{Y_1}\left(\frac{t}{\sqrt{n}}\right)\right)^n$.

(iii) The Taylor expansion of $\phi_{Y_1}(s)$ around $s = 0$ is $\phi_{Y_1}(s) = \phi_{Y_1}(0) + s\phi'_{Y_1}(0) + \frac{s^2}{2}\phi''_{Y_1}(0) + o(s^2)$. We know $\phi_{Y_1}(0) = 1$, $\phi'_{Y_1}(0) = i\,\mathbb{E}[Y_1] = 0$, and $\phi''_{Y_1}(0) = i^2\,\mathbb{E}[Y_1^2] = -\operatorname{var}[Y_1] = -1$. So, $\phi_{Y_1}(s) = 1 - \frac{s^2}{2} + o(s^2)$.

(iv) Substituting $s = t/\sqrt{n}$ yields $\phi_{Y_1}\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)$.

(v) Therefore, $\phi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n$.

(vi) As $n \to \infty$, this expression converges to $e^{-t^2/2}$, using the limit $(1+x/n)^n \to e^x$. We then note that $e^{-t^2/2}$ is the characteristic function of a standard normal distribution $\mathcal{N}(0,1)$.

(vii) By Lévy's continuity theorem, since the cfs converge to the cf of a $\mathcal{N}(0,1)$ distribution, $Z_n \xrightarrow{d} \mathcal{N}(0,1)$.

□

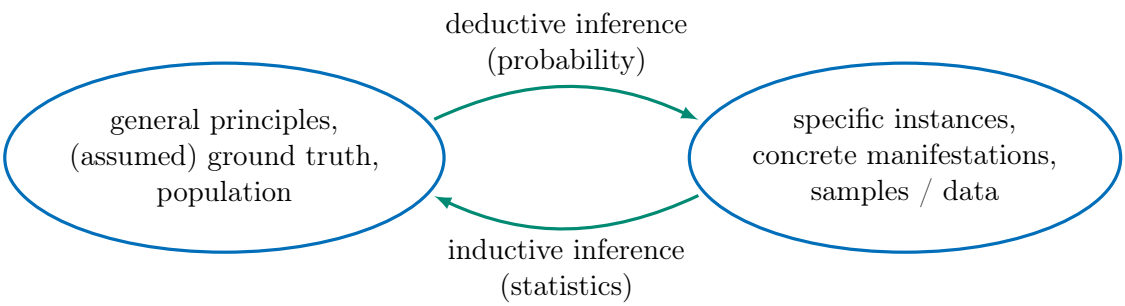
Remark 2.65 (significance of the clt).

- The central limit theorem is one of the most remarkable results in probability theory. The sum (or average) of a large number of iid random variables will have a distribution that is approximately normal, regardless of the underlying distribution of the individual variables (as long as mean and variance are finite). This is why the normal distribution appears so frequently in nature and in statistical applications.
- The clt implies that
$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)\,.$$
It is often also interpreted as
$$S_n \approx \mathcal{N}(n\mu, n\sigma^2)\,,$$
or equivalently that
$$\bar{X}_n \approx \mathcal{N}\Big(\mu, \frac{\sigma^2}{n}\Big)\,,$$
where we read $\approx$ as "approximately distributed like." That is, for large n, the sample sum is approximately distributed like a Gaussian with mean $n\mu$ and variance $n\sigma^2$ and the sample mean is approximately distributed like a Gaussian with mean μ and variance σ^2/n. Sometimes you find the clt sloppily written as $\bar{X}_n \xrightarrow{d} \mathcal{N}(\mu, \sigma^2/n)$. Note that this is mathematically incorrect, as convergence in distribution is only defined for a fixed target (that is, the right hand side must not depend on n). Also, in the limit $n \to \infty$, $\mathcal{N}(\mu, \sigma^2/n)$ becomes degenerate with a point mass on μ (which is formalized by the llns).

- These approximations are the foundation for many statistical inference procedures, such as constructing confidence intervals and hypothesis tests for means. For example, when you see the standard error reported as uncertainty intervals in plots, one is typically implicitly invoking the clt to assume that the distribution of whatever is plotted is Gaussian and shows the scaled standard deviation as a meaningful measure of the spread of this Gaussian. The $\sqrt{n}$ rate of convergence is an important key insight and statisticians are obsessed with "$\sqrt{n}$-rates" in all kinds of tasks.
- The version of the clt stated in theorem 2.64 assumes the random variables are iid. However, the clt holds under much more general conditions. For instance, the Lindeberg-Feller clt applies to sums of independent (but not necessarily identically distributed) random variables, provided that no single variable contributes too much to the total variance (formalized by the Lindeberg condition). There are also versions for certain types of dependent random variables (e.g., martingales, mixing sequences). This broad applicability underscores the fundamental nature of the normal distribution.
- The clt is often invoked as a blanket justification whenever you'd like to assume something to be Gaussian (which is then often convenient for computational purposes). Want you noise to be Gaussian? "There is a large number of unmeasured additive influences on our outcome variable whose sum, mumble mumble, clt, mumble, mumble... So it is reasonable to assume Gaussian noise."
- There are lots of nice visualizations out there to get an intuition for the central limit theorem. Besides the one on moodle, a quick first search took me here, here, or here.

3. Statistical inference

Overview 3.1 (from probability to statistics). In the previous chapters, we assumed knowledge of the underlying probability distribution (e.g., $X \sim \mathcal{N}(\mu, \sigma^2)$ with known μ, σ^2) and aimed to calculate probabilities of events or properties of random variables. This is often called *deductive inference*, moving from general principles (the model) to specific outcomes (data probabilities). Statistical inference, in contrast, typically involves *inductive inference*: we observe data (samples) and want to make statements or draw conclusions about the underlying (often unknown) probability distribution or its parameters that generated the data. We aim to learn from data.



Example 3.2 (what we want from statistical inference).

(i) Suppose we observe a sequence of n coin flips $x_1, \ldots, x_n \in \{0, 1\}$. Some tasks/questions/queries we may have (and statistical inference can solve) include:

a) What is the probability p of heads for this coin?

b) Can we say (with confidence?) whether the coin is fair ($p = 0.5$)? That is, can we test the hypothesis that the coin is fair against the alternative that it is biased?

c) What's a plausible range of values for p and how confident are we in that range?

(ii) An NVMe hard drive producer closes a contract with Apple in which they pledge that at most 1 out of 10,000 drives will fail within the first year. Apple orders 12 million of these drives for its new MacBook Pro model. It tracks 112,345 MacBooks sold within the first months and receives 15 complaints about failed hard drives (within one year of shipment). Can they sue the hard drive producer?

(iii) A bottling machine for 0.33 l bottles is certified by the vendor to be precise up to ± 2 ml. After selling 345,234 bottles of beer, you realize that you have brewed at least 500 liters more than you supposedly sold. Is the bottling machine a major suspect? Would measuring the exact amount in the next 100 filled bottles help?

(iv) Say someone decides to resolve Bertrand's paradox by drawing a circle on a table and dropping Mikado sticks "randomly" onto the circle 1 thousand times to get random chords. They find that 281 are in the favorable region whereas 719 are not. Does that allow us to say whether $P_1 = \frac{1}{3}$, $P_2 = \frac{1}{2}$, or $P_3 = \frac{1}{4}$ is "correct?"

(v) In a clinical study 46 participants receive a newly developed drug, whereas 234 patients in the control group receive the conventional treatment. In the treatment group 42 out of 46 survive, whereas 201 out of 234 survived in the control group. Should the new drug replace the conventional treatment?

(vi) In a survey conducted on a Thursday morning at Marienplatz, in which 239 people where interviewed, CSU had an approval rate of 79%. Can we conclude that they will obtain at least 2/3 of votes in the Bavarian state election (Landtagwahl) the following Sunday?

(vii) You measure CPU execution times for two different algorithms on a set of benchmark tasks on your laptop. Can we determine whether one algorithm is "significantly" faster than the other, or estimate the average difference in execution time.

(viii) The paper that presented the discovery of the Higgs boson states in its abstract:

> "Clear evidence for the production of a neutral boson with a measured mass of $126.0 \pm 0.4(\text{stat}) \pm 0.4(\text{sys})$ GeV is presented. This observation, which has a significance of 5.9 standard deviations, corresponding to a background fluctuation probability of 1.7×10^{-9}, is compatible with the production and decay of the Standard Model Higgs boson."

Have we discovered the Higgs boson or not?

(ix) Similarly, the paper announcing the first gravitational wave observation from LIGO states in the abstract (emphasis mine):

> "The signal was observed with a matched-filter signal-to-noise ratio of 24 and a false alarm rate estimated to be less than 1 event per 203 000 years, equivalent to a significance greater than 5.1σ. The source lies at a luminosity distance of 410^{+160}_{-180} Mpc corresponding to a redshift $z = 0.09^{+0.03}_{-0.04}$. In the source frame, the initial black hole masses are $36^{+5}_{-4}\,\mathrm{M}_\odot$ and $29^{+4}_{-4}\,\mathrm{M}_\odot$ [...] *All uncertainties define 90% credible intervals.*"

What does all that mean?

(x) As a data scientist within Amazon, you run a sentiment analysis of (verified) reviews of office equipment items. You find that reviews of buyers who self identify as male are on average 0.21% more critical compared to buyers who identify as female. Further tests declare the difference to be statistically significant. What recommendation do you give your manager?

Unfortunately, we will not be able to answer all of these questions fully. The goal of this chapter is to provide first glimpses and pointers into how statistical inference generally approaches such questions.

Overview 3.3 (how to approach this). Trying to tackle the questions in example 3.2 highlights the following aspects.

(i) We will still have to make some modeling assumptions. First and foremost, we have to decide what *not* to model, and what we deem possible. In particular, we will have to start with a "collection of possibilities." Concretely, we need a *statistical model*, which is a collection of distributions we deem possible so we can then decide how likely each of them is the ground truth (which is most likely?) given the data. We typically implicitly assume that the actual ground truth is within this collection (even though that itself is unlikely).

(ii) In reality, the statistical model (the set of possible truths) is often made up from many components and moving parts. Think of all the ways in which "noise" or "randomness" may enter the final observation of the Higgs particle. This is often not shown or visible in introductory examples.

(iii) The quantity (quantities) of interest may be "closer" or "further" from the underlying probability model. For example, the probability of heads p in the coin flip experiment is literally the one key parameter describing the underlying distribution. In contrast, the measured sentiment of written reviews is a complicated construction on top of the actual (random?) process of people expressing their impression and sentiment about products.

(iv) It feels "wrong" to provide final answers to most questions. Instead, one is almost forced to express oneself in "levels of confidence" rather than "certainties."

(v) The actual numbers involved play a big role.

3.1. Statistical models and basic concepts

Section goal. Introduce the fundamental concepts of statistical inference, including statistical models, statistics, parameters of interest, and parameter identifiability.

Definition 3.4 (statistical model). Let $(\Omega, \mathcal{A})$ be a measurable *sample space* and Θ a non-empty index set, which we call the *parameter space*. For a set of probability distributions $P_\Theta := (P_\theta)_{\theta\in\Theta}$ on $(\Omega, \mathcal{A})$, we call $(\Omega, \mathcal{A}, P_\Theta)$ a *statistical model*. Often we just call P_Θ the statistical model when the underlying sample space (and σ-algebra) are clear.

We call a statistical model *well-specified*, if there exists a $\theta \in \Theta$ such that P_θ is the true distribution of a random experiment and we then call θ *true parameter*. That is, ground truth is within your specified space of possibilities. Otherwise, it is called *mis-specified*.

If $\Theta \subseteq \mathbb{R}^k$ for some finite k, the model is called a *parametric model*. Otherwise, the model is called *non-parametric*. You will find different explicit definitions of non-parametric models in the literature. A common one is that a model is called non-parametric whenever "Θ is infinite-dimensional" (e.g., the set of all continuous distributions). Broadly speaking, non-parametric models are thought of as putting very little to no restrictions on the set of considered distributions. In this chapter, we primarily focus on parametric models.

Whenever for a random variable X we have $X \sim P_\theta$ for some $\theta \in \Theta$, we say that X *follows* or *is distributed according to* $(\Omega, \mathcal{A}, P_\Theta)$. We then write (in abuse of notation) $X \sim P_\Theta$.

Example 3.5 (parametric models). Relating to some of the settings in example 3.2:

(i) For the coin flips a reasonable statistical model is

$$(\Omega = \{0,1\}, \mathcal{A} = \mathcal{P}(\{0,1\}), (\mathrm{Ber}(p))_{p\in[0,1]}) ,$$

that is, the parameter space is $\Theta = [0,1]$ (and we denote $p = \theta$).

(ii) If we set one for female and zero for male, we can describe the random gender of a consumer with a Bernoulli distribution model: $(\{0,1\}, \mathcal{P}(\{0,1\}), (\mathrm{Ber}(p))_{p\in[0,1]})$.

(iii) Assume we buy many bags of 50 screws each. We can model the number of defective screws in any bag with a binomial distribution model

$$(\{0,\dots,50\}, \mathcal{P}(\{0,\dots,50\}), (\mathrm{Bin}(50,p))_{p\in[0,1]}) .$$

(iv) You have recorded the number of phone calls coming in at a call center for 20 consecutive 15-minute intervals. You could model these numbers by a Poisson distribution model

$$(\mathbb{N}, \mathcal{P}(\mathbb{N}), (\mathrm{Poi}(\lambda))_{\lambda\in\mathbb{R}_{>0}}) .$$

(v) On your way to work, you track the time you're waiting for the subway for every day for one year. The waiting time can be modeled with a uniform distribution model

$$(\mathbb{R}, \mathcal{B}, (\mathrm{Unif}(0,\theta))_{\theta\in\mathbb{R}_{>0}}) .$$

(vi) The university measures the lifetime of of 2000 newly installed light bulbs over 5 years. The lifetime can be modeled by an exponential distribution model

$$(\mathbb{R}, \mathcal{B}, (\mathrm{Exp}(\lambda))_{\lambda \in \mathbb{R}_{>0}}) .$$

(vii) For the bottling machine, we could interpret the ± 2 ml as strict bounds, resulting, for example, in the statistical model

$$(\Omega = [328, 332], \mathcal{B}|_\Omega, \mathrm{Unif}([328, 332]))$$

for which $|\Theta| = 1$, i.e., we're only allowing one possibility anyway. In this case, there is nothing to be inferred about the underlying distribution and most interesting questions we can ask go back to "probability theory," e.g., what is the probability of overfilling by 500 liters within n bottles?

Instead, we could also interpret ± 2 ml as follows: "deviations are (mumble, mumble clt) normally distributed and remain within 330 ± 2 ml in most cases." We could then assume

$$(\Omega = \mathbb{R}, \mathcal{B}, (\mathcal{N}(330, \sigma^2))_{\sigma^2 \in \mathbb{R}_{>0}})$$

and try to infer from our measurements what variance σ^2 actually is (or in which range it likely lies) and how that relates to the stated ± 2 ml. Here, the parameter space is the space of possible variances $\Theta = \mathbb{R}_{>0}$.

This model assumes the average filling amount is actually exactly 330 ml. Instead, we could also work with the statistical model

$$(\Omega = \mathbb{R}, \mathcal{B}, (\mathcal{N}(\mu, \sigma^2))_{(\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_{>0}}) ,$$

where the parameter space $\Theta = \mathbb{R} \times \mathbb{R}_{>0}$ contains both the mean and the variance.

Clearly, both models are "wrong," in that they would allow for negative amounts or amounts like 1 liter, which definitely do not fit into the bottle. Yet, they may be useful.

Can you argue for and against the case that they are well-specified?

(viii) Modern image generation methods often rely on "diffusion," a process in which a high-dimensional standard Gaussian distribution $\mathcal{N}(0, \mathbf{1}_{n \times n})$ for some large n (equaling 3 times the number of pixels for the three colors RGB), is transformed through a complex neural net with weights (parameters) $\theta \in \mathbb{R}^d$ for d being in the millions or even billions. The final "image distribution" is then given via (repeated) application of the change of variable formula for densities.

So in essence, we model images as $f_\theta(X)$ for $X \in \mathcal{N}(0, \mathbf{1}_{n \times n})$, where f_θ is the function that represents the (repeated) application of the neural net. Hence, we have the statistical model

$$(\Omega = \mathbb{R}^{n \times n}, \mathcal{B}^{n \times n}, (P_{f_\theta(X)})_{\theta \in \mathbb{R}^d}) .$$

Training a diffusion model to generate realistic images can then be viewed as inferring the parameters $\theta \in \Theta = \mathbb{R}^d$ that "best explain the observed images in the training set." (In practice that amounts to minimizing a loss, like the "mean squared error.")

Personally, I do not think that this view justifies the statement that "machine learning/artificial intelligence is just statistics on steroids," but one can see where it is coming from. Also refer to the following remark on what "inference" refers to in statistics versus machine learning.

(ix) Models can be constrained and defined in various ways. For example, in graphical models, we consider all distributions that have certain properties with respect to a given graph. In latent variable models, one typically specifies a latent distribution and a conditional of how the observed/state variables depend on the latents. One then gets the allowed distributions over the observed variables by marginalizing out the latents. There are also autoregressive, hierarchical, generalized linear, and many other models, that ultimately provide a certain set of distributions over the observed variables by defining or constraining different types of mechanisms and assumptions.

Remark ("inference" in stats vs. ml). In statistics, *inference* refers to the process of learning about unknown quantities—like parameters/properties of distributions—based on observed data, typically through estimation and uncertainty quantification. The "inference targets" are properties of the population distribution. In machine learning, *inference* often refers to the application of a trained model to new inputs, i.e., to producing new predictions. The inference target The "inference targets" are predictions on new inputs or newly generated outputs (images, text, etc) for generative models. You will often encounter terms like "inference time" (used synonymously for "test time" or "deployment time") and "inference cost" (the cost of making new predictions after training has finished).

This leads to confusion because what statisticians call *inference* amounts to "fitting the model" and "finding the best parameters." In machine learning, fitting or "finding the best weights" is called *training*. What machine learning calls *inference* corresponds to what statistics would call *prediction* (which machine learning also uses). The term *inference* is thus used in different stages of the modeling pipeline in each field.

This highlights a key difference between the two: statistics aims at inferring properties of an unknown ground truth whereas machine learning aims at inferring predictions (on novel inputs).

Definition 3.6 (statistic). Let $(\Omega, \mathcal{A}, (P_\theta)_{\theta\in\Theta})$ be a statistical model and let $(X_1, \ldots, X_n)$ be a random *sample*, i.e., a random vector on the product space $(\Omega^n, \otimes_{i=1}^n \mathcal{A})$. A *statistic* is a measurable function $T : \Omega^n \to \mathbb{R}^k$ for some $k \in \mathbb{N}_{>0}$ (that does not depend on the unknown parameter θ). Its value, $t = T(x_1, \ldots, x_n)$, is computed from a realized data sample $x = (x_1, \ldots, x_n)$. The key property of a statistic is that it is an observable quantity once the data is collected.

Example 3.7 (statistics). For a sample $X_1, \dots, X_n$ of RVRVs,

(i) the *sample mean* $\bar{X}_n := \frac{1}{n}\sum_{i=1}^n X_i$ is a statistic,

(ii) the *sample variance* $S_n^2 := \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ is a statistic,

(iii) the maximum value $X_{(n)} := \max(X_1, \dots, X_n)$ is a statistic,

(iv) the function $T(X_1, \dots, X_n) = \pi$ or $T(X_1, \dots, X_n) = X_6$ are also statistics.

Remark 3.8 (the n is implicit). A given statistic is only defined for a fixed n. Typically, the number of samples n stays variable and we assume that we have a given statistic for any n. This is straight forward for examples (i)-(iii). We then often just write T or T_n to make the n-dependence more explicit. However, example (iv) shows that for a given statistic there generally need not be "a version for any n." In practice, we only consider statistics where there is "a natural version for any n."

Definition 3.9 (parameter of interest). Often, we are not interested in the entire parameter $\theta \in \Theta$, but in a specific function of it. A *parameter of interest* is a function $\gamma : \Theta \to \mathbb{R}^k$. The goal of statistical inference is then to make statements about the value of $\gamma(\theta)$.

Example 3.10 (parameter of interest in Gaussians). In a Gaussian model

$$\left(\mathbb{R}, \mathcal{B}, (\mathcal{N}(\mu, \sigma^2))_{\mu\in\mathbb{R}, \sigma^2\in\mathbb{R}_{>0}}\right),$$

where $\theta = (\theta_1, \theta_2) = (\mu, \sigma^2)$, we might only be interested in the mean μ, so $\gamma(\theta) = \theta_1 = \mu$ or only in the variance $\gamma(\theta) = \theta_2 = \sigma^2$ or in the *coefficient of variation* given by $\gamma(\theta) = \sqrt{(\theta_2)}/\theta_1 = \sigma/\mu$. We will primarily consider $\gamma = \mathrm{id}$, i.e., our parameter of interest is all of θ.

Definition 3.11 (identifiability). A parameter of interest $\gamma(\theta)$ (which could also be the full parameter θ) is *identifiable* in a model $(\Omega, \mathcal{A}, P_\Theta)$ if $\gamma(\theta_1) \neq \gamma(\theta_2)$ implies $P_{\theta_1} \neq P_{\theta_2}$ for all $\theta_1, \theta_2 \in \Theta$. Equivalently, if $P_{\theta_1} = P_{\theta_2}$ implies $\gamma(\theta_1) = \gamma(\theta_2)$. If a parameter of interest is not identifiable, it is impossible to uniquely determine its value even with an infinite amount of data and we call it *unidentifiable*.

Example 3.12 (identifiability in Gaussians). Consider a RVRV $X = A + B$, where $A \sim \mathcal{N}(\mu_A, 1)$ and $B \sim \mathcal{N}(\mu_B, 1)$ are independent. Then X follows a statistical model with $P_\Theta = (\mathcal{N}(\theta_1 + \theta_2, 2))_{\theta\in\mathbb{R}^2}$ for $\theta = (\mu_A, \mu_B)$. Here, the parameter θ is not identifiable, because, for example, $(1, 2)$ and $(2, 1)$ yield the same distribution for X. However, the mean of X as parameter of interest $\gamma(\theta) = \mu_A + \mu_B$ is identifiable.

Definition 3.13 (likelihood function). Let $(X_1, \dots, X_n)$ be a random sample from a statistical model $(\Omega, \mathcal{A}, P_\Theta)$ where all $P_\theta \in P_\Theta$ have a pmf or density $p_\theta(\cdot) =: p(\cdot \mid \theta)$. Given observed data $x = (x_1, \dots, x_n)$, the *likelihood (function)* is the following function

$$L(\cdot \mid x) : \Theta \to \mathbb{R}_{\geq 0} : \theta \mapsto L(\theta \mid x) := p_\theta(x) = p(x \mid \theta) = p(x_1, \dots, x_n \mid \theta).$$

If $X_1, \dots, X_n$ are iid, then $L(\theta \mid x) = \prod_{i=1}^n p(x_i \mid \theta)$. The *log-likelihood function* is $\ell(\theta \mid x) := \ln(L(\theta \mid x))$.

Remark 3.14 (likelihood function interpretation).

(i) The likelihood function measures how "likely" a parameter value θ is to have produced the observed data x. Remember that it is a function of θ (not x) and it is specifically defined only for a given set of observations. It does not exist without reference to data, specifically, it is not a property only of the statistical model.

(ii) Maximizing $L(\theta \mid x)$ over θ, amounts to "finding the model (parameter) that is most likely to have generated the data x."

(iii) The likelihood function is *not* a probability density for θ.

(iv) It is one possible connection to go from "given the probability distribution, what can we say about possible observations" (which is probability theory) to "given observed data, what can we say about the underlying probability distribution (parameter)" (which is statistical inference).

(v) The newly introduced (and in principle unnecessary) notation $p(x \mid \theta)$ as well as $L(\theta \mid x)$ or $\ell(\theta \mid x)$ highlights that we are "conditioning" on the parameters or the data, respectively, which we consider fixed. That is, it just highlights "what is fixed and what varies."

(vi) Especially for iid RVs the log-likelihood is useful as it turns products into sums and constant pre-factors into constant summands. Since the logarithm is strictly monotonic, minimizing $L(\theta \mid x)$ over θ is the same as minimizing $\ell(\theta \mid x)$, where the latter is often more convenient.

Example 3.15 (likelihood for a normal distribution). Let $X_1, \dots, X_n$ be iid $\mathcal{N}(\mu, \sigma^2)$. The pdf for a single observation is $p(x_i \mid \mu, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp(-\frac{(x_i-\mu)^2}{2\sigma^2})$. The likelihood is

$$L(\mu, \sigma^2 \mid x) = \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} \exp\Big(-\frac{(x_i-\mu)^2}{2\sigma^2}\Big) = (2\pi\sigma^2)^{-n/2} \exp\Big(-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i-\mu)^2\Big) .$$

The log-likelihood is

$$\ell(\mu, \sigma^2 \mid x) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^n (x_i-\mu)^2 .$$

Remark 3.16 (revisiting implicit ns). In our definition of a statistical model, there is no n, but statistics is all about inference from n observations $x := (x_1, \dots, x_n)$ for some $n \in \mathbb{N}_{>0}$.

In practice, we primarily consider n iid RVRVs $X_1, \dots, X_n$. One possible viewpoint is to consider each X_i to (independently) follow a fixed statistical model $X_i \sim (\mathbb{R}, \mathcal{B}, P_\Theta)$

and they all have the same true parameter $\theta \in \Theta$. Another possibility is to let $X = (X_1, \ldots, X_n)$ follow a statistical model $(\mathbb{R}^n, \mathcal{B}^n, P_\Theta)$.

The latter approach is in various ways more general (need not assume iid), but the former is often more intuitive when we are actually talking about iid samples (which we will almost all the time). Hence, we opted for the former approach. Don't be surprised when you see $X \sim P_\Theta$ though where X already consists of n components.

3.2. Point estimation

Section goal. One common goal of statistical inference is to provide a single "best guess" for an unknown parameter of interest based on the observed data. This guess is called a point estimate.

Convention 3.17 (RV statistical model). Throughout this section, let $(\Omega, \mathcal{A}, P_\Theta)$ be a parametric statistical model and let $n \in \mathbb{N}_{>0}$. Following common notation, we typically assume that θ is the parameter of interest. However, all statements in the following also hold when replacing θ with a parameter of interest γ derived from θ. Finally, we assume that θ (γ) is identifiable in $(\Omega, \mathcal{A}, P_\Theta)$.

Definition 3.18 (estimator and estimate). An *estimator* for a parameter of interest γ is a statistic $\hat{\gamma} = T(X_1, \ldots, X_n)$. An *estimate* is the specific value $\hat{\gamma}_{\text{obs}} := \hat{\gamma}(x) = T(x_1, \ldots, x_n)$ obtained when the estimator is applied to a particular realization x of the data. As before, we assume this statistic to exist for any n, and then sometimes write $\hat{\gamma}_n$ to make the dependence on n explicit. Again, the full parameter is a valid parameter of interest $\gamma(\theta) = \theta$ and we then write $\hat{\theta}$.

Example 3.19 (can I win?). At the yearly CIT graduation celebration you can buy a lottery ticket to win a TUM branded hoodie. There is a total of 10 winning tickets. To get a feeling for your chances of winning, you're interested in the total number of tickets $N \in \mathbb{N}_{>0}$. Let's buy a ticket and try to estimate it.

Since you draw the ticket yourself, assume the ticket number is Laplace distributed $X \sim (\mathbb{N}_{>0}, \mathcal{P}(\mathbb{N}_{>0}), (\text{Lap}_{[N]})_{N \in \mathbb{N}})$. Here, the parameter space is $\Theta = \mathbb{N}$. You draw the number $x \in \mathbb{N}_{>0}$ (here $n = 1$, so $x = x_1$). What should be our guess for N? One possible estimator for N is $\hat{N}(X) = X$, i.e., just the number you've drawn. It may seem intuitively unlikely that you actually drew the ticket with the largest number, so you may feel more comfortable with the estimator $\hat{N}(X) = X + 15$ or maybe $\hat{N}(X) = \text{round}(X \cdot 1.2)$. Maybe instead you expect to have drawn the number in the middle, so another possible choice is $\hat{N}(X) = 2X$, twice the number you've drawn. These are all valid estimators.

To get a better feeling, you buy two more tickets, so you now have the data/observations $x = (x_1, x_2, x_3)$ (assume N is so large that these are still essentially independent draws from $\text{Lap}_{[N]}$). Is $\hat{N}(x) = \max\{x_1, x_2, x_3\}$ a reasonable estimate for N? What about

$\hat{N}(x) = \max\{x_1, x_2, x_3\} +$ empirical variance? What about $\hat{N}(x) = 1000$? (Again, these are all valid estimates (plugging in the actual values) or estimators (plugging in the RVs).)

We better find ways of deciding whether estimators are "good" or "bad."

Definition 3.20 (properties of estimators). Let $\hat{\theta}$ be an estimator for the parameter θ. The quality of this estimator can be assessed according to various properties.

(i) **Bias:** The *bias* of an estimator $\hat{\theta}$ is $\text{bias}[\hat{\theta}] = \mathbb{E}[\hat{\theta}] - \theta$. An estimator is *unbiased* if $\text{bias}[\hat{\theta}] = 0$, i.e., $\mathbb{E}[\hat{\theta}] = \theta$ for all $\theta \in \Theta$. (What is the expectation over?)

(ii) **Consistency:** An estimator $\hat{\theta}_n$ is *consistent* if $\hat{\theta}_n \xrightarrow{P} \theta$. It is *strongly consistent* if $\hat{\theta}_n \xrightarrow{a.s.} \theta$.

(iii) **Variance:** The variance of an estimator is $\text{var}[\hat{\theta}] = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$. For multi-variate θ, this turns into $\text{cov}[\hat{\theta}, \hat{\theta}]$.

(iv) **Mean squared error (mse):** The *mean squared error* of an estimator $\hat{\theta}$ is

$$\text{mse}[\hat{\theta}] = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{var}[\hat{\theta}] + (\text{bias}[\hat{\theta}])^2 \,.$$

This equality is known as the *bias-variance decomposition*. This is often taken to mean that there is an inherent tradeoff (bias-variance-tradeoff) between bias and variance. The argument goes as follows. Simple estimators/models (few parameters) often have large bias—they simply cannot fully capture θ. In other words, they "underfit the data." While complex estimators/models (many parameters) can bring down the bias, they increase the variance—they "move more to capture the observations." In other words, they "overfit the data." This argument, while plausible and nicely visualized in polynomial regression, does not necessarily hold and thus does not imply the presence of a necessary tradeoff. Modern ML has shown that even extremely complex overparameterized models can result in low variance. Once can reduce both, bias and variance, and thus the mse simultaneously.

In practice, especially in machine learning, one seeks an estimator that minimizes the mse.

For multi-variate θ the mse becomes

$$\text{mse}[\hat{\theta}] = \mathbb{E}[\|\hat{\theta} - \theta\|_2^2] \,.$$

(v) **Asymptotic normality:** An estimator $\hat{\theta}_n$ is *asymptotically normal* if its distribution, when properly centered and scaled, converges to a normal distribution. Formally, if there exists a quantity $V(\theta)$ such that

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, V(\theta)) \,.$$

The term $V(\theta)$ is the *asymptotic variance*. This property is fundamental for constructing approximate confidence intervals and hypothesis tests for large sample sizes, as it implies $\hat{\theta}_n$ approximately follows $\mathcal{N}(\theta, V(\theta)/n)$.

(vi) **Efficiency:** Efficiency compares the variance of unbiased estimators. An estimator is a *minimum variance unbiased estimator (mvue)* if it achieves the lowest possible variance among all unbiased estimators.

Example 3.21 (properties of sample mean). Let $X_1, \dots, X_n$ be iid RVRVs with mean μ and finite variance $\sigma^2 > 0$. More precisely, assume $X_1, \dots, X_n \sim P_\theta$ iid, where $\theta = (\mu, \sigma^2, \dots) \in \Theta$ is the true parameter in our statistical model and μ is the mean of P_θ and σ^2 is the variance of P_θ. Note that neither P_θ (nor any other distribution in our model) need to be Gaussian. There could be more parameters in θ describing other properties of the distribution P_θ as well.

Consider the estimator $\hat{\mu} = \bar{X}_n = \frac{1}{n}\sum X_i$ for the parameter of interest $\gamma(\theta) = \mu$.

(i) The sample mean $\bar{X}_n$ is an unbiased estimator of μ, $\mathbb{E}[\bar{X}_n] = \mu$.

(ii) By the wlln and slln, $\bar{X}_n \xrightarrow{P} \mu$ and $\bar{X}_n \xrightarrow{a.s.} \mu$. So $\bar{X}_n$ is strongly consistent.

(iii) The variance decreases in n as $\mathrm{var}[\bar{X}_n] = \sigma^2/n$. Note that this is also what we would expect from the clt.

(iv) Since $\hat{\mu}$ is unbiased, $\mathrm{mse}[\bar{X}_n] = \mathrm{var}[\bar{X}_n] = \sigma^2/n$.

(v) By the clt, the sample mean $\bar{X}_n$ is an asymptotically normal estimator with asymptotic variance $V(\theta) = \sigma^2$.

(vi) The sample mean $\bar{X}_n$ is efficient for Gaussian models, but need not be efficient for other distributions.

Example 3.22 (sample variance estimator). Let $X_1, \dots, X_n$ be iid RVRVs with mean μ and variance σ^2. The *sample variance* $S_n^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ is an unbiased estimator of σ^2, i.e., $\mathbb{E}[S_n^2] = \sigma^2$. The division by $n-1$ is known as *Bessel's correction*. The "naive" estimator $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ is biased, as $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2 < \sigma^2$. Both are consistent estimators of σ^2.

Example 3.23 (biased/consistent estimators). Let $X_1, \dots, X_n$ be iid RVRVs with mean μ and finite variance $\sigma^2 > 0$.

- Biased but consistent: Consider the estimator $\hat{\mu}_n = \bar{X}_n + \frac{1}{n}$, where $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. Then $\mathbb{E}[\hat{\mu}_n] = \mathbb{E}[\bar{X}_n + \frac{1}{n}] = \mu + \frac{1}{n}$, i.e., $\hat{\mu}_n$ is biased. By the WLLN (theorem 2.61), $\bar{X}_n \xrightarrow{P} \mu$. The term $\frac{1}{n} \to 0$ as $n \to \infty$ and thus $\hat{\mu}_n$ is consistent.
- Unbiased but not consistent: Now consider the estimator $\tilde{\mu}_n = X_1$. Then $\mathbb{E}[\tilde{\mu}_n] = \mathbb{E}[X_1] = \mu$ and $\tilde{\mu}_n$ is unbiased. However, since $P(|\tilde{\mu}_n - \mu| \geq \epsilon) = P(|X_1 - \mu| \geq \epsilon)$ and $\mathrm{var}[X_1] = \sigma^2 > 0$, X_1 is not a constant equal to μ (almost surely). Therefore, $\tilde{\mu}_n$ is not consistent.

Example 3.24 (mvue for a uniform distribution). Consider an iid sample $X_1, \dots, X_n$ from the model $(\mathbb{R}, \mathcal{B}, (\mathrm{Unif}(0,\theta))_{\theta>0})$, i.e., $\Theta = \mathbb{R}_{>0}$. This is the model from part example 3.5(v) where we aim to infer the maximum waiting time for the subway. We can now phrase this formally as constructing an estimator $\hat{\theta}$ for θ that ideally has "good properties."

(i) **Naive sample mean:** We already know that $\bar{X}_n$ is an unbiased, strongly consistent, asymptotically normal estimator for $\mathbb{E}[X_i] = \theta/2$, so (by linearity of expectation) $\hat{\theta}_1 := 2\bar{X}_n$ has the same properties as an estimator for θ. Its variance is

$$\mathrm{var}[\hat{\theta}_1] = \mathrm{var}[2\bar{X}_n] = 4\,\mathrm{var}[\bar{X}_n] = 4\frac{\mathrm{var}[X_1]}{n} = \frac{4}{n}\frac{\theta^2}{12} = \frac{\theta^2}{3n}\,.$$

(ii) **A "better" estimator:** In the exercises we'll see that the "maximum likelihood estimator" (see definition 3.27) $\hat{\theta}_{\mathrm{mle}} = X_{(n)} = \max\{X_1, \dots, X_n\}$ has $\mathbb{E}[X_{(n)}] = \frac{n}{n+1}\theta$, i.e., $\hat{\theta}_{\mathrm{mle}}$ is biased. By rescaling, we obtain $\hat{\theta}_2 := \frac{n+1}{n}X_{(n)}$ as an unbiased estimator. To find $\mathrm{var}[\hat{\theta}_2]$, we first need $\mathrm{var}[X_{(n)}]$. Using the pdf found in the exercises, we compute

$$\mathbb{E}[X_{(n)}^2] = \int_0^\theta y^2 \frac{ny^{n-1}}{\theta^n}\,dy = \frac{n}{\theta^n}\int_0^\theta y^{n+1}\,dy = \frac{n}{n+2}\theta^2\,.$$

So, $\mathrm{var}[X_{(n)}] = \mathbb{E}[X_{(n)}^2] - (\mathbb{E}[X_{(n)}])^2 = \frac{n}{n+2}\theta^2 - \left(\frac{n}{n+1}\theta\right)^2 = \frac{n\theta^2}{(n+2)(n+1)^2}$ and thus

$$\mathrm{var}[\hat{\theta}_2] = \frac{\theta^2}{n(n+2)}\,.$$

Since for any $n > 1$, we have $n(n+2) > 3n$, $\mathrm{var}[\hat{\theta}_2] < \mathrm{var}[\hat{\theta}_1]$. Hence, the estimator $\hat{\theta}_1 = 2\bar{X}_n$ is unbiased but not the mvue. We also say that $\hat{\theta}_2$ is more efficient than $\hat{\theta}_1$. One can show that $\hat{\theta}_2$ is actually the unique mvue for θ. The sample mean is often "good," but sometimes not the "best."

Example 3.25 (James-Stein estimator). Let $X \sim \mathcal{N}(\mu, I_d)$ be a single observation from a d-dimensional standard normal distribution, where we wish to estimate the unknown mean $\mu \in \mathbb{R}^d$. The standard (unbiased) sample mean estimator $\hat{\mu}(X) = X$ (which is also the mle) has mse

$$\mathrm{mse}[\hat{\mu}] = \mathbb{E}[\|X - \mu\|^2] = \ldots = d\,.$$

For dimensions $d \geq 3$, the biased *James-Stein shrinkage estimator* is defined by

$$\hat{\mu}_{\mathrm{js}} = \left(1 - \frac{d-2}{\|X\|^2}\right)X\,.$$

It shrinks the observation X towards the origin. The remarkable result is that it has uniformly lower risk than the mle: $\mathrm{mse}[\hat{\mu}_{\mathrm{js}}] < d$ for all $\mu \in \mathbb{R}^d$.

This is paradoxical: regardless of where the mean of the Gaussian actually is, we improve our estimate by shrinking towards the origin. One can also show that shrinking in any

other direction improves the estimate as well—even more surprising. Please see the wiki page for more details on why this makes sense.

Remark 3.26 (constructing estimators). So far, it seemed like we just need to manually "come up with" estimators and then check which properties they satisfy to decide whether they're good or bad. However, there are also general procedures to construct estimators with certain properties and there there is a large literature in statistics on when and how one can obtain mvues or blues (best *linear* unbiased estimators) etc. So this appears more like "guesswork" than it often is in practice. However, finding "better" estimators for very specific settings is still actively pursued by researchers in statistics. We'll now see one general procedure to construct estimators with certain properties.

Definition 3.27 (maximum likelihood estimation (mle)). The *maximum likelihood estimator (mle)* $\hat{\theta}_{\text{mle}}$ is the value of $\theta \in \Theta$ that maximizes the likelihood function $L(\theta \mid x)$, i.e.,

$$\hat{\theta}_{\text{mle}}(x) \in \operatorname*{arg\,max}_{\theta \in \Theta} L(\theta \mid x) = \operatorname*{arg\,max}_{\theta \in \Theta} \ell(\theta \mid x) \,.$$

Maximizing the log-likelihood is often mathematically simpler. If $\ell(\theta \mid x)$ is differentiable and the maximum is in the interior of Θ, $\hat{\theta}_{\text{mle}}$ can often be found by solving the *likelihood equation* $\nabla_\theta \ell(\theta \mid x) = 0$ for θ.

Example 3.28 (mle for Gaussian). From example 3.15, we know that for an iid sample $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ we have

$$\ell(\mu, \sigma^2 \mid x) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2 \,.$$

To maximize this over $\theta = (\mu, \sigma^2)$, we compute

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2\sigma^2}\sum_{i=1}^n 2(x_i - \mu)(-1) = \frac{1}{\sigma^2}\left(\sum_{i=1}^n x_i - n\mu\right) \,.$$

Setting this to zero gives $\sum_{i=1}^n x_i - n\mu = 0$, which yields

$$\hat{\mu}_{\text{mle}} = \frac{1}{n}\sum_{i=1}^n x_i = \bar{x}_n \,.$$

For σ^2 we compute

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^n (x_i - \mu)^2 \,.$$

Setting this to zero gives $-\frac{n}{\sigma^2} + \frac{1}{(\sigma^2)^2}\sum(x_i - \mu)^2 = 0$, which yields $\sigma^2 = \frac{1}{n}\sum(x_i - \mu)^2$ and plugging in $\hat{\mu}_{\text{mle}}$ for μ gives

$$\hat{\sigma}^2_{\text{mle}} = \frac{1}{n}\sum_{i=1}^n (x_i - \bar{x}_n)^2 \,.$$

As we have noted before, $\hat{\mu}_{\text{mle}}$ is unbiased (sample mean), but $\hat{\sigma^2}_{\text{mle}}$ is biased.

Example 3.29 (mle for common distributions). Let $X_1, \dots, X_n$ be an iid sample from the respective distributions. Here we list maximum likelihood estimators for the parameters of a few relevant distributions. In general, no closed form expression for mle need to exist so sometimes mle need to be computed numerically.

(i) For a Bernoulli model $\mathrm{Ber}(p)$, we have $\hat{p}_{\mathrm{mle}}(X) = \frac{1}{n}\sum X_i = \bar{X}_n$.

(ii) For a Poisson model $\mathrm{Poi}(\lambda)$ we have $\hat{\lambda}_{\mathrm{mle}}(X) = \bar{X}_n$.

(iii) For an exponential model $\mathrm{Exp}(\lambda)$ we have $\hat{\lambda}_{\mathrm{mle}}(X) = \frac{1}{\bar{X}_n}$.

(iv) For a Gaussian model $\mathcal{N}(\mu, \sigma^2)$ we found that $\hat{\mu}_{\mathrm{mle}}(X) = \bar{X}_n$ and $\hat{\sigma}^2_{\mathrm{mle}}(X) = \frac{1}{n}\sum(X_i - \bar{X}_n)^2$.

(v) For a uniform model $\mathrm{Unif}(0, \theta)$ you found (on the exercises) that $\hat{\theta}_{\mathrm{mle}}(X) = X_{(n)} = \max\{X_1, \dots, X_n\}$.

Remark 3.30 (properties of mle). While in general, we cannot give many guarantees for mle for finite n (see example 3.24), under fairly mild regularity conditions (which hold for most common models and we assume to always hold) mle have desirable asymptotic properties as $n \to \infty$.

(i) **Consistency:** $\hat{\theta}_{\mathrm{mle}} \xrightarrow{P} \theta$.

(ii) **Asymptotic normality:** $\sqrt{n}(\hat{\theta}_{\mathrm{mle}} - \theta) \xrightarrow{d} \mathcal{N}(0, V(\theta))$, where $V(\theta) = I(\theta)^{-1}$ is the Fisher information matrix, an extremely important object in statistics which we won't have time to cover.

(iii) **Asymptotic efficiency:** mle are asymptotically efficient, meaning they achieve the lowest possible variance among a large class of estimators as $n \to \infty$.

(iv) **Invariance:** If $\hat{\theta}_{\mathrm{mle}}$ is the mle of θ, and g is a function, then $g(\hat{\theta}_{\mathrm{mle}})$ is the mle of $g(\theta)$ (provided g does not reduce the dimensionality of Θ). For instance, for a normal distribution, the mle of the standard deviation σ is $\sqrt{\hat{\sigma}^2_{\mathrm{mle}}}$.

Example 3.31 (variance estimation). We continue from example 3.22 and compare the two estimators $S_n^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ for σ^2. The former is an unbiased estimator for the variance in any model. The latter is the mle for the variance in a Gaussian model. Both are estimators for σ^2 in any model (remember that "anything is a valid estimator").

(i) **Deriving the expectation:** We first compute the expectation of the sum of squares. A key algebraic identity is

$$\begin{aligned}
\sum_{i=1}^{n}(X_i - \bar{X}_n)^2 &= \sum_{i=1}^{n}((X_i - \mu) - (\bar{X}_n - \mu))^2 \\
&= \sum_{i=1}^{n}(X_i - \mu)^2 - 2(\bar{X}_n - \mu)\sum_{i=1}^{n}(X_i - \mu) + n(\bar{X}_n - \mu)^2 \\
&= \sum_{i=1}^{n}(X_i - \mu)^2 - 2n(\bar{X}_n - \mu)^2 + n(\bar{X}_n - \mu)^2 \\
&= \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X}_n - \mu)^2 \,.
\end{aligned}$$

Taking the expectation and using linearity, we get

$$\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X}_n)^2\right] &= \sum_{i=1}^{n}\mathbb{E}[(X_i - \mu)^2] - n\,\mathbb{E}[(\bar{X}_n - \mu)^2] \\
&= \sum_{i=1}^{n}\operatorname{var}[X_i] - n\operatorname{var}[\bar{X}_n] \\
&= n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2 \,.
\end{aligned}$$

(ii) **Bias of the estimators:** The expectation of the mle is $\mathbb{E}[\hat{\sigma}^2] = \frac{1}{n}\,\mathbb{E}\left[\sum(X_i - \bar{X}_n)^2\right] = \frac{n-1}{n}\sigma^2$, so it is biased. The expectation of the sample variance is

$$\mathbb{E}[S_n^2] = \frac{1}{n-1}\,\mathbb{E}\left[\sum(X_i - \bar{X}_n)^2\right] = \sigma^2 \,,$$

so it is unbiased. This is where Bessel's correction comes from. Does this mean that S_n^2 is better?

(iii) **mse for normal data:** Now assume $X_i \sim \mathcal{N}(\mu, \sigma^2)$. In this case, one can compute (we skip this) $\operatorname{var}[S_n^2] = \frac{2\sigma^4}{n-1}$. Since S_n^2 is unbiased, this gives

$$\operatorname{mse}[S_n^2] = \operatorname{var}[S_n^2] = \frac{2\sigma^4}{n-1} \,.$$

For $\hat{\sigma}^2$ one can compute (we skip it)

$$\operatorname{mse}[\hat{\sigma}^2] = \operatorname{var}[\hat{\sigma}^2] + (\operatorname{bias}[\hat{\sigma}^2])^2 = \frac{2(n-1)\sigma^4}{n^2} + \left(-\frac{\sigma^2}{n}\right)^2 = \frac{(2n-1)\sigma^4}{n^2} \,.$$

Since $\frac{2}{n-1} > \frac{2n-1}{n^2}$ for all $n > 1$, $\hat{\sigma}^2$ has better mse than S_n^2.

This is a classic example supporting the bias-variance trade-off idea: accepting a small bias leads to a reduction in variance that is large enough to improve the overall mse of the estimator.

Overview 3.32 (outlook on point estimation). Maximum likelihood estimation provides a general principle for constructing estimators with good asymptotic properties. However, there is a rich classical theory for finding and evaluating estimators, especially optimal estimators for finite samples, that we do not have time to cover. This framework provides a powerful toolkit for understanding the limits of estimation and for systematically constructing estimators with optimal properties. Here is a brief glimpse into some of the central concepts and their relationships.

At the heart of this theory lies the *Fisher information*

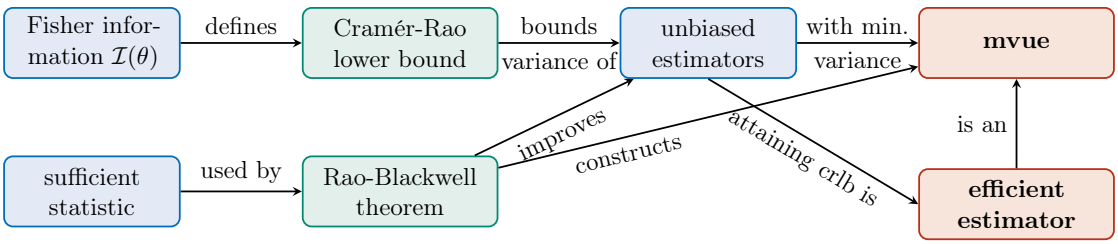
$$\mathcal{I}(\theta) = \mathbb{E}\Big[\Big(\frac{\partial}{\partial\theta}\ell(\theta \mid x)\Big)^2 \mid \theta\Big] = -\,\mathbb{E}\Big[\frac{\partial^2}{\partial\theta^2}\ell(\theta \mid x) \mid \theta\Big]\,,$$

which quantifies how much information an observable random variable X carries about an unknown parameter θ. (The second equality only holds under certain regularity conditions.) So the Fisher information is the variance of the *score* $\partial/\partial\theta\ell(\theta \mid x)$, which can also be written as the negative expected second derivative (Hessian for multi-variate θ). The Fisher information sets a fundamental limit on the precision of estimators via the *Cramér-Rao lower bound (crlb)*, which states that for any unbiased estimator $\hat{\theta}$, its variance is bounded by $\mathrm{var}[\hat{\theta}] \geq 1/\mathcal{I}(\theta)$.

This bound gives rise to the notion of efficiency. An unbiased estimator is called *efficient* if its variance attains the crlb and is thus an mvue. While every efficient estimator is an mvue, an mvue might exist even if no estimator reaches the crlb (making lots of room for further research).

To construct such optimal estimators, the concept of sufficiency is key. A statistic $T(X)$ is *sufficient* if it captures all information about θ from the sample. The *Rao-Blackwell theorem* provides a method to improve any unbiased estimator by conditioning it on a sufficient statistic, a process that often leads to an mvue.

A related but more restrictive concept is the *best linear unbiased estimator (blue)*, which is the mvue within the class of linear estimators. The famous Gauss-Markov theorem states that the ordinary least squares estimator is the blue for linear regression models under certain assumptions. The interplay of these concepts is illustrated below.



3.3. Hypothesis testing

Section goal. Hypothesis testing provides a formal framework for making decisions about claims on population (true) parameters based on sample data/observations. We formulate a "null hypothesis" (H_0), often representing a default state, no effect, or a statement to be falsified, and an "alternative hypothesis" (H_1), representing the claim we wish to support. Then we use the observed data to decide (ideally with high confidence) whether we can reject the null hypothesis.

Overview 3.33 (hypothesis testing motivation). A *statistical hypothesis* is a claim about the parameters of a statistical model. In hypothesis testing, we typically consider two competing, mutually exclusive hypotheses.

- The *null hypothesis*, denoted H_0, is a statement of no effect or no difference, often representing a default or status quo position. It consists of a subset $\Theta_0 \subseteq \Theta$ of the parameter space.
- The *alternative hypothesis*, denoted H_1 (or H_A), is a statement that contradicts the null hypothesis, representing the claim we wish to find evidence for. It specifies a subset $\Theta_1 \subseteq \Theta$ such that $\Theta_0 \cap \Theta_1 = \emptyset$.

For a sample $X = (X_1, \dots, X_n) \sim P_\Theta$, the goal is then to decide whether $(\Omega, \mathcal{A}, P_{\Theta_0})$ is misspecified, that is, whether we can decide that the true parameter is not in Θ_0. That would mean that the true distribution (parameter) could not have been in P_Θ and we can reject the null hypothesis. This is a binary decision based only on the sample X, so a test maps observed data x to $\{0, 1\}$—whether the null should be rejected or not. To design such a decision rule, we typically consider a real-valued statistic $T(X)$ for which we can derive its distribution (approximately) assuming the null is true. That is, for an observed sample, we can compute a real number $T(x)$ and we know which distribution this value would come from, if the null hypothesis were true. With this knowledge, we can then judge how likely the value $T(x)$ of the actually observed data $x = (x_1, \dots, x_n)$ is under this distribution. When this appears exceedingly unlikely (according to some pre-selected fixed threshold), we reject the null.

Definition 3.34 (building blocks & properties of hypothesis tests). Let $(\Omega, \mathcal{A}, P_\Theta)$ be a statistical model and let $\Theta = \Theta_0 \mathring{\cup} \Theta_1$ be a partition of the parameter space.

(i) A hypothesis is *simple* if $|\Theta_0| = 1$ (or $|\Theta_1| = 1$) and we then denote the single element by θ_0 (or θ_1). Otherwise ($|\Theta_{0/1}| > 1$), it is called *composite*. Often, there is also a special $\theta_{0/1} \in \Theta_{0/1}$ for composite statements, e.g., $\Theta_0 = \{\theta \in \Theta \mid \theta > \theta_0\}$.

(ii) For $X \sim P_\Theta$, the *null hypothesis H_0 is valid* if $(\Omega, \mathcal{A}, P_{\Theta_0})$ is well-specified. Otherwise, the *alternative H_1 is valid*.

(iii) A *statistical test* is a measurable map

$$\varphi : \Omega^n \to [0,1]\,,$$

which is interpreted as a decision rule for whether to reject the null or not. If $\varphi(\mathcal{X}) \subseteq \{0,1\}$, we call the test *non-randomized*, otherwise we call it *randomized*. We focus primarily on non-randomized tests.

(iv) For non-randomized tests, by convention we interpret $\varphi(x) = 1$ as rejecting the null. The set $\varphi^{-1}(\{1\}) = \{x \in \Omega^n \mid \varphi(x) = 1\}$ is called *rejection region*.

- Rejecting the null hypothesis H_0 when it is actually true, i.e., $\varphi(x) = 1$ albeit P_{Θ_0} well-specified, is called *type I error*.
- Failing to reject the null hypothesis H_0 when it is actually false, i.e., $\varphi(x) = 0$ albeit P_{Θ_1} well-specified, is called a *type II error*.

	decision	
true state	fail to reject H_0	reject H_0
H_0 is true	correct decision	type I error
H_1 is true	type II error	correct decision

(v) The *significance level* or *size* of a test φ is the maximum probability of a type I error

$$\alpha := \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\varphi]\,.$$

A test φ is called *test at (significance) level α* for $\alpha \in [0,1]$ if $\sup_{\theta\in\Theta_0} \mathbb{E}_\theta[\varphi] \leq \alpha$.

(vi) The *power function* of a test is

$$G_\varphi : \Theta \to [0,1],\ \theta \mapsto \mathbb{E}_\theta[\varphi]\,.$$

For $\theta \in \Theta_1$, we call $G_\varphi(\theta)$ the *power of φ against (the alternative) θ*. The power is the probability with which the alternative $\theta \in \Theta_1$ is recognized, when it is true. The probability of a type II error at $\theta \in \Theta_1$ is then $\beta(\theta) := 1 - G_\varphi(\theta)$.

Example 3.35 (hypotheses).

(i) For a coin with unknown probability of heads p, we might want to test if it is fair. We set up the hypotheses $H_0 : p = 0.5$ (a simple hypothesis) versus $H_1 : p \neq 0.5$ (a composite hypothesis).

(ii) Let μ be the average recovery time with a new drug. The current standard is $\mu_0 = 10$ days. To test if the new drug is better, we might formulate $H_0 : \mu \geq 10$ (no improvement or worse) versus $H_1 : \mu < 10$ (improvement). Both are composite. A simpler setup often used is $H_0 : \mu = 10$ versus $H_1 : \mu < 10$.

Remark 3.36 (testing in practice).

(i) There is a deliberate asymmetry in type I and type II errors when designing a test. In hypothesis testing, we primarily want to be able to detect "non-default situations." That is, we want to detect when we should reject the null. Thus, in practice we first fix a desirable significance level α (which determines the rejection region) and then look for a test with (a) $G_\varphi(\theta) \leq \alpha$ for all $\theta \in \Theta_0$, that then (b) maximizes $G_\varphi(\theta)$ for all $\theta \in \Theta_1$. That is, we demand an upper bound on type I errors, and within that requirement seek to then minimize type II errors.

(ii) For a fixed sample size n, there is an inherent trade-off between α and β. Decreasing the probability of a type I error (decreasing α) will generally make the rejection region smaller. Intuitively, with smaller α, more extreme observations are required to reject the null. At the same time, lowering α—being more conservative in rejecting the null—increases the probability of a type II error $\beta(\theta)$ for at least some alternatives $\theta \in \Theta_1$ and thus decreases the power against certain alternatives.

(iii) Note that not rejecting the null does not amount to accepting the null or having found strong evidence for the null. The absence of evidence against the null is not the same as evidence in favor of the null. A test can have weak power regardless of the significance level.

(iv) For randomized tests, when $\varphi(x) \in (0,1)$ one typically decides to reject the null with probability $\varphi(x)$. That is, the binary rejection decision is a single draw from a $\mathrm{Ber}(\varphi(x))$-distributed RV. Randomized hypothesis tests are relevant in the study and construction of "uniformly most powerful" tests, see definition 3.38.

Convention 3.37. From now on, we always consider a statistical model $(\Omega, \mathcal{A}, P_\Theta)$, a partition $\Theta = \Theta_0 \mathring{\cup} \Theta_1$ into the null hypothesis $H_0 : \Theta_0$ and alternative $H_1 : \Theta_1$, and a test φ for testing H_0 against H_1.

Definition 3.38 (uniformly most powerful test (ump)). Let $\alpha \in [0,1]$. We call φ *(uniformly) most powerful test (ump) at level* α, if for any other test ψ of H_0 against H_1 at level α we have

$$G_\varphi(\theta) \geq G_\psi(\theta) \quad \text{for all } \theta \in \Theta_1 \,.$$

Definition 3.39 (unbiased test). Let $\alpha \in [0,1]$. We call φ *unbiased at level* α, if

$$G_\varphi(\theta_0) \leq \alpha \leq G_\varphi(\theta_1) \quad \text{for all } \theta_0 \in \Theta_0,\, \theta_1 \in \Theta_1 \,.$$

Theorem 3.40 (Neyman-Pearson Lemma, 1932). *Consider a binary statistical model $(\Omega, \mathcal{A}, (P_{\theta_0}, P_{\theta_1}))$ in which we test the simple null hypothesis $H_0 : \{\theta_0\}$ against the simple alternative $H_1 : \{\theta_1\}$. Let $L(\theta \mid x)$ be the likelihood function. For a given significance level $\alpha \in (0,1)$, the most powerful test is the* likelihood ratio test (lrt)*, which rejects H_0 if and only if*

$$\frac{L(\theta_1 \mid x)}{L(\theta_0 \mid x)} > k \,,$$

where the constant k is chosen such that the test has size α (implying that such a test exists).

Example 3.41 (testing the mean of a Gaussian). Let $X_1, \dots, X_n$ be an iid sample from a $\mathcal{N}(\mu, \sigma^2)$ distribution with unknown mean μ but known variance, i.e., $\mu = \theta \in \Theta = \mathbb{R}$. We test a simple null $H_0 : \{\mu_0\}$ against a simple alternative $H_1 : \{\mu_1\}$, where we assume $\mu_1 > \mu_0$ for concreteness. The likelihood ratio is

$$\begin{aligned}\Lambda(x) &:= \frac{\exp\left(-\frac{1}{2\sigma^2}\sum(x_i-\mu_1)^2\right)}{\exp\left(-\frac{1}{2\sigma^2}\sum(x_i-\mu_0)^2\right)} \\ &= \exp\left(\frac{1}{2\sigma^2}\left[\sum(x_i-\mu_0)^2 - \sum(x_i-\mu_1)^2\right]\right) \\ &= \exp\left(\frac{1}{2\sigma^2}\Big(2(\mu_1-\mu_0)\sum x_i + n(\mu_0^2-\mu_1^2)\Big)\right) .\end{aligned}$$

The test rejects H_0 if $\Lambda(x) > k$ for some k. With the monotonicity of the logarithm, we look for

$$\frac{1}{2\sigma^2}\Big(2(\mu_1-\mu_0)\sum x_i + n(\mu_0^2-\mu_1^2)\Big) > \ln(k) .$$

With $\mu_1 > \mu_0$, we rearrange to isolate the sample mean $\bar{x}_n = \frac{1}{n}\sum_{i=1}^n x_i$ to get $\bar{x}_n > c$ for some constant c (depending on k). Hence, the ump for this problem depends only on the sample mean! To achieve size α, we look for c such that

$$P_{\mu_0}(\bar{X}_n > c) = \alpha .$$

Under H_0, $\bar{X}_n \sim \mathcal{N}(\mu_0, \sigma^2/n)$ and with standardized $Z = (\bar{X}_n - \mu_0)/(\sigma/\sqrt{n}) \sim \mathcal{N}(0,1)$, we are looking for c such that

$$P\left(Z > \frac{c-\mu_0}{\sigma/\sqrt{n}}\right) = \alpha .$$

For this, we simply rewrite

$$\frac{c-\mu_0}{\sigma/\sqrt{n}} = z_\alpha \qquad \Longrightarrow \qquad c = \mu_0 + z_\alpha\frac{\sigma}{\sqrt{n}} .$$

where z_α is such that $P(Z \geq z_\alpha) = \alpha$ of the standard normal distribution. Note that z_α is the the $1-\alpha$-quantile of the standard normal. This is a common convention in testing that t_α denotes the $1-\alpha$-quantile of some random variable T.

In summary, the ump is to reject H_0 whenever $\bar{x}_n > \mu_0 + z_\alpha\frac{\sigma}{\sqrt{n}}$. This is called the *one-sided z-test*. (If we had assumed $\mu_1 < \mu_0$, the rejection region would be of the form $\bar{x}_n < c'$.)

Viewpoint 3.42 (constructing tests in practice). So far, theorem 3.40 is the only example we have seen for constructing tests φ and example 3.41 demonstrated how this can work. The view of a test as $\varphi : \mathcal{X} \to \{0,1\}$ is powerful for theoretical analysis, but in practice, one does not typically try to just write down the decision rule φ directly. Instead, the following steps can be extracted from example 3.41:

(i) Define a real-valued *test statistic* $T : \mathcal{X} \to \mathbb{R}$, which can be any statistic. When testing a mean, taking the sample mean is a good starting point. When testing a variance, a sample variance may be a good starting point. These can get more sophisticated though. The goal is to reduce the high-dimensional sample x to a single number $T(x)$.

(ii) Try to choose T such that you can (approximately) derive the distribution of $T(X)$ if the null were true. In example 3.41, we had $T_n(X) = \bar{X}_n$ and directly knew that $T(X) \sim \mathcal{N}(\mu_0, \sigma^2/n)$ under the null. In other settings, we may invoke the clt to assume an approximate normal of our test statistic or find other ways to approximate $T(X)$ under $X \sim P_{\theta_0}$ (where $\theta_0 \in \Theta_0$).

(iii) Define the rejection region R via quantiles of $T(X)$ under the null. For example, when for the observed sample x we find $t_{\text{obs}} = T(x) > t_\alpha$ (which, remember is the $1-\alpha$-quantile of $T(X)$), we reject the null. (Instead we could define the rejection region to be outside both the $\alpha/2$ and $1-\alpha/2$ quantiles.) The value t_α is also called *critical value* and the rejection region *critical region*. For composite hypothesis with $|\Theta_0| > 1$, where there is no single distribution of $T(X)$ under the null, remember that $\alpha = \sup_{\theta \in \Theta_0} G_\varphi(\theta)$, so this is what determines R. In many settings, where $\Theta \subset \mathbb{R}$ is an interval, α is monotonic in θ and only one of the boundary values of Θ_0 has to be considered (the *null boundary*).

(iv) Define the final test as

$$\varphi(x) = \chi_R(T(x)) := \begin{cases} 1 & \text{if } T(x) \in R \quad (\text{reject } H_0)\,, \\ 0 & \text{if } T(x) \notin R \quad (\text{don't reject } H_0)\,. \end{cases}$$

The central tasks in constructing a test are choosing an appropriate statistic $T(X)$ and determining the rejection region R by studying the distribution of $T(X)$ under the null hypothesis to ensure the test has the desired size α.

Definition 3.43 (one/two-sided tests). Let $\Theta \subset \mathbb{R}$.

- A *one-sided* (or *one-tailed*) test is used for directional hypotheses, such as $H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$. The rejection region is entirely in one tail of the distribution of the test statistic.
- A *two-sided* (or *two-tailed*) test is used for non-directional hypotheses, such as $H_1 : \theta \neq \theta_0$. The rejection region is split between the two tails of the distribution.

For a given α, a one-sided test is more powerful for detecting an effect in the specified direction, but has no power to detect an effect in the opposite direction.

Example 3.44 (z-test for a normal mean (again)). Let $X_1, \ldots, X_n$ be an iid sample from $\mathcal{N}(\mu, \sigma^2)$, where σ^2 is known. We want to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$. Following

example 3.41, we use as test statistic is the standardized sample mean

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \, .$$

Under H_0, $Z \sim \mathcal{N}(0,1)$. We reject H_0 when the observed z_{obs} is "too far in either tail of $\mathcal{N}(0,1)$." Concretely, for a fixed level $\alpha \in (0,1)$, we reject if $|z_{\text{obs}}| > z_{\alpha/2}$, where $z_{\alpha/2}$ is the $1-\alpha/2$-quantile of $\mathcal{N}(0,1)$.

For example, if $\alpha = 0.05$, then $z_{\alpha/2} \approx 1.96$. So we reject H_0 if $|z_{\text{obs}}| > 1.96$.

Example 3.45 (one-sample t-test for mean). If σ^2 in the previous example is unknown, we estimate it with the sample variance S_n^2. The test statistic becomes $T = \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}}$. Under H_0, this statistic follows a Student's t-distribution with $n-1$ degrees of freedom, $T \sim t_{n-1}$. The rejection region for a two-sided test is $|T_{obs}| > t_{n-1,\alpha/2}$, where $t_{n-1,\alpha/2}$ is the $1-\alpha/2$-quantile of the t_{n-1} distribution.

Example 3.46 (common Gaussian tests). Let $X_1, \ldots, X_n$ be iid from $\mathcal{N}(\mu, \sigma^2)$ and $\alpha \in (0,1)$. We denote the upper α-quantile (i.e., the $1-\alpha$-quantile) of a distribution by a subscript, e.g., z_α is the value such that $P(Z > z_\alpha) = \alpha$ for $Z \sim \mathcal{N}(0,1)$.

(i) **One-sided z-test for the mean** with known σ^2. To test $H_0 : \mu \leq \mu_0$ against $H_1 : \mu > \mu_0$, we use the statistic

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1) \quad \text{(under } H_0 \text{ boundary)} \, ,$$

and reject H_0 if $z_{\text{obs}} > z_\alpha$.

(ii) **Two-sided z-test for the mean** with known σ^2. To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we use the same statistic Z and reject H_0 if $|z_{\text{obs}}| > z_{\alpha/2}$.

(iii) **One-sided t-test for the mean** with unknown σ^2. To test $H_0 : \mu \leq \mu_0$ against $H_1 : \mu > \mu_0$, we use the statistic

$$T = \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}} \sim t_{n-1} \quad \text{(under } H_0 \text{ boundary)} \, ,$$

where S_n is the sample standard deviation. We reject H_0 if $t_{\text{obs}} > t_{n-1,\alpha}$.

(iv) **Two-sided t-test for the mean** with unknown σ^2. To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we use the same statistic T and reject H_0 if $|t_{\text{obs}}| > t_{n-1,\alpha/2}$.

(v) **One-sided χ^2-test for the variance** with unknown μ. To test $H_0 : \sigma^2 \leq \sigma_0^2$ against $H_1 : \sigma^2 > \sigma_0^2$, we use the statistic

$$C = \frac{(n-1)S_n^2}{\sigma_0^2} \sim \chi^2_{n-1} \quad \text{(under } H_0 \text{ boundary)} \, ,$$

and reject H_0 if $c_{\text{obs}} > \chi^2_{n-1,\alpha}$. For a left-sided test ($H_1 : \sigma^2 < \sigma_0^2$), we reject if $c_{\text{obs}} < \chi^2_{n-1,1-\alpha}$.

(vi) **Two-sided χ^2-test for the variance** with unknown μ. To test $H_0 : \sigma^2 = \sigma_0^2$ against $H_1 : \sigma^2 \neq \sigma_0^2$, we use the same statistic C and reject if $c_{\text{obs}} < \chi^2_{n-1,1-\alpha/2}$ or $c_{\text{obs}} > \chi^2_{n-1,\alpha/2}$.

(vii) **Two-sample t-test for means** with unknown but equal variances. Let $X_1, \dots, X_{n_X}$ and $Y_1, \dots, Y_{n_Y}$ be two independent iid samples from $\mathcal{N}(\mu_X, \sigma^2)$ and $\mathcal{N}(\mu_Y, \sigma^2)$. To test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X \neq \mu_Y$, we use the statistic

$$T = \frac{\bar{X}_{n_X} - \bar{Y}_{n_Y}}{S_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} \sim t_{n_X+n_Y-2} \quad (\text{under } H_0)\,,$$

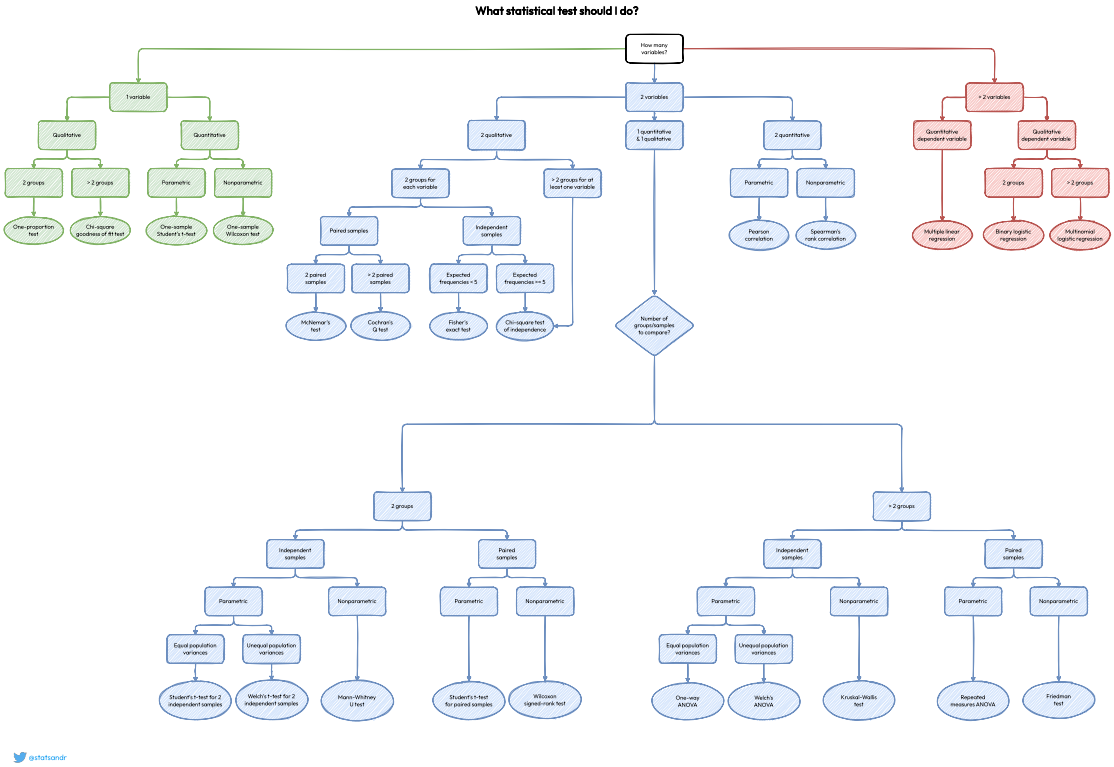
where $S_p^2 = \frac{(n_X-1)S^2_{n_X} + (n_Y-1)S^2_{n_Y}}{n_X+n_Y-2}$ is the pooled sample variance. We reject H_0 if $|t_{\text{obs}}| > t_{n_X+n_Y-2,\alpha/2}$.

Note that ultimately, for many of these tests we only need the distribution of the test statistic under the null. Due to the clt, when n is large, we can relax the assumption to X_i being iid with mean μ and finite variance $\sigma^2 > 0$ (need not be normal) to find $\bar{X}_n - \mu/(\sigma/\sqrt{n})$ approximately distributed like $\mathcal{N}(0,1)$ and we can still use the corresponding tests.

Here's an overview of these standard tests. Note that one/two-sided versions of these tests use the same statistics and null distributions but adjust the rejection region to one or both tails.

name	$\boldsymbol{H_0}$	**statistic**	**null distr.**
single mean μ			
z-test (σ^2 known)	$\mu = \mu_0$	$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}$	$\mathcal{N}(0,1)$
t-test (σ^2 unknown)	$\mu = \mu_0$	$T = \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}}$	t_{n-1}
single variance σ^2			
χ^2-test (μ unknown)	$\sigma^2 = \sigma_0^2$	$C = \frac{(n-1)S_n^2}{\sigma_0^2}$	χ^2_{n-1}
comparing means μ_X, μ_Y			
two-sample t-test (equal variances)	$\mu_X = \mu_Y$	$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}}$	$t_{n_X+n_Y-2}$

Remark 3.47 (testing zoo). As you can imagine there is a large zoo of different statistical tests for different questions, different distributional assumptions, and prior knowledge. You'll find many flowcharts (e.g., this one) and guides on which test to use in which situation online. Many of them are robust approximations and they all have different gotchas, which you will now be able to understand.



Definition 3.48 (p-value). The viewpoint 3.42 and definition 3.43 give rise to the idea of a *p-value*, which is the probability, assuming the null hypothesis H_0 is true, of observing a test statistic at least as extreme as the one computed from the sample data. The direction of "extremeness" is determined by the alternative H_1. For an observed test statistic t_{obs}:

- if $H_1 : \theta > \theta_0$, p-value $= P_{\theta_0}(T(X) \geq t_{\text{obs}})$.
- if $H_1 : \theta < \theta_0$, p-value $= P_{\theta_0}(T(X) \leq t_{\text{obs}})$.
- if $H_1 : \theta \neq \theta_0$, p-value $= P_{\theta_0}(T(X)\text{as or more extreme than } t_{\text{obs}})$.

The test is then $\varphi(x) = 1$ if the p-value is less than or equal to the chosen significance level $\alpha \in [0, 1]$.

Viewpoint 3.49 (falsification vs. confirmation). The logic of hypothesis testing is rooted in the philosophy of falsification, popularized by Karl Popper. We cannot prove a scientific theory (or H_0) true, because we can never observe all possible instances. However, we can find evidence that falsifies it. Hypothesis testing adopts this principle: we never "accept" H_0. We either find enough evidence to "reject" it in favor of H_1, or we "fail to reject" it due to insufficient evidence. This distinction is crucial. Failing to find evidence against a claim is not the same as finding evidence for it.

Caveat 3.50 (p-value misinterpretations). The p-value is one of the most misinterpreted concepts in statistics.

- The p-value is *not* the probability that the null hypothesis is true, i.e., p-value $\neq P(H_0 \text{ is true} \mid \text{data})$. This is a very common mistake. The p-value is calculated *assuming* H_0 is true.
- A large p-value (e.g., $p > 0.05$) does not mean H_0 is true. It only means the data are not inconsistent with H_0. It could also mean the test had low power to detect a true effect.
- Statistical significance (a small p-value) does not imply practical significance. A tiny effect can be statistically significant with a large enough sample size, but may be irrelevant in practice.
- In practice, instead of comparing the p-value to a pre-chosen size α, one directly reports p-values and says a test was *statistically significant* if p-value ≤ 0.05 or *highly statistically significant* if p-value ≤ 0.001 (where 0.05 and 0.001 are typical default values the community has converged on). This further suggests the p-value as a continuous indicator of "how true the claim is." It is not.
- Other fields have much stricter "p-value standards." In physics, a discovery is often considered real when it is 5σ or 6σ, which means it is outside of 5 or 6 standard deviations of the normal distribution, which amounts to a p-value close to $2.86 \cdot 10^{-7}$. In genome wide association studies, thresholds of $5 \cdot 10^{-8}$ for rejection are not uncommon.

The American Statistical Association (ASA) published a statement in 2016 to address the widespread misuse of p-values. In another statement they said

> *We conclude, based on our review of the articles in this special issue and the broader literature, that it is time to stop using the term "statistically significant" entirely. Nor should variants such as "significantly different," "$p \leq 0.05$," and "nonsignificant" survive, whether expressed in words, by asterisks in a table, or in some other way.*

Remark 3.51 (p-hacking and the replication crisis). *p-hacking* (or data dredging) is the practice of reanalyzing data in many different ways until a statistically significant result is found. This can involve trying different statistical tests, removing "outliers," or testing many different hypotheses. Such practices dramatically increase the rate of false positives and have contributed to the replication crisis in many scientific fields, where published findings fail to be replicated by independent researchers. Preregistering study designs and analysis plans is one proposed remedy.

Remark 3.52 (summary of hypothesis tests). The general procedure for developing a hypothesis test is as follows.

(i) Formulate the null (H_0) and alternative (H_1) hypotheses based on the research question. This includes statistical modeling (writing down your model).

(ii) Choose an appropriate test statistic $T(X)$ that is expected to behave differently under H_0 and H_1. Often this is based on an estimator for the parameter of interest.

(iii) Determine the sampling distribution of the test statistic *under the assumption that H_0 is true*. This is the most critical and often most difficult step.

- For some models (like the Gaussian), the exact distribution is known.
- For large samples, the distribution can often be approximated using the clt.
- In other cases, simulation methods (like permutation tests or bootstrapping) can be used to approximate the null distribution.

(iv) Specify the significance level α and determine the rejection region R based on $T(X)$ under the null distribution.

(v) Collect data, calculate the observed value of the test statistic t_{obs}, and make a decision: reject H_0 if $t_{\text{obs}} \in R$.

Overview 3.53 (outlook on hypothesis testing).

(i) While the Neyman-Pearson lemma provides a constructive way of designing umps for simple hypotheses, for composite hypotheses umps rarely ever exist, except for simple one-sided examples in exponential families, etc. In many real-world settings, the quality of hypothesis tests is evaluated empirically in simulation studies that mimic the expected data characteristics.

(ii) When using statistical tests to make decisions to change something, e.g., switch to a new medication for a given indication, or introduce a new design on a large online platform, it is important to ensure high power of the test. Hence, especially for real-world studies on humans, one typically performs *power analysis* up front to determine which sample size n will be required to achieve a desired level of power (at fixed size α).

(iii) We have covered the frequentist (or Neyman-Pearson) approach to hypothesis testing. This is the dominant paradigm in many scientific fields. However, other approaches exist. *Bayesian hypothesis testing*, for instance, does not use p-values. Instead, it computes the *Bayes factor*, which is the ratio of the marginal likelihood of the data under H_1 to that under H_0. This allows for quantifying evidence in favor of H_0 or H_1.

(iv) A crucial issue in modern applications is *multiple testing*. When many hypotheses are tested simultaneously (e.g., in genomics or A/B testing online), the probability of making at least one type I error inflates dramatically. Procedures like the *Bonferroni correction* or methods to control the *false discovery rate (FDR)* are essential to address this problem.

(v) The framework we discussed, particularly tests for simple hypotheses (like the z-test or t-test), forms the foundation for more complex statistical models, including analysis of variance (ANOVA), chi-squared tests, and regression analysis, where we test hypotheses about model parameters or their relationships.

3.4. Interval estimation

Section goal. While a point estimate provides a single "best guess" for a parameter, it conveys no information about the uncertainty of that guess. An *interval estimate*, such as a confidence interval, provides a range of plausible values for an unknown parameter, accompanied by a level of confidence.

Definition 3.54 (confidence interval). A $100(1-\alpha)\%$ *confidence interval (ci)* for a parameter of interest $\gamma(\theta) \in \mathbb{R}$ is an interval with random endpoints $(L(X), U(X))$, computed from the sample $X = (X_1, \dots, X_n)$, such that

$$P_\theta(L(X) \leq \gamma(\theta) \leq U(X)) \geq 1 - \alpha \quad \text{for all } \theta \in \Theta\,.$$

The value $1-\alpha$ is the *confidence level*. The key is that the interval is random, while the parameter is treated as a fixed, unknown constant.

Remark 3.55 (interpretation of confidence intervals). The frequentist interpretation of a 95% confidence interval is as follows: If we were to repeat our sampling procedure a large number of times, each time constructing a 95% confidence interval, then about 95% of these randomly generated intervals would contain the true, fixed parameter value. It is *incorrect* to say that for a specific, calculated interval, there is a 95% probability that the true parameter lies within it. Once calculated, the interval is fixed, and the true parameter is either in it or not. The probability statement applies to the procedure, not a particular outcome.

Example 3.56 (ci for normal mean, known variance). Let $X_1, \dots, X_n$ be iid $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known. The quantity $Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$ is a *pivotal quantity* because its distribution does not depend on the unknown parameter μ. We can find a critical value $z_{\alpha/2}$ such that $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1-\alpha$. Rearranging the inequality for μ yields

$$P\left(\bar{X}_n - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha\,.$$

Thus, the $100(1-\alpha)\%$ CI for μ is $\bar{X}_n \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$.

Example 3.57 (ci for normal mean, unknown variance). If σ^2 is unknown, we use the pivotal quantity $T = \frac{\bar{X}_n - \mu}{S_n/\sqrt{n}}$, which follows a Student's t-distribution with $n-1$ degrees

of freedom. The $100(1-\alpha)\%$ CI for μ is

$$\bar{X}_n \pm t_{n-1,\alpha/2} \frac{S_n}{\sqrt{n}} ,$$

where $t_{n-1,\alpha/2}$ is the critical value from the t_{n-1} distribution. The t-distribution has heavier tails than the normal, accounting for the extra uncertainty of estimating σ^2.

Example 3.58 (ci for a proportion). Let $X_1, \ldots, X_n$ be iid $\mathrm{Ber}(p)$. Let $\hat{p} = \bar{X}_n$. For large n, the clt implies that $\frac{\hat{p}-p}{\sqrt{p(1-p)/n}}$ is approximately distributed like $\mathcal{N}(0,1)$. By replacing the unknown p in the standard error with its estimate $\hat{p}$, we get the *Wald interval*, an approximate $100(1-\alpha)\%$ CI for p:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} .$$

This approximation performs poorly for small n or when p is near 0 or 1. Other methods like the Wilson score interval or the Clopper-Pearson interval provide better coverage properties in those cases.

Remark 3.59 (duality between cis and hypothesis tests). There is a direct correspondence between two-sided hypothesis tests and confidence intervals. A $100(1-\alpha)\%$ confidence interval for a parameter θ consists of all values θ_0 for which the null hypothesis $H_0 : \theta = \theta_0$ would *not* be rejected at significance level α. Conversely, if a test rejects $H_0 : \theta = \theta_0$, then θ_0 will not lie in the $100(1-\alpha)\%$ CI. This duality provides two perspectives on the same inferential problem.

— Start Philosophical Interlude —

3.5. Bayesian vs. Frequentist Inference

Overview 3.60 (two schools of thought). The methods discussed so far—mle, hypothesis testing with confidence levels or p-values, and confidence intervals—belong to the frequentist school of statistics. This school treats parameters as fixed, unknown constants and defines probability as the long-run frequency of outcomes in (hypothetical) repeated experiments. Bayesian inference offers an alternative perspective, treating parameters as random variables about which we can have beliefs that are updated with data.

Definition 3.61 (Bayesian statistical model). A (parametric) Bayesian statistical model has the following constituents.

(i) A *likelihood* $p(x \mid \theta)$, which is the same as in the frequentist model, describing the data generating process for a given parameter θ.

(ii) A *prior distribution* $\pi(\theta)$, which represents our belief about the parameter θ *before* observing the data. It is a probability distribution over the parameter space Θ.

Definition 3.62 (posterior distribution). Using Bayes' rule, the prior belief is updated with the observed data x to form the *posterior distribution* $\pi(\theta \mid x)$ according to

$$\pi(\theta \mid x) = \frac{p(x \mid \theta)\pi(\theta)}{p(x)} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} .$$

The denominator, $p(x) = \int_\Theta p(x \mid \theta)\pi(\theta)d\theta$, is the *marginal likelihood* of the data and acts as a normalizing constant. The posterior distribution $\pi(\theta \mid x)$ encapsulates all our knowledge about θ after seeing the data. We often work with the unnormalized posterior $\pi(\theta \mid x) \propto p(x \mid \theta)\pi(\theta)$.

Definition 3.63 (Bayesian point estimators). Point estimates for θ can be derived from the posterior distribution in different ways. Common choices include the

- *posterior mean*, $\hat{\theta}_{\text{mean}} = \mathbb{E}[\theta \mid x] = \int \theta\pi(\theta \mid x)d\theta$, an estimator that minimizes the posterior expected squared error.
- *posterior median*, the value $\hat{\theta}_{\text{med}}$ such that $P(\theta \leq \hat{\theta}_{\text{med}} \mid x) = 0.5$. It minimizes the posterior expected absolute error.
- *maximum a posteriori (map)* estimate, $\hat{\theta}_{\text{map}} = \arg\max_\theta \pi(\theta \mid x) = \arg\max_\theta p(x \mid \theta)\pi(\theta)$, the mode of the posterior. This is perhaps the most common choice for a point estimator derived from Bayesian inference and is akin to the mle when not considering a prior.

Definition 3.64 (credible interval). A $100(1-\alpha)\%$ *credible interval* for a parameter θ is an interval (l, u) such that

$$P(l \leq \theta \leq u \mid x) = \int_l^u \pi(\theta \mid x)d\theta = 1 - \alpha .$$

Unlike frequentist confidence intervals, the boundaries l, u are fixed numbers and we now indeed interpret the credible interval as the range in which the ture parameter lies with probability $1 - \alpha$, given the data and the model. A common choice is the *equal-tailed interval*, where the lower and upper bounds are the $\alpha/2$ and $1 - \alpha/2$ quantiles of the posterior distribution. This is in line with the posterior representing our final beliefs about the true parameter value.

Remark 3.65 (confidence vs. credible intervals). The interpretation of a credible interval is direct and intuitive: "Given the data, there is a 95% probability that the true parameter θ lies in this specific interval." This is fundamentally different from the frequentist confidence interval, where the probability statement is about the procedure of generating intervals, not about a specific interval covering the fixed parameter. The confidence interval is a statement about the data, while the credible interval is a statement about the parameter. The fact that it is a common pitfall to interpret frequentist confidence

intervals as if they were Bayesian credible intervals often fuels arguments that we should just always use Bayesian analysis and credible intervals instead of frequentist methods.

Viewpoint 3.66 (the philosophical divide and practical convergence). The core philosophical difference lies in the definition of probability and the nature of parameters. Frequentists view parameters as fixed constants and probability as a long-run frequency. Bayesians view parameters as random variables representing states of knowledge and probability as a degree of belief. In practice, for large sample sizes and under non-informative priors, Bayesian and frequentist results are often numerically similar. For example, the credible interval for a normal mean with a flat prior is identical to the frequentist confidence interval. However, the interpretations remain distinct. The choice between approaches often depends on the specific problem, prior knowledge available, and the desired interpretation of the results.

Specifically, frequentist inference typically has the goal of building methods with frequency guarantees (in repeated sampling), whereas Bayesian inference aims at analyzing degrees of belief. This divide has been well formulated in this stackexchange reply and you can read about common fallacies around confidence intervals in this paper.

— End Philosophical Interlude —

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A. Additional material

A.1. Set theory

Definition A.1 (*(Proper) sub/super-set*). Let X, Y be sets.

- X *subset* of Y (and Y *superset* of X), $X \subseteq Y$ (or $Y \supseteq X$) $:\Leftrightarrow \forall\, x \in X\colon\ x \in Y$
- *Equality of sets*:

$$X = Y \ :\Leftrightarrow\ (X \subseteq Y) \wedge (Y \subseteq X)$$

- X a *proper subset* of Y (and Y *proper superset* of X), $X \subsetneq X$ ($Y \supsetneq X$) $:\Leftrightarrow X \subseteq Y \ \wedge\ X \neq Y$
- *Relation properties of "$\subseteq$"*:

$$X \subseteq X \qquad \text{(reflexivity)}$$
$$(X \subseteq Y) \ \wedge\ (Y \subseteq Z) \ \Rightarrow X \subseteq Z \qquad \text{(transitivity)}$$

Example A.2 (notation).

- $\{x\}$ is the set containing the single element x
- $\{a, b, \ldots, *, \odot\}$ is the set consisting of the elements $a, b, \ldots, *, \odot$
- The *empty subset* $\emptyset$ is the set containing no element.

Definition A.3 (power set). Let X be a set. The *power set of X*, $\mathcal{P}(X)$ (or 2^X) is the set of subsets of X.

Example A.4 (power set of $\emptyset$). We have:

(i) $\mathcal{P}(\emptyset) = \{\emptyset\}$, i.e., the power set of the empty set is the set with the only element being $\emptyset$.

(ii) $\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$

(iii) $\mathcal{P}(\{*, \odot\}) = \{\emptyset, \{*\}, \{\odot\}, \{*, \odot\}\}$

Remark A.5 (Notation 2^Ω). The power set of a set Ω is also often denoted by 2^Ω, which usually stands for the set of all mappings $\Omega \to \{0,1\}$. There is a one to one mapping to $\mathcal{P}(\Omega)$ by interpreting each map as deciding for each element whether it is included (1) or excluded (0) in the subset.

Definition A.6 (*(relative) complement, intersection, disjoint, union*). Let X be a set, $A, B \subseteq X$ subsets.

- the *(relative) complement of B in A* is

$$A \setminus B := \{x \in X \mid (x \in A) \wedge (x \notin B)\}$$

- the *complement of A* is (if the superset is clear from context)

$$A^{\mathsf{c}} := X \setminus A$$

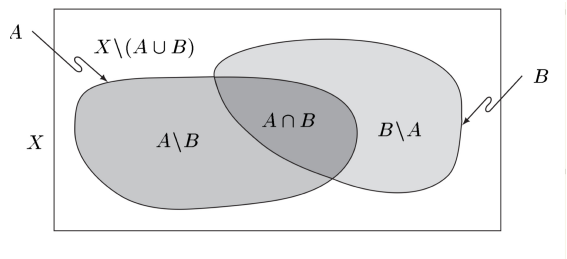
- the *intersection* of A and B is

$$A \cap B := \{x \in X \mid (x \in A) \wedge (x \in B)\}$$

- A and B are called *disjoint* if $A \cap B = \emptyset$
- the *union* of A and B is

$$A \cup B := \{x \in X \mid (x \in A) \vee (x \in B)\}$$

Remark (*Venn diagrams*). One can represent set-relationships by *Venn diagrams* as the one shown below. These are however not suitable for proofs but only to provide intuition.



Definition A.7 (family of sets, index set). Let $I \neq \emptyset$ and A_α a set for each $\alpha \in I$. Then $\{A_\alpha \mid \alpha \in I\}$ is called a *family of sets* or *collection of sets* and I an *index set* for this family. This definition does not exclude that $A_\alpha = A_\beta$ for $\alpha, \beta \in I$ with $\alpha \neq \beta$, that is, the same set can be contained in the family multiple times. Also, $A_\alpha = \emptyset$ for some $\alpha \in I$ can happen, i.e., some members of the family can be empty sets.